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Andrea Andrenelli,
Przemyslaw Kowalski,
Clarisse Legendre,
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Nathan Alan-Lee

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This paper assesses the economic impacts of export restrictions on critical raw materials (CRMs) through case studies of recent measures targeting cobalt, lithium, and nickel in the Democratic Republic of the Congo, Argentina, Zimbabwe, and Indonesia. Using data from the OECD Inventory of Export Restrictions on Industrial Raw Materials and mixed quantitative–qualitative analysis, it examines effects on trade flows, production patterns, and downstream processing, alongside environmental, social, and governance (ESG) implications. The impact of such measures on domestic value addition depended on broader domestic investment conditions. It was generally concentrated in the primary processing steps of the value chain or accrued largely to foreign investors and came with important trade-offs in terms of costs for trading partners, environmental goals, and supply security in global markets. The study highlights the importance of international cooperation to promote stable supply in global markets and sustainable development.

Key words: Critical raw materials, Downstream processing, ESG, Trade policy, Resource-rich economies, Supply security

JEL codes: F13, Q32, Q37, O13, O19

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Key findings

With the energy transition underway, and production of some critical raw materials (CRMs) essential for this transition concentrated while demand is widespread, concerns have been growing about the potential impact of export restrictions on the availability and prices of CRMs in global markets.

Based on descriptive statistical analysis of trade and production data and drawing on the literature on environmental, social and governance (ESG) effects, this report explores three cases of export restrictions, on cobalt, lithium and nickel, to explore their impacts, benefits and limitations in relation to domestic processing and trade. Key findings include:

- Export bans introduced by the Democratic Republic of the Congo and Indonesia appear to have resulted in a significant reduction in exports of primary forms of cobalt and nickel and were correlated with investments in domestic downstream capacity leading to larger downstream production.
- That said, impacts tended to be concentrated in the immediate next processing stages and involved mainly Chinese investors who already controlled these segments of the value chain. The existing literature also suggests that only a part of the value added from the expansion of downstream processing activities accrued to the host economies. In addition, some of these projects had negative ESG effects.
- Argentina's export taxes have helped the country raise the tax revenue but also seem to have played a role in reducing exports. This highlights the potential trade frictions that may emerge if more countries rely on export taxes. This may lead to longer-term impacts for the countries themselves, both in terms of impacts on the development of the industry domestically and in creating incentives for other countries to speed up development of alternatives as they lose faith in global markets as a source of supply.
- While more difficult to assess, trade and domestic economic and ESG effects of Zimbabwe's more recent export ban on primary lithium appear qualitatively similar to those in the DRC and Indonesia.

The domestic economic effects of export restrictions are strongly conditioned by a range of other factors, including those related to the broader investment climate. This, and the growing use of export restrictions, underscores a need for a better understanding of, and efforts to address, the concerns of source countries. Moreover, while export restrictions can in some circumstances have positive effects on domestic processing or fiscal revenue, these can occur at the expense of trading partners and environmental objectives, and could also contribute to further concentration of supply. With no country having all the minerals needed for all purposes and all countries being affected by reduced access to CRMs, there is a strong case for plurilateral or multilateral co-operation to restrain the use of export restrictions and devise more mutually beneficial alternatives.

Emerging inter-governmental agreements and partnerships on CRMs, which can also involve the private sector, such as the Minerals Security Partnership and the European Union's raw materials diplomacy partnerships, can play a role in ensuring that the stable and resilient supply of critical minerals occurs hand-in-hand with value addition and wider economic development in resource-rich developing countries. Greater investment, including in infrastructure and in compliance with ESG standards, such as that pursued through the OECD Forum on Responsible Mineral Supply Chains, can help deliver the economic benefits of greater domestic value addition and promote more sustainable supply without compromising the stability of global markets.

Introduction

With the energy transition underway and rapidly progressing digitalisation, reliance on raw materials which are critical for renewable and digital technologies (hereafter "critical raw materials", CRMs) is raising new challenges. One such challenge is that current reserves, extraction and processing of several CRMs are concentrated geographically, including in terms of ownership or control, while demand is widespread and expected to grow rapidly. This means both that international trade and investment will play an increasingly important role in securing CRMs and technology deployment in the foreseeable future, and that the policies of individual producers may have large spillover effects.

Against this background, one concern is the growing use of export restrictions. These restrictions can have negative effects on international markets and their use, particularly by resource-rich developing and emerging economies, has increased almost fivefold since 2009 (OECD, 2025[1]; Kowalski and Legendre, 2023[2]). Some countries using export restrictions maintain that these measures help them meet domestic policy objectives, such as attracting foreign investment, promoting domestic processing, and reducing negative environmental and social effects of mining and processing of CRMs. Yet, there are debates about whether export restrictions are the best policy instruments to sustainably deliver the intended development objectives. In addition, resort to export restrictions by some countries can create incentives for other countries to also impose them, resulting in spiralling[1] negative effects on international CRM markets.

OECD countries, faced with the challenge of the energy transition, have strong interests in ensuring secure access to competitively priced CRMs. Some OECD countries, such as Australia, Canada and Chile, hold important reserves of CRMs. Many OECD firms hold technologies and capital needed for further exploration and exploitation of CRMs and have experience in implementing CRM-related projects, including according to high environmental, sustainability and governance (ESG) standards. However, the majority of reserves for some CRMs and most production capacity remains in countries currently beyond the OECD Membership.[2] Important reserves of key CRMs, such as cobalt, lithium and nickel, for example, are held by countries that see the growing demand for these minerals as an important economic development opportunity, which would benefit from international cooperation and investment. Strengthening cooperation on CRMs between OECD and resource-rich countries outside the OECD is therefore a possible way forward.

Demand for CRMs is also taking place against a backdrop of growing geopolitical tensions and strategic rivalries. Prior to its invasion of Ukraine and the resulting economic sanctions, the Russian Federation (hereafter "Russia") was an important supplier for some CRMs (e.g. cobalt, nickel, magnesium). More importantly, the People's Republic of China (hereafter "China") has emerged as a major owner of reserves, producer and processor of several CRMs, thanks to both a significant expansion of domestic production and investments in mining and processing activities domestically and in other countries (often supported by government policies).[3] China has also been one of the most prominent users of export restrictions on CRMs in recent years.

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[1] Raising demand for, and prices of, CRMs create incentives for resource-rich countries to use export restrictions to lower prices for domestic users and attract new processing activities. At the same time, these restrictions create also incentives for other producing countries to introduce similar restrictions, putting yet more upward pressure on international prices and ultimately creating more incentives to restrict exports.

[2] It should be noted that Indonesia, an important source of CRMs and the subject of the case study on nickel in this report, has requested to become an OECD Member and its accession roadmap was adopted by the OECD Council on 29 March 2024. In June 2025, on the occasion of the 2025 OECD Ministerial Council Meeting, Indonesia submitted its Initial Memorandum, marking a significant step forward towards accession to the OECD.

[3] Supporting the Chinese industry in expanding abroad and improving China's security of raw materials supplies are some of the objectives of China's Belt and Road Initiative, a strategic government programme and an integral part of China's planning system [e.g. European Commmission (2024[81])]. In addition, Chinese foreign investments in energy and non-energy mining sectors have been driven mainly by Chinese state-owned enterprises (Moses et al., 2024[85]).

Gathering evidence on the international and domestic effects of export restrictions can help motivate and inform international cooperation on CRMs. It is important to better understand the benefits and limitations of export restrictions for achieving economic development goals, as well as to draw on a more granular mapping of how CRMs cross borders to identify where potential bottlenecks for world markets may exist.

The work presented in this report uses the data from the OECD Inventory of Export Restrictions on Industrial Raw Materials and complements the recent stocktaking of the incidence and evolution of export restrictions on key CRMs by Kowalski and Legendre (2023[2]), undertaking exploratory case studies of selected economic effects of export restrictions on cobalt, lithium and nickel introduced recently by the Democratic Republic of Congo (hereafter DRC), Argentina and Zimbabwe, and Indonesia. The principal methodology employed is a descriptive quantitative and qualitative assessment of the evolution of the export restrictions, as well as the impact on trade flows and production of primary and processed products containing the three minerals in the selected countries. This report also draws on findings from literature on ESG effects associated with the sector and the introduction of export restrictions.

The objective of the analysis is to shed more light on the following questions: Do export restriction measures have significant trade effects? For resource-rich countries which choose to introduce export restrictions, are these measures the best levers to achieve the identified domestic policy objectives? How could co-operation with importing countries help to find win-win approaches to achieve domestic objectives while minimising negative effects on international markets?

1. Cobalt

1.1. Stylised facts on cobalt production and export restrictions

1.1.1. Key features of cobalt production and mining

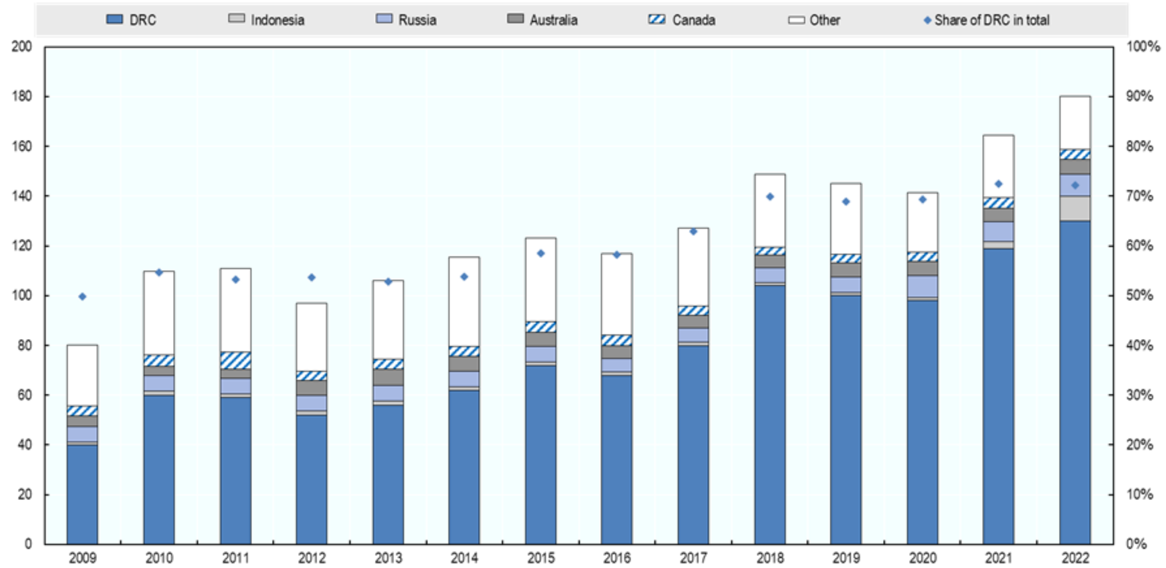
The extraction and mining of cobalt is highly concentrated in one country: the Democratic Republic of the Congo (USGS, 2024[3]; Petavratzi, Gunn and Kresse, 2019[4]). This concentration is due to the fact that cobalt is mainly mined as a by-product of copper or nickel (Mudd et al., 2013[5]). The African Copperbelt region, located between the DRC and Zambia, is the largest sediment-host copper area in the world and a major mining site for copper where supergene ores have particularly high cobalt concentrations, making their extraction particularly economically viable (Decrée et al., 2010[6]).

This is also why the DRC is home to around 50% of the reserves[4] of cobalt globally, and to more than 70% of the world's cobalt production in both 2021 and 2022 (Figure 1.1). Beyond the DRC, major sites are found in Indonesia (6% of world production), Russia (5%), Australia (3%) and Canada (2%), with production reportedly occurring also in an additional eight countries in 2022 [in order of importance at the time of writing: Philippines, Cuba, Papua New Guinea, Madagascar, Türkiye, Morocco, China, United States (US)]. Reserves show similar, albeit lower, levels of concentration, with the DRC (53%) leading, followed by Australia (20%), other areas [which is likely to indicate Zambia, (8%), Indonesia (8%) and Cuba (7%)], as well as an additional nine countries.

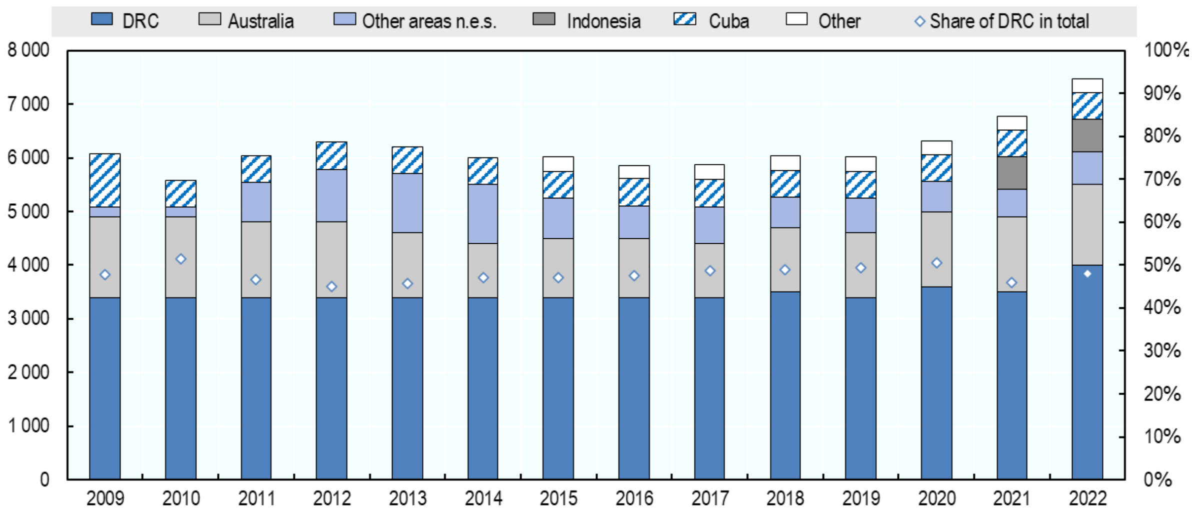
[4] According to the USGS definition, reserves are the part of an identified mineral resource that could be economically extracted or produced in a given year (USGS, 2024[86]). It follows that there exist deposits of a given mineral which are not being counted as reserves in the context of currently available extraction technologies or prevalent market prices of the mineral or prices of inputs which are used in extraction. Such deposits may graduate to the status of reserves if relevant technology or economic conditions change. An example of this may be seen in the identification of 120 million tonnes of cobalt in polymetallic nodules and crusts on the floor of the Atlantic, Indian, and Pacific Oceans Congo (USGS, 2024[3]) These deposits have been identified, but environmental concerns and technological limitations make them non-viable for the time being.

Figure 1.1. The DRC accounts for a sizeable and growing share of Cobalt mined globally, and holds about 50% of global reserves

a. Annual mine production of Cobalt in thousand metric tonnes (primary axis) and share of DRC in annual production (secondary axis)



b. Global reserves of Cobalt in thousand metric tonnes (primary axis) and share of DRC in total known reserves (secondary axis)



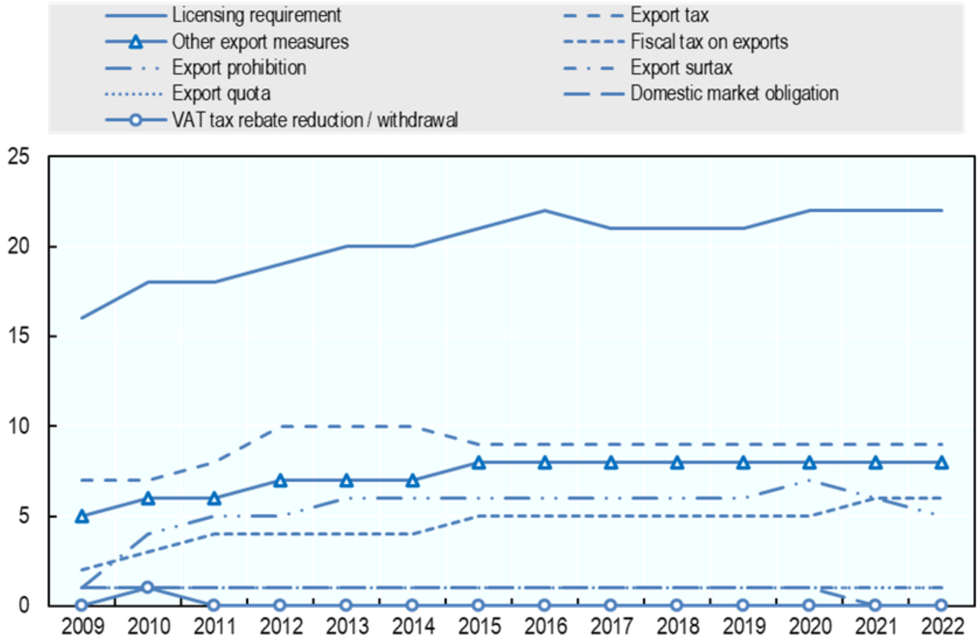
Note: "Other areas" is likely to include Zambia. Refer to the United States Geological Survey Statistics and Information for the definition of mine production and reserves.
Source: USGS.

1.1.2. Export restrictions on cobalt

According to the *OECD Inventory of Export Restrictions on Industrial Raw Materials* (thereafter "the OECD Inventory", see also Annex A), 33 countries were maintaining export restrictions on at least one form of cobalt in 2022. This number has remained relatively stable over the period 2012-2022, but in the prior period 2009-2012, the number of countries having at least one form of export restriction increased from 24 to 32. Licensing requirements and export taxes were the most widely adopted measures, followed by "other export measures" (e.g. additional charges or certificates related to outgoing shipments of the mineral), fiscal taxes on exports and outright export prohibitions, currently in place for waste and scrap of cobalt in five countries.

Figure 1.2. Licensing requirements are the most common type of export restriction, followed by fiscal measures and export prohibitions

Number of countries adopting export restriction on at least one cobalt product, by type of restriction



Note: Measures applying to at least one form of cobalt (e.g. ores and concentrates, waste and scrap, or semi-manufactured cobalt). See the methodological note for a description of the different types of measures registered in the database.
Source: OECD (2025[1]).

Export restrictions were not equally frequent across all forms of cobalt. One possible explanation – which is also often mentioned in the context of export restrictions on other minerals – is discrimination between stages of processing with the objective of, for instance, promoting exports of downstream processed cobalt by restricting exports of cobalt in primary forms. The OECD Inventory highlights that waste and scrap of cobalt was the form of cobalt with the greatest number of restrictions in 2022, typically adopted with the objective of monitoring exports of hazardous waste and complying with environmental regulations.[5] Waste and scrap of cobalt faces licensing requirements in 19 countries, export taxes in nine and "other export measures" in eight (Figure 1.3, Panel a). For other forms of cobalt, the total number of restrictions gradually decreases with stages of processing, starting with 13 export restrictions in place in 2022 for ores and

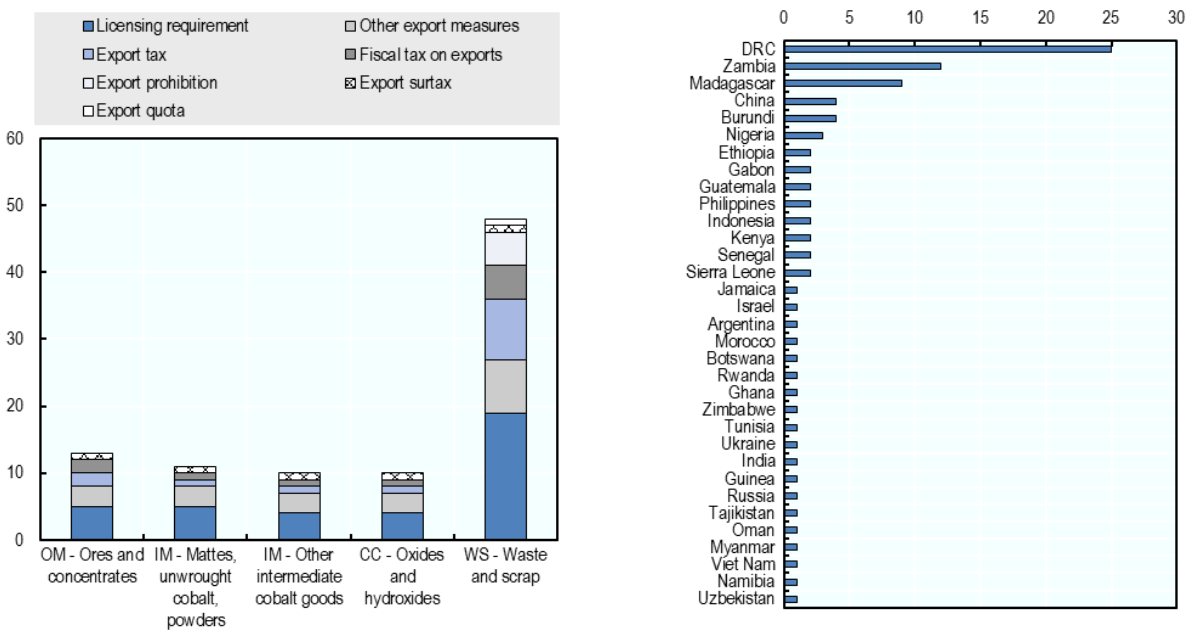
[5] Higher incidence of export restrictions on waste and scrap of minerals is a phenomenon not restricted to cobalt but applies to most minerals. This reflects countries' environmental concerns (about ensuring appropriate processing waste and scrap in destination countries) but also, increasingly, the desire to keep the already processed, energy efficient, and environmentally friendly forms of minerals in the domestic economy for use in the circular economy [e.g. Kowalski and Legendre (2023[2])]. Note that the Inventory excludes export controls that have been implemented to comply with international conventions and agreements that limit the trade of certain goods, and, regarding metallic waste and scrap, this includes the Basel Convention on the Control of Transboundary Movements of Hazardous Wastes and their Disposal.

concentrates, 11 for unwrought cobalt and powders, and ten for metals and chemical compounds. In terms of the geographical distribution of those measures, the DRC, Zambia and Madagascar are the leading adopters, followed by China, Burundi and Nigeria (Figure 1.3, Panel b). In 2022, the DRC applied five types of export restrictions (export surtaxes, export tax, fiscal tax on exports, licensing requirement, other export measures) on each one of the five forms of cobalt recorded in the Inventory.

Figure 1.3. Waste and scrap of cobalt attracts the greatest number of export restrictions, while the DRC, Zambia, and Madagascar were the leading adopters of export restrictions on cobalt in 2022

a. Number of export restrictions faced by cobalt at different stages of processing

b. Indicator of export restriction adoption by country



Note: Figure 1.3, Panel a counts once each country-type of export restriction recorded in the database (e.g. one export tax in Argentina, one licensing requirement in Burundi, one export tax in Burundi, although there may be more than one licensing requirement). The indicator of export restriction adoption by country in Figure 1.3, Panel b reflects the extent to which different types of export restrictions are applied on cobalt at different stages of production. For instance, the DRC applied in 2022 five types of export restrictions (Export surtaxes, Export tax, Fiscal tax on exports, Licensing requirement, Other export measures) on each one of the five forms of cobalt recorded in the export restrictions database.
Source: OECD (2025[1]).

1.2. Mapping trade in cobalt products

Assessing the impact of export restrictions on international trade in cobalt requires understanding the geography of trade in this mineral as it moves along international value chains. As for many other CRMs, cobalt crosses international borders in several different forms, first as ore or concentrate (where the primary forms of minerals are classified) and then as metal and intermediate products, chemical compounds, in batteries or magnets and in waste and scrap. The geography of flows in processed products differs from that of primary products (ores, concentrates), with firms located in several different economies – including non-resource-rich economies – taking part in transformation of the mineral.

The degree to which different forms of cobalt and other CRMs are identifiable in trade statistics depends, in part, on how they are classified in the Harmonized System (HS) nomenclature of the World Customs Organization (WCO). Cobalt products are relatively well tracked in the HS codes, and can be distinctively identified as ores and concentrates, intermediate cobalt products (e.g. powders, unwrought cobalt, mattes), oxides and hydroxides (i.e. chemical compounds) as well as in waste and scrap (Table 1.1).

Beyond the HS nomenclature, national customs nomenclatures can add further precision for the identification of additional forms of cobalt traded across borders. For instance, in the product code that tracks trade in inorganic chemical chlorides, Japan, the European Union, United Kingdom, and the United States differentiate chlorides of cobalt from other chlorides (e.g. those involving tin, iron or magnesium) (Table 1.1). This more granular classification enables identification of flows of cobalt chemical compounds that would otherwise not be visible in 6-digit HS codes. This level of detail helps to more accurately map the participation of these countries in world cobalt trade (Annex Box A.B.1).

Table 1.1. The HS nomenclature offers visibility across different forms of cobalt, while national nomenclatures of G7 countries add precision for cobalt chemical compounds

"x" indicates that cobalt product is separately identifiable in the relevant nomenclature

Forms	Cobalt products	Harmonized System nomenclature	Canada nomenclature	Japan nomenclature	EU and UK nomenclature	United States nomenclature
OM	Ores and concentrates	x (260500)	x	x	x	x
IM	Unwrought cobalt, mattes, powders	x (810520)	x	x	x	d
IM	Other intermediate forms	x (810590)	x	x	x	x
CC	Oxides and hydroxides	x (282200)	x	d	x	x
CC	Chlorides			x	x	x
CC	Sulphates				x	x
CC	Acetates			x		x
WS	Waste and scrap	x (810530)	x	x	x	x

Note: The European Union and the United Kingdom share the same customs nomenclature. Cobalt forms refer to: OM: ores and concentrates; CC: chemical compounds; IM: metals and semi-manufactured articles; WS: waste and scrap. "d" indicates that national customs codes provide further detail or distinction within the product category (e.g. for cobalt oxides and hydroxides, nomenclatures distinguish oxides from hydroxides in the Japanese nomenclature; for intermediate forms of cobalt, alloys are separately identified in the US nomenclature).
Source: Own calculation based on the 2022 revision of the Harmonised System and the latest available version of G7 customs nomenclatures in Q1 2024.

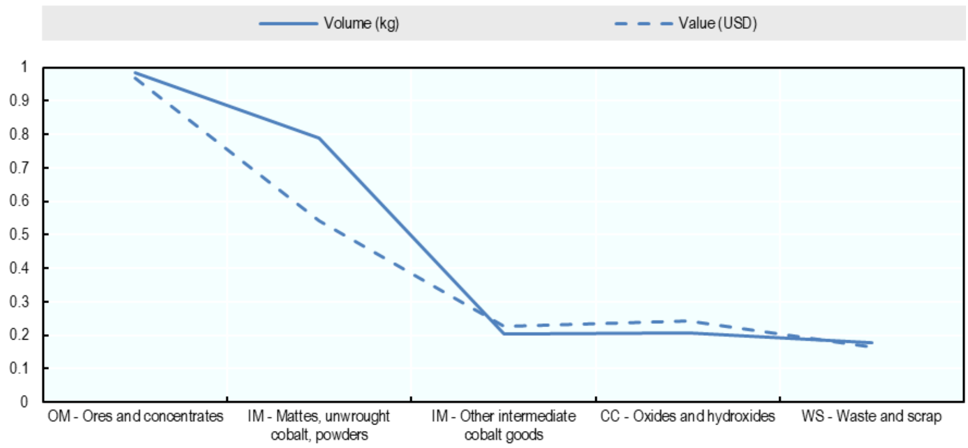
The detailed HS nomenclature-based classification of cobalt along its value chain (Table 1.1) also helps to more accurately describe global trade flows. At the global level, cobalt is characterized by large volumes of primary forms shipped from the DRC to China, and lower concentrations in trade flows along its value chain. For primary forms, exports from the DRC to China alone accounted for about 91% of the global volume and 83% of the global value of world trade in cobalt ores and concentrates in 2022.[6] Similarly, trade is highly concentrated for semi-processed goods like unwrought cobalt and cobalt powders, where exports from the DRC to China account for 87.5% of the total volume and 72% of the total value of world trade in the product.[7] These patterns can be linked to the dominant role of China as refiner of cobalt, reportedly accounting for more than 80% of global cobalt chemicals refining capacity (OECD, 2016[7]; Gulley, McCullough and Shedd, 2019[8]). Concentration is much lower for other intermediate forms of cobalt, chemical compounds, and waste and scrap, reflecting the participation of a broader number of countries in the downstream stages of the cobalt value chain (Figure 1.4).

[6] Estimate based on product code 260500 and on the UN COMTRADE database (using imports as reported flow).

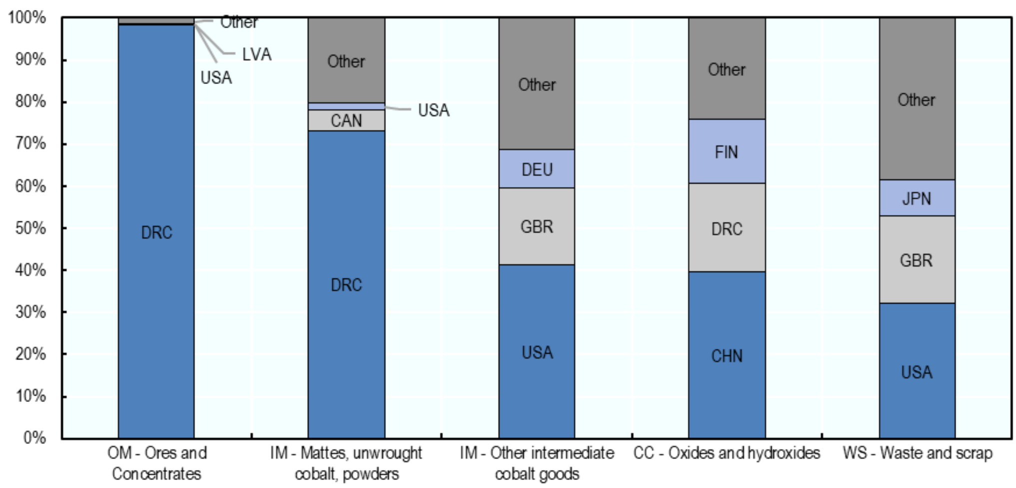
[7] HS code 810520.

Figure 1.4. The concentration of trade in cobalt products declines with the degree of processing

a. HHI index of concentration in world exports, computed at the country level, 2022



b. Top three exporters by cobalt product (value), 2022



Note: The Herfindahl–Hirschman Index (HHI) is a measure of market concentration. In this case, it is obtained by squaring the share of each country's exports in total world exports in each HS 6-digit code and taking the sum of this number. a. Products refer to, from left to right, OM - Ores and Concentrates (HS 260500), IM - Mattes unwrought cobalt, powders (HS 810520), IM - Other intermediate cobalt goods (HS 810590), CC- Oxides and hydroxides (HS 282200), WS - Waste and scrap (HS 810530). Intra-EU trade is not included. b. Products follow the same order as graph A with the hs-6-digit codes.
Source: UN COMTRADE.

1.3. The potential impact of export restrictions on trade in cobalt products

The potential impact of export restrictions on cobalt trade needs to be assessed against the main features of the value chain specific to this mineral. Cobalt is characterized by concentration in the DRC for reserves, production and exports of upstream cobalt forms; by a strong bilateral trade relationship between the DRC and China; and weak direct exports from producers of raw forms of cobalt to G7 countries. This section explores the potential impact of export restrictions on trade in cobalt focusing on measures adopted by the DRC – the world's largest supplier of cobalt ores – to promote greater domestic processing in the industry.

1.3.1. The potential impact of export restrictions on exports of cobalt from the DRC

With around 50% of the world reserves, more than 70% of world production and more than 90% of world exports of cobalt ores and concentrates, export restrictions adopted by the DRC are highly relevant for supply in world markets. The OECD Inventory highlights that all forms of cobalt products are subject to various export restrictions in the DRC, including licensing requirements, export taxes, export surtaxes, fiscal taxes on exports and other export measures (i.e. the obligation for firms to repatriate a fixed a share of export earnings to the DRC).

The Inventory records three main changes in the export restriction regime over 2009-22: the increase in the share of export earnings to be repatriated by mining companies from 40% to 60% in 2018; the decrease in fiscal taxes on exports in 2019 from USD 150 to USD 100 (loosening of the export restriction),[8] and the introduction, between 2017 and 2020, of an export prohibition on primary forms of cobalt (Box A.B,2 and Table A.B.1.2). With the exception of the export prohibition adopted in 2017, the adopted measures do not discriminate between different forms of cobalt.

The 2017 export prohibition on cobalt ores offers a valuable event that can help shed light on the potential impact of export restrictions in promoting the shift from exports of minerals in primary forms to processed or semi-processed forms.[9] The measure was adopted with the objective of promoting domestic value creation by firms located in the DRC, by prohibiting exports of unprocessed forms of cobalt. The measure is reported to have expired in August 2020.[10]

Trade statistics suggest that the export prohibition is likely to have accelerated an existing trend in the DRC away from exports of cobalt ores and concentrates and towards greater exports of cobalt mattes, powders and unwrought forms (Figure 1.5). Cobalt ores and concentrates went from accounting for about 60-74% of the total volume (i.e. weight, as measured in metric tonnes) of exports in the three years prior to the ban (2014-2016) to just 7% in 2022. Conversely, exports of mattes, unwrought cobalt and powders grew from 20-37% of the total volume of exports in the three years prior to the ban to 91% of the total volume of exports in 2022.[11] A similar trend can be observed in the total value of cobalt exports, although changes in the prices of cobalt make it harder to discern a distinct shift in 2017 (Figure 1.5).

[8] This refers to a tax for the authorisation of exports out of the DRC, see Arrêté interministériel n° 2019 - 0001 et 2019 - 009 du 22 février 2019.

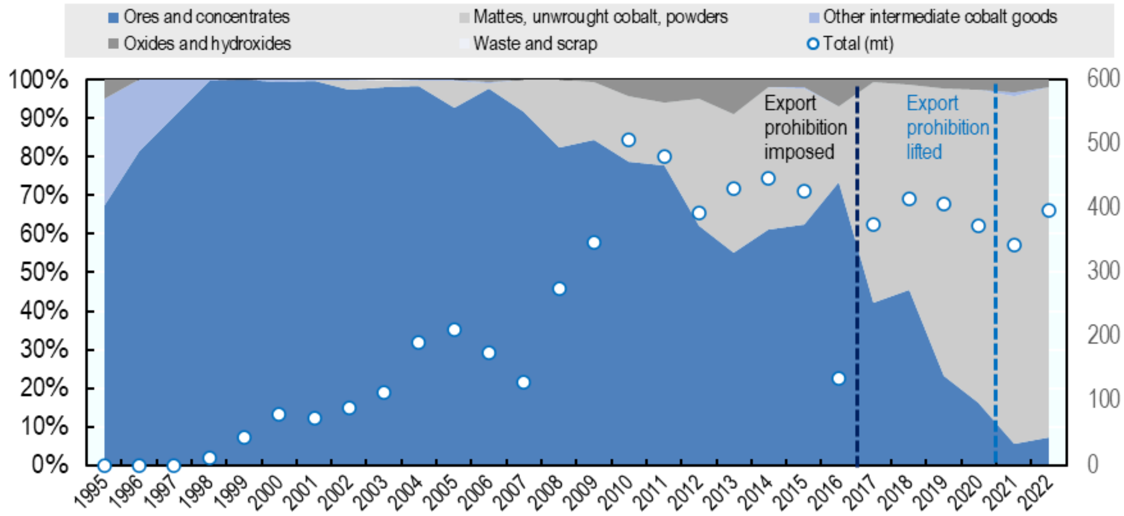
[9] While changes to fiscal measures are also interesting in term of their potential to affect the costs of exporting, their effect is more difficult to disentangle from other factors that could also have impacted the DRC's exports of cobalt over the period.

[10] See OECD (2025[1]) and Arrêté interministériel n°0129/CAB.MIN/MINES/01/2017.

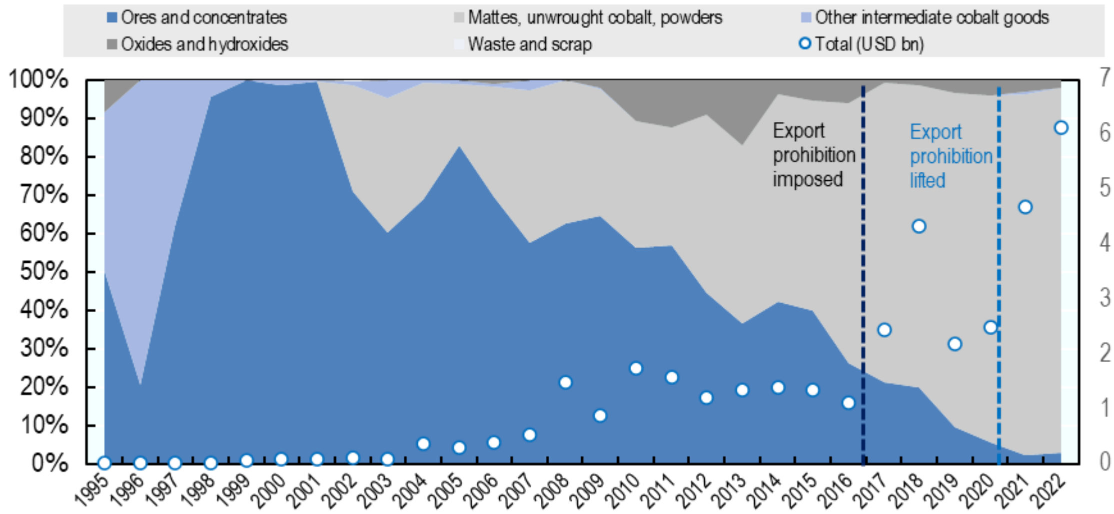
[11] This could reflect the greater use of whole ore leach (WOL), which has reportedly become common practice in the DRC, to provide improved metal recoveries following crushing and grinding (Petavratzi, Gunn and Kresse, 2019[4])

Figure 1.5. The export prohibition on cobalt ores resulted in a shift of exports away from ores and towards unwrought cobalt forms, powders and mattes

a. Shares of different forms of cobalt in total cobalt exports from the DRC (left axis) and total volume of cobalt goods exports in thousand metric tonnes (right axis)



b. Shares of different forms of cobalt in total cobalt exports from the DRC (left axis) and total value of cobalt goods exports in USD billion (right axis)



Note: Dark blue dashed line indicates the introduction of the measures while the blue line refers to its removal.
Source: UN COMTRADE, based on mirror import data.

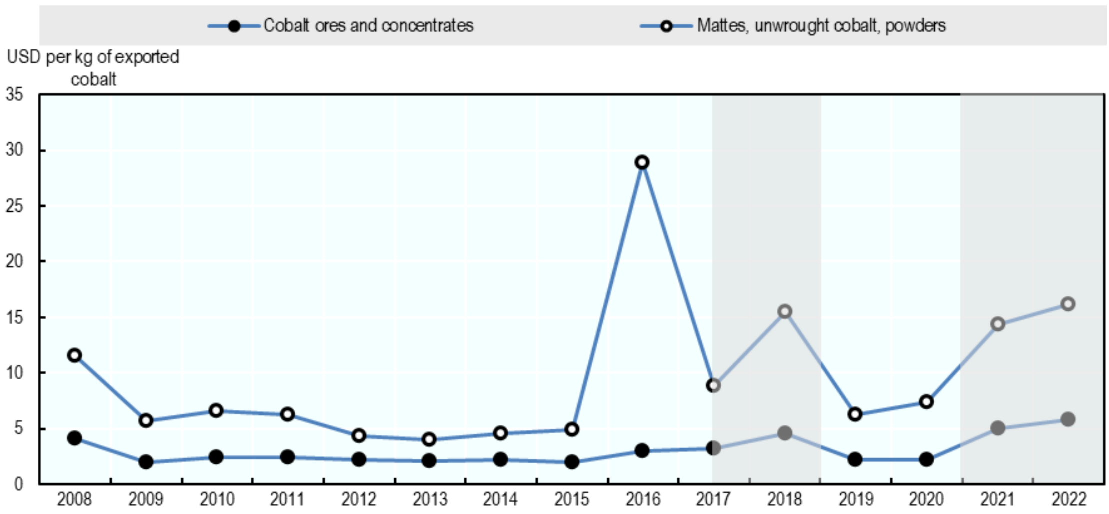
The change in the composition of exports contributed to an increase in the price paid per kg of cobalt exported from the DRC, with other reported effects including greater domestic processing, lower shipping costs and more waste deposited in the country[12] (Gulley, 2022[9]). Unwrought forms of cobalt, powders and mattes tended to be exported from the DRC at average unit values (i.e. value per kg) that are two to three times greater than those of cobalt ores and concentrates (for instance equal to USD 6.3 per kg compared to USD 2.2 per kg for ores in 2019) – Figure 1.6, meaning that the change in composition of trade is likely

[12] As highlighted elsewhere, waste and scrap can also be an opportunity as it can consist of already processed, energy efficient, and environmentally friendly forms of minerals for potential use in the circular economy (Kowalski and Legendre, 2023[2]).

to also have increased export tax revenues generated in the DRC. This difference in price proxies also increased further in periods of price hikes such as 2017-2018 and 2021-2022, indicating that these semi-processed products may be more likely to have capitalised on external price changes.[13]

Figure 1.6. Mattes, powders and unwrought cobalt are exported from the DRC at higher unit values than cobalt ores and concentrates

Unit value of cobalt exported from the DRC, by form. Grey areas denote a high cobalt price environment



Note: Unit values of imports form the DRC for HS codes 260500 and 810520, computed as the total value of trade divided by total quantity in kg.
Source: UN Comtrade.

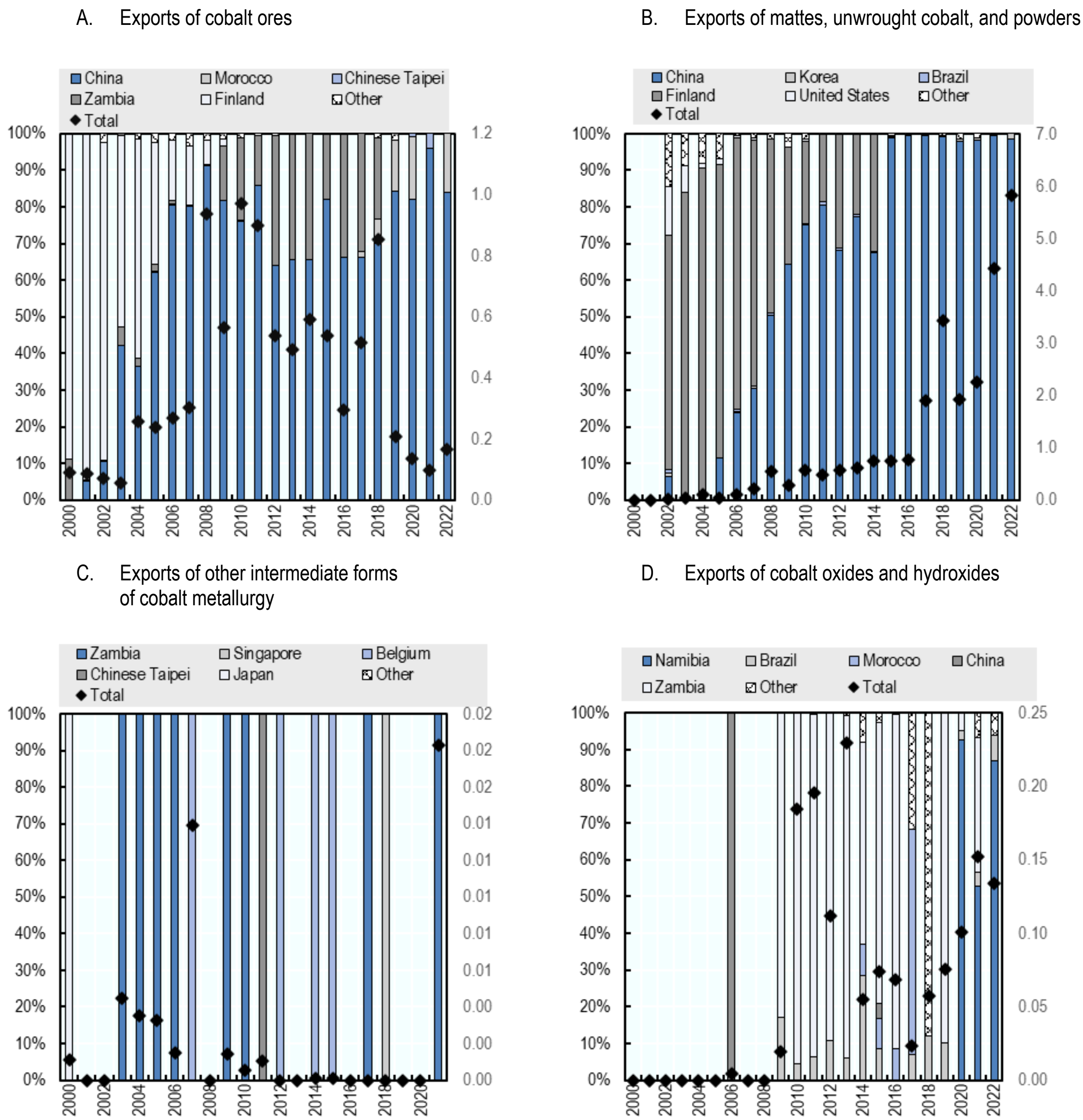
However, the period of greater exports of semi-processed cobalt also corresponded with a period of sustained high concentration of exports to China. Trade statistics reveal that, in the years between 2015 and 2022, China consistently accounted for between 97.9% to 99.6% of the value of exported mattes, unwrought cobalt and powders, with hardly any of this semi-processed cobalt reaching any other destination market (Figure 1.7, Panel b). This concentration is even higher than for exports of cobalt ores and concentrates (Figure 1.7, Panel a), meaning that more processing in the DRC coincided with most of the processed cobalt being exported to China and highlighting the strong vertical links between this processing activity and further refining activities in China.

Second, the export prohibition appears to have had limited impact beyond the increase in upstream processing of cobalt from ores and concentrates to unwrought cobalt, mattes and powders. Exports of other metal forms of cobalt – which were historically low – also became less frequent and never exceeded USD 200 000 in value of annual exports (Figure 1.7, Panel c). Exports of cobalt oxides and hydroxides – forms produced in chemical refining processes (Petavratzi, Gunn and Kresse, 2019[4]) – also increased, starting from 2017, in USD terms (Figure 1.7, Panel d), although they remained a small share of the DRC's total cobalt exports (Figure 1.5).

[13] The spike in the unit value of exports of semi-processed cobalt forms in 2016 is due to an increase in the unit value of exports to China in this product from USD 4.8 per kg of cobalt in 2015 to USD 29 per kg in 2016 (a unit value that is similar to that of exports to, for example, Germany or the Czech Republic), with the value then decreasing again to USD 8.8 per kg in 2017. It is difficult to explain this increase in price paid for this particular cobalt form, but possible reasons include that China imported in that year more processed forms of cobalt within the HS 6-digit code 810520 or higher prices were paid for the scarce volume of cobalt available from the DRC in that particular year (the total volume of exports to all destinations dropped from 427 metric tonnes of exported cobalt goods in 2015 to 134 in 2016). The year 2016 also corresponded to a year of greater awareness on the use of child labor in cobalt mines, which may have resulted in lower volumes of trade in the product (Amnesty International, 2016[79]).

Figure 1.7. China imports more than 98% of the DRC's mattes, powders and unwrought cobalt, highlighting limited geographical diversification

Left axes indicates shares of exports by destination, right axes indicate value in USD billion



Source: UN COMTRADE.

The lack of impacts of comparable magnitude on exports of chemical forms of cobalt is particularly noteworthy in the context in which industry demand has shifted in the past decade from cobalt in metal forms (used in superalloys, permanent magnets, high-speed steels, etc.) to cobalt chemicals, primarily used in the production of batteries (Gulley, 2022[9]).[14] The lack of similar effects beyond primary production steps is consistent with evidence from geological surveys indicating that "[m]ost of the African copper-

[14] It should be noted that cobalt metal of high quality ("class I" for approximation) is also used in batteries production (Schmidt, Buchert and Schebek, 2016[10]).

cobalt sulfide concentrate is processed "in-house" to produce an impure cobalt hydroxide. This is then shipped to China's chemical refineries, where a range of cobalt salts including cobalt oxide and hydroxide are produced for use in multiple applications" (Petavratzi, Gunn and Kresse, 2019[4]).

The distinction between semi-processed, intermediate cobalt, which consists of "mattes, unwrought cobalt and powders", and which is likely to include products that underwent leaching, smelting or crunching processes as upstream processes,[15] and "refined cobalt", including cobalt chemical compounds that are closer to their downstream use in batteries, is hence important for the purpose of assessing the impact of this export restriction (Schmidt, Buchert and Schebek, 2016[10]; Petavratzi, Gunn and Kresse, 2019[4]; van den Brink et al., 2020[11]). While the export prohibition appears to have played a role in promoting processing at the early stages of the value chain, it does not appear to have boosted refining capacity in the country, with this activity largely expanding in China instead over recent years (Gulley, 2022[9]).

Beyond export restrictions, it is also important to consider other significant factors that are correlated with the adoption of the restrictions and that may also explain the transition in upstream cobalt processing – and the lack of greater refining – observed in the DRC over the past decade. These factors include:

- In 2007 – the year after the reported imposition, or threat of imposition, of the first export prohibition on cobalt ores and concentrates, the DRC signed with China a USD 5-6 billion "minerals for infrastructure" agreement (Gulley, 2022[9]). Chinese state-owned banks agreed to provide USD 5-6 billion in loans for DRC infrastructure while a Chinese state-owned mining consortium received the rights to multiple mineral concessions in the country. Since then, Chinese firms intensified purchases of additional cobalt assets in the DRC through the 2010s.
- This has led to Chinese firms owning or having stakes in 15 of the 19 cobalt-producing mines in the DRC in 2020 (Basakaran, 2023[12]). Greater Chinese investment in the DRC also resulted in vertically integrated structures, with business units of Chinese "fine refiners" operating in the DRC before minerals are exported to China (Gulley, McCullough and Shedd, 2019[8]). This – in conjunction with the adoption of the export prohibition – is likely to play an important role in explaining greater observed values of exports of cobalt intermediate goods and their concentration in the Chinese market.
- China is reported to be the dominant player in cobalt refining, accounting for as much as 80% of the world's cobalt chemical refining capacity (OECD, 2016[7]; Gulley, McCullough and Shedd, 2019[8]). This is likely to play an important role in explaining the DRC's concentration of exports in intermediate cobalt forms for refining in China.

In addition, several qualitative sources highlight that greater extraction and production of semi-processed cobalt resulted in adverse ESG effects in the DRC. Box A.B.3 includes a brief literature review of the ESG challenges that have been documented in this context.

1.4. Cobalt: Key findings and tentative conclusions

Export restrictions on primary forms of cobalt appear to have helped the DRC – the world's largest supplier of cobalt – to transition from exporting cobalt ores and concentrates to exporting processed intermediate forms of the mineral in the period 2017-2022. These intermediate forms were exported at higher unit values than ores and concentrates. However, these exports are also highly concentrated in the Chinese market, which absorbed about 99% of exports by value between 2017 and 2022. The measures also do not appear to have similarly boosted exports of more refined forms of cobalt like chemical compounds – which are used intensely in production of batteries – and other products of cobalt metallurgy.

While export restrictions hence appear to have been a tool for greater upstream domestic processing, they are also not in themselves sufficient for building a full, resilient and diversified cobalt value chain at the technology frontier, which points to the importance of additional domestic factors in shaping the impact of

[15] In light of the values of trade being highest in this product, and contextual information highlighting the importance of crude cobalt hydroxide as processed intermediate product in the DRC, it is possible that this code captures this intermediate good (Cobalt Institute, 2020[80]). More analysis is needed to understand how more specific forms of cobalt are classified in the HS system, and in particular in codes 810520 and 282200 respectively.

these measures. In addition, the use of export restrictions went hand-in-hand with greater FDI, which may be an important factor explaining greater domestic processing of cobalt in the DRC.

Several policy levers beyond export restrictions need to be in place to make sure that the DRC can contribute to resilient value chains for cobalt: from improvements to the reliability of energy supply and a more stable policy environment for foreign investment, to policies that can support greater domestic value creation and value chain diversification. The dominant role of China as world refiner of cobalt also contributes to the concentration of international cobalt markets, highlighting the potential benefits of greater geographical diversification in refining capacity in the DRC and beyond and the potential for win-win solutions that both support domestic development objectives in the DRC and foster more resilient global supply chains for this mineral.

The development of a domestic cobalt refining industry provides some clear benefits for the DRC. Not only would this enable the export of products with higher domestic value-added content, but it would also foster opportunities for more and higher skilled jobs. Yet, the growth of industry complexity would also need to be met with the strengthening of state institutions and the effective creation and application of legal and regulatory governance. For the time being, the cobalt industry in the DRC remains subject to practices of concern across the ESG spectrum. Looking forward, the future development of the cobalt industry governance in the DRC is likely to be influenced by China's rapidly growing role in the mining sector, and in international governance regimes.

2. Lithium

2.1. Stylised facts on lithium production and export restrictions

2.1.1. Key features of lithium production and mining

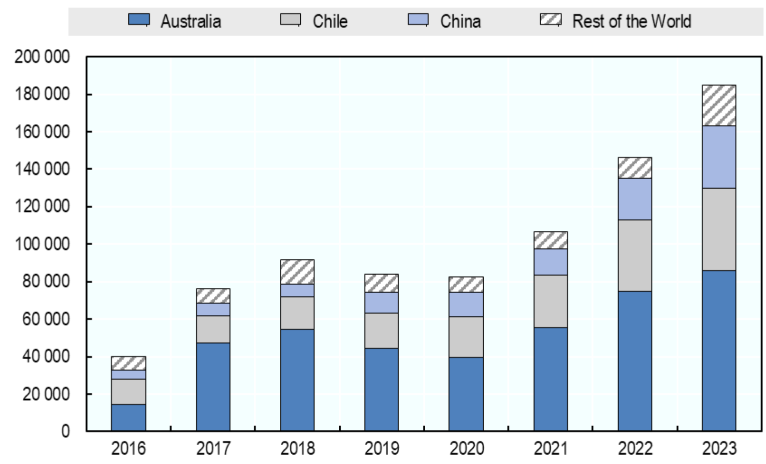
Until recently, lithium was mostly used for ceramics, glasses, and other industrial applications and li-ion batteries accounted for less than 30% of lithium demand (McKinsey & Company, 2022[13]). Today, over 85% of lithium demand comes from the battery sector (Mehdi, 2024[14]). Due to the ongoing energy transition and a shift towards electrification in vehicles, lithium production is projected to increase significantly, from approximately 500,000 metric tonnes of lithium carbonate equivalent (LCE) in 2021 to some 3-4 million metric tonnes in 2030. This surge is largely attributed to an increase in manufacturing of li-ion batteries driven by the rapid growth of electric vehicle (EV) sales – which is set to triple from 10.5 million in 2022 to over 31 million in 2027 and is expected to more than double to over 74.5 million units in 2035 (EV Volumes, 2023[15]).[16] As a result of this strong forecasted demand, the global price of lithium surged before 2022 (see Annex Figure A.C.1 below).

The extraction and mining of lithium is highly concentrated in three countries: Australia, Chile, and China (Figure 2.1); together these countries account for more than 65% of global reserves (USGS, 2024[16]).The high lithium content in the reserves of these countries makes extraction more economically viable with a higher yield per unit of ore or brine. In addition, all three countries have well established mining operations and industries. Other minor producers of lithium in 2023 included Argentina (6% of world production) and the United States (5%), with smaller production in Brazil, Canada, Namibia, Portugal, and Zimbabwe.

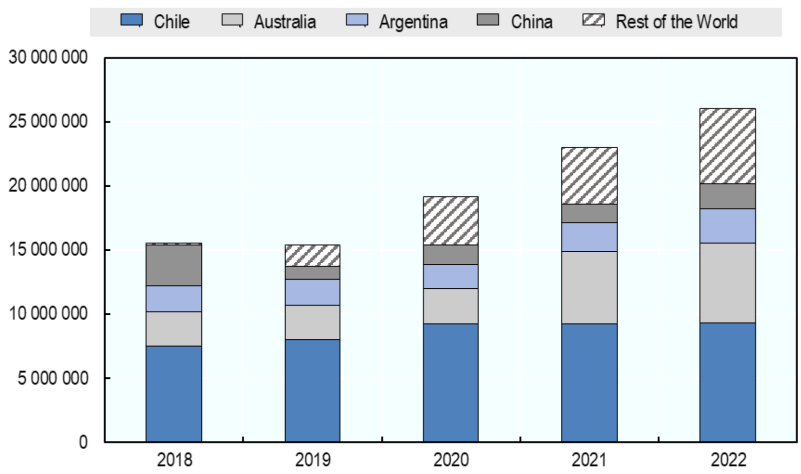
[16] Note also that, over the last decade, a variety of government support policies for EVs were instituted in key markets globally which helped to stimulate demand. Similarly, supply-side drivers (e.g. EV sales targets) also played a role.

Figure 2.1. Australia, Chile, and China account for most of the lithium produced globally, and together hold more than 65% of global reserves

Panel A. Annual production of lithium in metric tonnes of lithium content



Panel B. Reserves of lithium in metric tonnes



Source: USGS.

2.1.2. Export restrictions on lithium

According to the OECD Inventory, only two countries used export restrictions on lithium in the analysed period: Argentina and Zimbabwe (Figure 2.2). Argentina used an export tax on lithium, fluctuating between 0% (0% in the period 2016-18) and 12% from 2009 to 2022. This measure, mandated by the federal government, is part of a comprehensive policy applying levies to a wide range of commodities – including extractives (e.g. precious metals) and agricultural products – to generate fiscal revenues.[17]

Zimbabwe, a minor lithium exporter, has used a licensing requirement since 2002 for monitoring and control of exports of several minerals including lithium, gold, silver, and platinum. In 2022, Zimbabwe banned exports of primary forms of lithium to promote the processing of lithium ore into lithium carbonate and develop activities to create more value added in the supply chain. Shortly after, several Chinese-

[17] See https://www.argentina.gob.ar/normativa/nacional/decreto-1060-2020-345886/texto.

owned companies[18] invested more than USD 1 billion acquiring and developing lithium projects in Zimbabwe (Gbadamosi, 2023[17]) and Chinese-owned processing plants are now shipping lithium carbonate to China for further processing. In 2023, Zimbabwean spodumene[19] concentrate exports to China were reported to have increased nearly five-fold to 177 000 tonnes, compared to 38 000 tonnes in 2022 (South China Morning Post, 2024[18]). According to the government of Zimbabwe, fiscal revenues generated from the export of lithium grew from USD 1.8 million in 2018 to USD 209 million in 2023 as these Chinese-driven mining and processing projects took off (TRT Afrika, 2023[19]).

In addition, almost a year after Zimbabwe's adoption of export restrictions on unprocessed forms of lithium (June 2023), Namibia restricted exports not only of lithium but also of various other unprocessed minerals (Mining & Energy, 2023[20]). The declared aim of the ban was to promote the beneficiation of the products domestically and enhance the local refining industry (Ya Ngulu, 2023[21]). In the context of the surging global demand for energy efficient technologies, the country was striving to strengthen its production capacity and secure its role in international CRM supply chains and the renewable energy sector, with plans to produce battery-grade materials locally. The decision prompted a surge in foreign investment in the local processing of minerals (mainly by mining firms from China, Australia, Canada and South Africa) (US Department of State, 2024[22]). According to the Bank of Namibia, China's direct investment inflows accounted for 26.9% of country's total FDI, ranking as the largest source of investment (Ngula, 2024[23]).[20]

This chapter begins by discussing challenges related to tracking lithium products in international trade statistics. It then presents a simple descriptive statistical analysis of the impact of Argentina's taxes on the country's lithium product exports and the resulting fiscal revenue, while also commenting on the possible ESG effects of these measures. Although a similar export impact analysis is not conducted for Zimbabwe—due to the short time between the introduction of the latest restrictions and the completion of this study—Annex C includes a discussion of expert commentary on the early ESG effects of those measures.

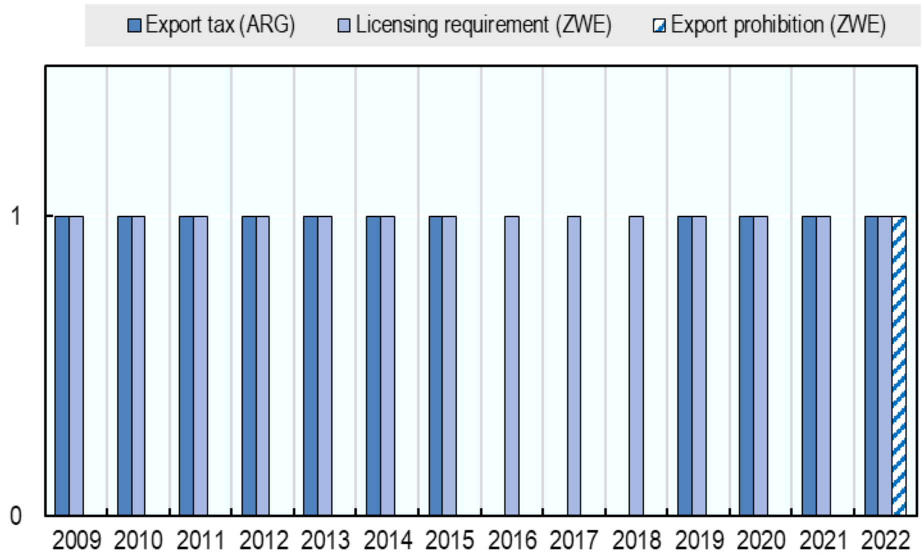
[18] Chinese firms involved in lithium projects in Zimbabwe include Zhejiang Huayou Cobalt, Sinomine Resource Group, Chengxin Lithium Group, Yahua Group, and Canmax Technologies.

[19] Spodumene is one of the most important lithium-bearing minerals obtained through hard-rock mining. Lithium can also be obtained from lithium-rich brine sources which is an entirely different extraction process from spodumene mining. The process relies on solar evaporation, where the water is removed, leaving behind concentrated lithium salts, which can be processed into lithium carbonate or lithium hydroxide. Brine extraction is cheaper and more environmentally favourable in regions with dry climates but takes longer to produce lithium and requires significant water resources. Major brine resources are located in the "Lithium Triangle" of South America (Chile, Argentina, Bolivia) and certain parts of China and the United States.

[20] Some commentators in the European Union have criticised Namibia's move as being against the spirt of the Memorandum of Understanding the EU which was signed with Namibia only one year earlier (November 2022) and which aimed at ensuring both local benefits and EU's secure access to rare earth minerals (Reilly, 2024[84]).

Figure 2.2. Argentina and Zimbabwe maintain export restrictions on lithium

Number of countries adopting export restriction on at least one lithium product, by type of restriction



Note: Measures applying to at least one form of lithium (e.g. ores and concentrates, waste and scrap, or semi-manufactured lithium). See the methodological note for a description of the different types of measures registered in the database.
Source: OECD (2025[1]).

2.2. Mapping trade in lithium products

Assessing the impact of export restrictions on international trade in lithium requires understanding the geography of trade along international value chains. Lithium crosses international borders in several different forms, first as spodumene then in more processed chemical compounds such as carbonates, and oxide and hydroxide, and finally in cells for batteries or electrical accumulators. The geography of flows in processed products differs from that of geological reserves, with firms located in several different countries – including non-resource-rich economies – taking part in the transformation process.

The degree to which different forms of lithium are identifiable in trade statistics depends, in part, on how such products are classified in the HS code. In contrast to cobalt (Section 1.2), lithium products have limited visibility in the HS classification, where they are mainly identifiable as chemical compounds, and batteries and electrical accumulators which contain lithium-based components,[21] but not as ores or separate waste and scrap categories. This is particularly problematic for primary forms of lithium, with spodumene, one of the primary lithium-bearing minerals which can be traded across borders, typically classified with “other mineral substances” under the HS code *253090 Mineral substances; not elsewhere classified in chapter 25*, and not separately identifiable.

Beyond the HS code, some national customs nomenclatures add further precision for additional forms of lithium traded across borders. This is for example the case for primary lithium forms (i.e. spodumene) which the EU nomenclature identifies under a separate code (Table 2.1). The more granular definitions in the HS code of the G7 countries enables a closer look at their participation in lithium trade (Box A.C.1).

[21] Note, however, that lithium is just one of many critical minerals that are used in battery production. Lithium, which is used for manufacturing of EV battery cathodes, accounts for approximately just above 3% of the weight of an average EV battery, while cobalt accounts for 4.3% and nickel for 15.7% (Source: The key minerals in an EV battery - MINING.COM).

Table 2.1. The HS nomenclature offers visibility across different forms of Lithium, while national nomenclatures allow for the identification of upstream forms

"x" indicates that lithium product is separately identifiable in the relevant nomenclature

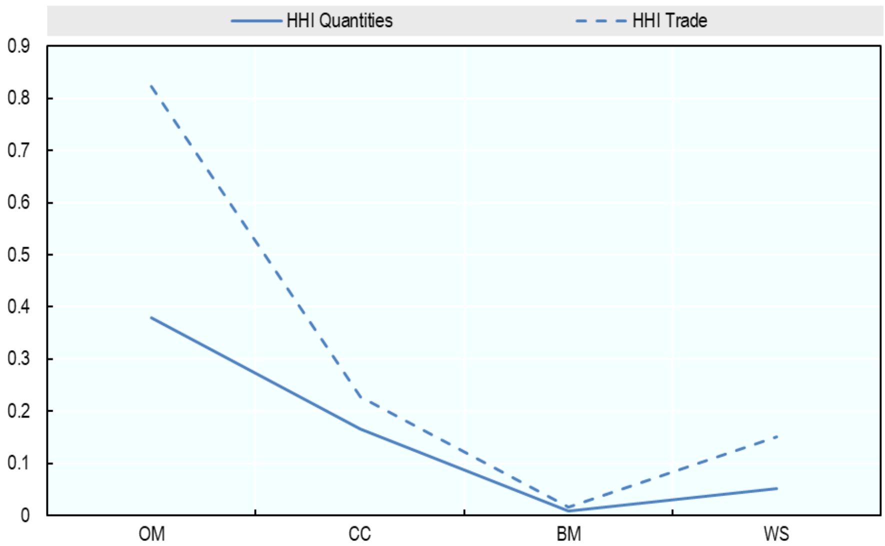
Forms	Lithium products	Harmonized system nomenclature	Canada nomenclature	Japan nomenclature	EU and UK nomenclature	United States nomenclature
OM	Spodumene				x	
CC	Carbonates	X (283691)	x	x	x	d
CC	Oxides and hydroxides	X (282520)	x	x	x	x
CC	Sulfonates	X (290433)	x	x	x	x
BM	Niobate wafers				x	
BM	Cells and batteries	X (850650)	d	x	d	d
BM	Electric accumulators	X (850760)	d	x	x	d
BM	Other accumulators		x			

Note: The European Union and the United Kingdom share the same customs nomenclature. Lithium forms refer to: OM: ores and concentrates; CC: chemical compounds; BM: batteries and magnets, as well as other goods or components for intermediate or final use. "d" indicates that national customs codes provide further detail or distinction within the product category (e.g. distinguishing lithium batteries depending on their size in Canada's customs nomenclature).
Source: Own calculation based on the 2022 revision of the Harmonised System and the latest available version of G7 customs nomenclatures.

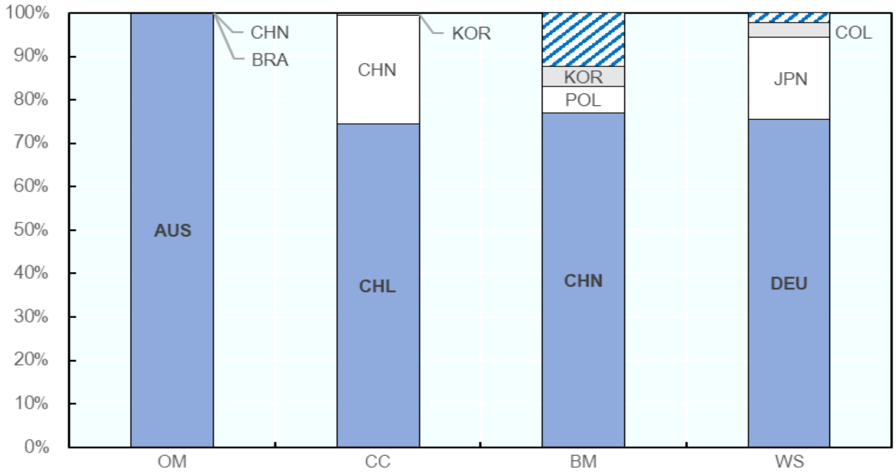
The concentration of trade in lithium falls along the processing value chain (Figure 2.3, Panel A). At the global level, large volumes of spodumene are shipped from Australia, with exports from Australia to China accounting for more than 90% of the global value of trade in lithium ores in 2022. While still concentrated, exports of more processed forms of lithium, such as carbonates, and oxides and hydroxides were slightly more diversified, with exports from Chile and China accounting for 74% and 25% respectively of global exports by value (Figure 2.3, Panel B). Trade concentration was much lower for even more processed forms of lithium – i.e. batteries and magnets, and waste and scrap – reflecting the participation of a larger number of countries in the downstream stages of the lithium value chain. China accounted for over 70% of the global value of exports of batteries and magnets in 2022. Poland and Korea accounted for a substantial part of the remaining exports, with the remaining shares distributed across other countries. Germany and Japan were the largest exporters of battery waste and scrap accounting together for 95% of global exports of these products.

Figure 2.3. The concentration of trade in lithium products declines with the level of processing

a. HH index computed over bilateral trade pairs, 2022



b. Major contributors to the HHI (value), 2022



Note: Herfindahl–Hirschman (HH) index calculated for 4 different forms of lithium products in 2022. HS-6-digit codes for ores and concentrates (OM) are: 253090; chemical compounds (CC): 283691, 282520, 290433; batteries and magnets (BM): 382499, 850650, 850760; and waste and scrap (WS): 854913, 854914.
Source: UN Comtrade.

2.3. The potential impact of export restrictions on trade in lithium products: The case of Argentina and Zimbabwe

2.3.1. Argentina's export taxes

Argentina is a relatively small exporter of lithium, accounting for 3% of global lithium exports by value in 2022. However, it could arguably play a larger role in the future, as it accounts for 10% of global reserves. Lithium in Argentina is obtained from lithium-rich brines and mainly produced in two forms: lithium chloride and lithium carbonate (Table 2.2). Lithium chloride, which accounted for a rather small share of Argentina's lithium production (12% over 2016-21), is used in the manufacture of lithium-based metals. Lithium carbonate, which accounted for the largest share of lithium production, is used for manufacture of li-ion batteries and is almost exclusively exported (Table 2.2). Until recently, there was no domestic battery industry to further process lithium into manufactured products in Argentina.[22] The top destinations for the country's lithium exports were battery production powerhouses China, the United States, and Japan (Figure 2.4).

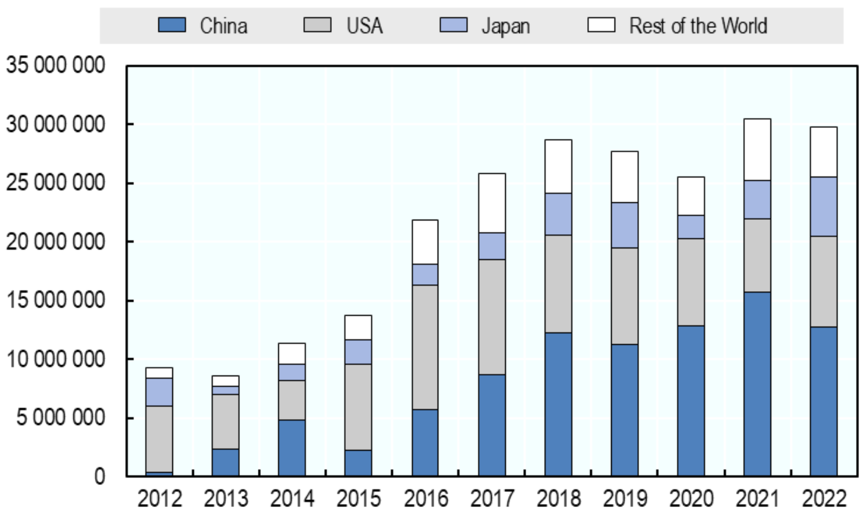
Table 2.2. Argentina exports almost all its production of lithium carbonate.

Year	Lithium chloride (production in metric tonnes of LCE)	Lithium carbonate (production in metric tonnes of LCE)	Exports lithium carbonate (in metric tonnes)	Export-to-production ratio of lithium carbonate
2016	4 501	24 409	21 829	89%
2017	3 907	26 559	25 832	97%
2018	4 344	29 707	28 658	96%
2019	3 715	29 994	27 664	92%
2020	4 195	26 911	25 544	95%
2021	3 231	28 520	30 485	107%

Note: Exports of lithium chloride are not observable because they are reported under an HS code that includes more products (HS code 282739). The export-to-production ratio of lithium carbonate can be above 100% as stocks of lithium carbonate can be exported.
Source: USGS and UN COMTRADE.

Figure 2.4. Top export destinations of lithium from Argentina have consistently been China, United States, and Japan

Exports of lithium carbonate from Argentina to main trading partners (in kg)



Note: Exports of lithium carbonate from Argentina (HS code 283691).
Source: UN COMTRADE.

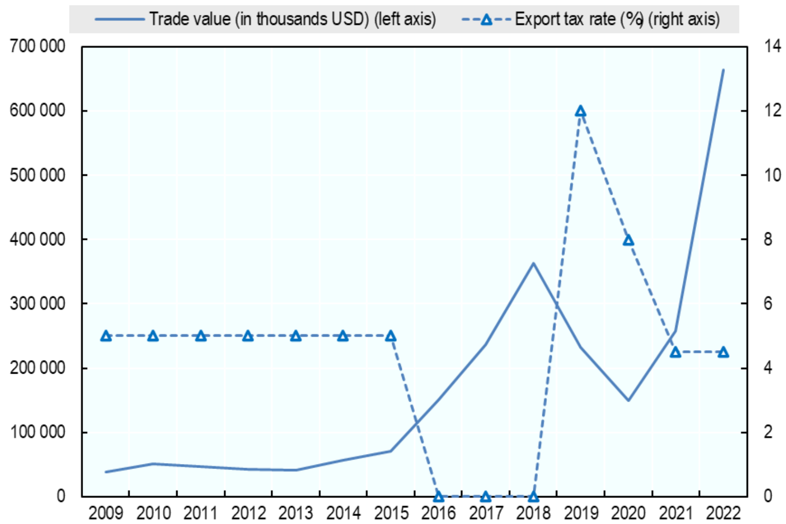
At first glance, the relationship between export taxes in Argentina and lithium exports reveals a pattern suggesting that such restrictions may have had a trade effect. The export tax on lithium fluctuated between 0% and 12% from 2009 to 2022; in 2016, it fell from 5% to 0% and the value of lithium exports surged by 113% the same year (Figure 2.5, Panel a). Such a large increase in the value of exports can only be partly explained by the price rise (15%) observed over the same period (Figure A C.1). In addition, the volume of lithium exports (i.e. net weight in kg) also significantly increased by 58% (Figure 2.5, Panel b), meaning that exports also increased in real terms (i.e. keeping prices constant).

[22] The first domestic battery factory in Argentina opened in late 2023 (Heredia, 2023[87]).

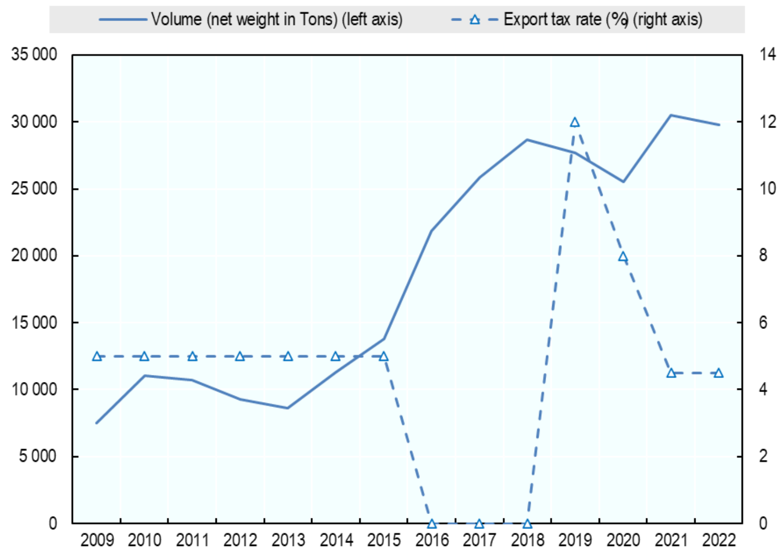
In 2019, when the tax was re-introduced and set at 12%, the value of lithium exports fell by 36% – while the price decreased by only 4.8% over the same period. The volume of lithium exports also decreased by 3.5%, showing a (albeit smaller) decrease in real terms as well. In 2020, the export tax was reduced from 12% to 8%. While exports in both value and volume also decreased (by 35.7% and 7.6% respectively) this was in the context of a significant price decline of 41.2%, related to the global COVID-19 pandemic. In 2021, the tax was decreased further from 8% to 4.5% and exports increased by 72% in value and 19% in quantity; at the same time, the lithium price soared by 421%. In 2022, the tax rate was left unchanged while exports continue to grow, probably partly driven by a continued upward trend in price.

Figure 2.5. In Argentina, export taxes and lithium exports tend to mirror each other

a. Exports in value



b. Exports in volume

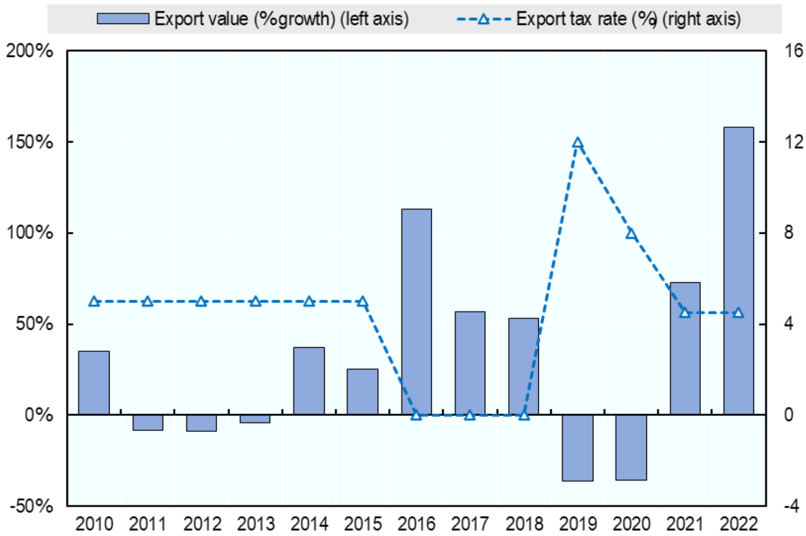


Note: Exports of lithium carbonate from Argentina (HS code 283691).
Source: UN COMTRADE and OECD (2025[1]).

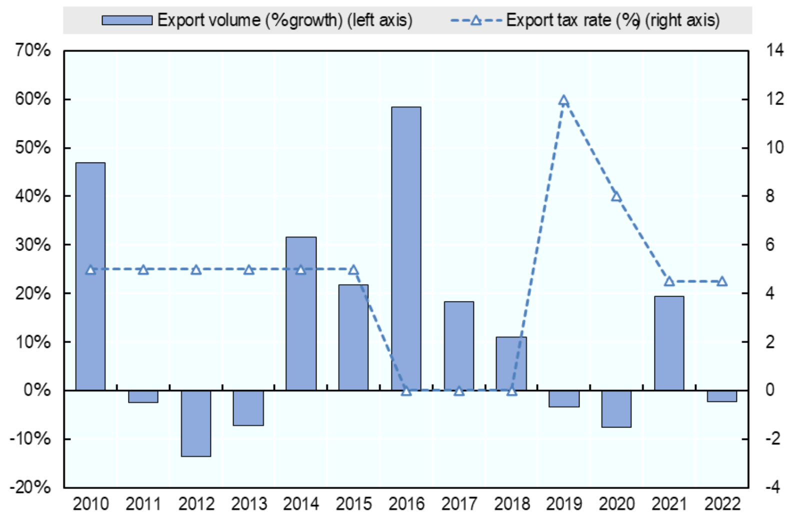
In sum, the export tax and exports (in value and quantity) appear to roughly mirror each other, with lithium exports increasing when export restrictions decrease and vice versa. That said, many factors could explain the observed variation in lithium exports and only a full econometric analysis may allow for an analytical counterfactual to isolate the effect of export taxes on lithium exports. However, this descriptive statistical analysis suggests that drops in the export tax rate coincide with positive growth rates in (both value and volume) lithium exports, apart from 2020 which could be considered an outlier because of the global pandemic. Figure 2.6 illustrates the correlations discussed above in changes instead of in levels.

Figure 2.6. A decrease in the export tax rate coincides with a positive growth rate in lithium exports

a. Exports in value



b. Exports in volume

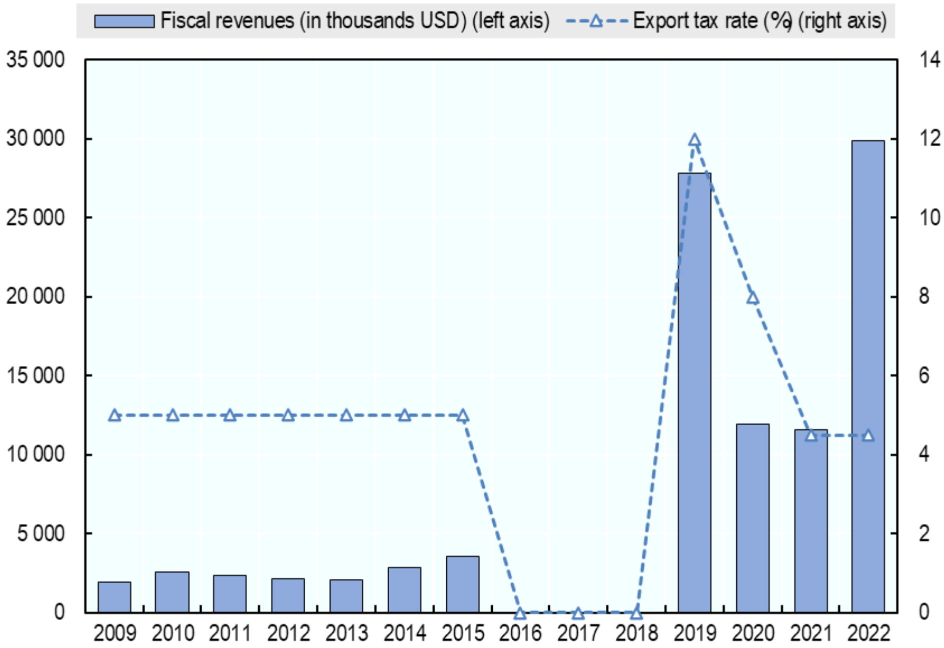


Note: Exports of lithium carbonate from Argentina (HS code 283691).
Source: UN COMTRADE and OECD (2025[1]).

Finally, calculations – which simply multiplied tax rates by the value of exports – suggest that fiscal revenues generated from the tax generally followed the swings in the export tax rate. Mechanically, when the tax was set to zero in 2016-18, no revenues were generated (Figure 2.7). In 2019, the estimated revenues from the tax increased to their highest level in the investigated period despite the significant actual decrease in exports (both in value and volume) discussed previously. In 2020, the estimated revenues were driven downward because exports (both in value and volume) also decreased. In 2021, fiscal revenues from the tax remained constant while the tax rate was decreased and exports increased. In 2022, the estimated fiscal revenues from the tax increased because exports increased in value. Just as with the effect on exports, further analyses would be needed to precisely estimate the net effect of export taxes on fiscal revenues. However, these simple calculations suggest that this net effect could be positive as – as the data point for 2019 in Figure 2.7 shows – the combination of a much higher tax rate and reduced value of exports can still result in an increased fiscal revenue. That said, there might be negative longer-term impacts on the development of the industry.

Beyond these economic effects, Annex Box A.C.2 discusses existing qualitative evidence on the possible indirect ESG effects of Argentina's export taxes.

Figure 2.7. Fiscal revenues follow the level of the export tax on lithium export



Note: Estimated fiscal revenues, calculated by multiplying the tax rate by export values.
Source: UN COMTRADE and OECD (2025[1]).

2.3.2. Zimbabwe's export ban

Zimbabwe has the largest lithium reserves and mines in Africa and ranks sixth in terms of global lithium deposits (Kamba, 2024[24]). Lithium in Zimbabwe can be found mostly in hard-rock mineral deposits, such as spodumene and petalites. These minerals are mined and further processed to produce compounds, such as lithium carbonates and lithium hydroxides (IFC, 2024[25]). As it has only been two years since the implementation of export controls, it remains premature to draw conclusions regarding the impacts on export flows of lithium products from Zimbabwe. However, it is noteworthy that total lithium exports increased significantly, reaching USD 209 million in 2023, largely driven by production by foreign companies based in Zimbabwe (Smith and Kemp, 2023[26]). Processing plants began using more of their capacities following the ban, yet the impact of their activities post-prohibition remains limited, as the price fall in 2023 and the country's domestic conditions forced them to reduce production and dismiss workers (Chingono, 2024[27]).

As discussed above, spodumene is not currently distinguished in the HS classification and it is therefore difficult to assess the impact of Zimbabwe's export ban on exports of this product. However, according to Tang (2024[28]), in 2023 Zimbabwe still held a 9% share of China's spodumene import market share despite the ongoing export ban. Exports to China reached 393 000 metric tonnes, which marked an increase of almost 570% from the previous year. Regarding lithium chemical compounds (lithium carbonates and lithium hydroxides or oxides), the available information is also limited, but official trade statistics reported for HS codes 283691 and 282520 indicate a notable increase in Zimbabwe's exports of more processed forms a few months after the adoption of the ban. Given the proximity to the time of the measure, it is unclear whether this rise is a direct consequence of the ban.

As previously noted, Chinese mining companies were operating prior to 2022 and began investing heavily in lithium processing plants and infrastructure in early 2023 (Tsico Agric, 2023[29]). Thus, it may still be too soon for the ban to have led to this rapid export of solely processed lithium forms, as exports of these products to China were already taking place from the period when the country was dynamically developing its refining industry. Meanwhile, China's lithium carbonate and hydroxide global production capacity totalled 1.1 million metric tonnes per year and 700 000 metric tonnes per year in 2023, representing increases of 83% and 94%, respectively, compared to 2022 (Tang, 2024[30]).

The implementation of the ban was also correlated with an increase in inward FDI in Zimbabwe's natural resources sector.[23] In 2023, the country attracted a total of USD 588 million in FDI, which was the highest inflow in five years, representing a 49% increase from 2022, and outpacing the 22% growth seen in Southern Africa as a whole (UNCTAD, 2024[31]).[24] China continued to be the leading investor, with 369 licensed investors (mainly in the mining and mineral processing sectors, and lithium in particular), representing 60% of all investors in 2023 and marking a 154% increase from 2022 (ZIDA, 2023[32]). Other main source countries of FDI included Russia, Iran, the United Kingdom, the United Arab Emirates, India, and South Africa, but also Botswana and other members of Southern African Development Community (SADC). Annex Box A.C.3 discusses relevant ESG dimensions relating to the lithium mining industry in Zimbabwe.

2.4. Lithium: Key findings and tentative conclusions

The vast majority of global lithium exports are not subject to export restrictions, with only two countries, neither of which are major producers, applying restrictions through export taxes or licensing to lithium exports: Argentina and Zimbabwe. Therefore, at present, export restrictions do not appear to represent a major issue for trade in lithium products.

However, relevant preliminary analysis from Argentina – where export taxes on lithium have been applied at a fluctuating rate – suggests that lithium exports may have decreased as export taxes increased. In some years, the estimated fiscal revenues generated by the tax have increased in parallel with the tax rate even as exports decreased, suggesting that the net effect on fiscal revenues is positive. That said, given that lithium is a commodity exchanged on global markets and for which Argentina is not a major producer, decreased exports and increased taxes may also have hurt domestic producers as they would not be able to easily pass the tax increase on to foreign countries. The exact economic effects of such measures could be more thoroughly investigated in the future, including through more in-depth qualitative and econometric analyses.

While it is too early to estimate the full effect of Zimbabwe's export ban on primary forms of lithium, it can already be seen that this decision prompted foreign companies, notably Chinese companies, to invest in local production in Zimbabwe of more refined forms of lithium. Export revenues surged significantly, more than tripling between 2022 and 2023, which seems to be the result of increased exports volume of processed lithium.[25] However, such export restrictions may have more limited effects in the long run as other underlying challenges such as low-quality infrastructure, skills shortages, and a lack of adequate

[23] The increase in FDI has also likely been facilitated by the implementation of the 2018 "open for business" policy which included the establishment of the Zimbabwe Investment and Development Agency (ZIDA) as a single point of contact in 2020 (US Department of State, 2024[83]).

[24] Although FDI inflows into Zimbabwe have risen, they remain volatile due to external factors such as the implications for FDI of the COVID-19 pandemic, political instability and global geopolitical tensions (UNCTAD, 2024[31]). Factors specific to Zimbabwe such economic sanctions, the country's high debt levels and concerns over corruption, environmental, social, and governance (ESG) also play a role.

[25] Zimbabwe's exports data for 2023 were not yet available at the time of writing.

financial, technical, and human capital, which are fundamental drivers of investment (World Bank, 2023[33]), affect efforts to move up the value chain. In this example, some commentators argue that the main beneficiaries might well be Chinese companies, which invested in Zimbabwe for decades and now run the majority of Zimbabwe's mines and which therefore have a strategic advantage in expanding domestic processing compared to potential rivals (Gbadamosi, 2023[17]). Meanwhile, intensifying ESG concerns related to the recent expansion of mineral mining and processing industry are being documented in the literature.

In the near future, lithium production is likely to diversify further with new projects opening in a wide range of countries including in OECD Members where lithium mines projects are at the exploration stage in Austria, Canada, France, Mexico, Portugal, Spain, the United Kingdom, and the United States. Around the world, lithium mining companies recorded a 50% increase in capital expenditure to ramp up the scale of their existing operations in 2022 (IEA, 2023[34]).[26] Exploration spending for lithium projects also rose by 90% in 2022, mostly in Canada and Australia (IEA, 2023[34]). Exploration activities were also expanding in Africa and Brazil, although the decline in lithium prices which started in 2023, continued through 2024 and stabilised at multi-year lows in the first half of 2025 is already resulting in postponing or canceling of some of the planned projects.

Although necessarily partial, the results of this section shed light on the potential trade frictions that may emerge if more countries relied on export restriction as the industry develops in new areas. Having said that, for lithium exports, this report examined the impact of export restrictions in Argentina and Zimbabwe, two countries that are not the major producers of lithium. The results derived from Argentina and Zimbabwe may not extend to other countries who are major producers of lithium in the global markets if those countries would implement export restrictions, as they would likely be in a better position to pass on the costs of these increases on to lithium importers.

3. Nickel

3.1. Stylised facts on nickel production and export restrictions

3.1.1. Mining and uses of nickel

Nickel, notably in alloy form, is used intensely in the stainless-steel and the "super-alloys" industries. Nickel is also valued for its energy density, full recyclability and other properties sought after by the fast-growing batteries and EV industries.[27] Currently, close to 70% of global demand for nickel is accounted for by the stainless-steel industry, products of which are mainly used in food, petrol and chemical industries, as well as transport and construction. The batteries industry accounts for approximately only 5% of current demand but is also the most rapidly growing source of demand and is predicted to have a growing influence on nickel markets in future.[28]

[26] Companies based in China nearly doubled their investment spending in 2022 (IEA, 2023[34]).

[27] Nickel is the most high-performing, cost-efficient and safest material for use in li-ion batteries: its high purity helps increase the capacity of batteries (and hence EV range); it is capable of handling high discharge rates (which determine vehicle acceleration); and itis attractive in terms of safety (i.e. high temperatures at which it remains stable). However, nickel-free battery chemistries such as lithium iron phosphate (LFP) are becoming more popular and this may reduce nickel consumption in battery manufacturing.

[28] It is also easier to substitute nickel for other materials in the stainless-steel industry than it is to substitute for it in the batteries industry. This is because, due to its energy density properties, nickel has no known substitute.

There are two principal classes of (primary) nickel, distinguished by purity, and normally obtained from different kinds of nickel deposits in different geographical locations with different geological characteristics. *Class 1* products are high purity, obtained typically from magmatic *sulfide* deposits (where the principal ore mineral is pentlandite). *Class 2* products are the lower purity products obtained typically from *laterite* deposits (where the principal ore minerals are nickeliferous limonite and garnierite):

- *Sulfide resources* (estimated at some 118 mln metric tonnes) are concentrated in South Africa (28.1%), Canada (18.6%), Russia (17.3%), Australia (10.1%), China (5.1%) and United States (4.3%) (Nickel Institute, 2016[35]). [29]
- *Laterite resources* (estimated at some 178 mln metric tonnes) can be mostly found in Indonesia (18.7% of reserves), Australia (17.7%), Philippines (10.3%), Cuba (9.1%), New Caledonia (8.4%) and Brazil (8.3) (Trinitian Green Energy Metals, 2023[36]).

Class 1 products are used mainly, though not only, in the production of cathodes of electrical batteries (e.g. those used in EV) while *Class 2* products are typically used in production of stainless steel and superalloys.[30] Production of *Class 2* products tends to have higher environmental and biodiversity footprints.[31] While it is possible to obtain *Class 1* products from laterite deposits, this can entail higher greenhouse gas emissions and a more significant environmental impact. Chinese investors operating in the nickel industry in Indonesia are reported to have significantly advanced the technology to obtain Class 1 products from laterite resources, opening the possibility for Indonesia to tap into the market for EV batteries. However, the process of purification from low to high grades remains highly energy intensive and generates large amounts of environmentally harmful residues. In Indonesia, the environmental footprint is reportedly very high as the principal source of energy for purification of nickel is coal and tailings from nickel purification have until relatively recently[32] been reported to be deposited into the ocean (Tritto, 2023[37]).

The current volume of global primary nickel production is about 2.5 mln metric tonnes. Demand is projected to increase to 10 mln metric tonnes by 2050 in the IEA's most ambitious energy transition scenario (IEA, 2021[38]). However, some analysts point out that the indices of pressure on resources — calculated as ratios of expected cumulated demand to known reserves — are relatively favourable for nickel (around 50-60% as compared to 78-90% for copper). This means that nickel, even when taking into account only the currently known economically-viable deposits, is available in relatively sufficient quantities and that pressures on international nickel markets are likely to be smaller relative to other minerals (Hache, 2021[39]).

Historically, nickel also has a reputation for high price volatility. Nickel prices are influenced by a myriad of factors, but industry analysts report that in recent years they were shaped by commercial and industrial policies of key primary nickel producing countries such as Indonesia and the Philippines. The recent historical high of international nickel prices was during the Global Financial Crisis of 2008-09, and prices

[29] These are current reserves, but economic viability is one of the criteria for estimating reserves and thus incentives for search and/or approval of deposits are influenced by market prices, which have shown remarkable volatility in recent years. Some projects not deemed viable at low prices may be viable at higher prices, affecting the geography of reserves and ultimately production.

[30] Even though the two kinds of reserves produce nickel of different purity which is in turn destined for quite different uses, the existing international standard for reporting of international trade data (i.e. the HS code at 6-digit level) does not distinguish between nickel ores and concentrates obtained from these different deposits. Moreover, the only HS product category encompassing relatively unprocessed nickel products (*HS260400 - Nickel ores & concentrates*, see Annex Table A D.1) covers both ores and concentrates, that is, it does not distinguish between the direct product of mining and the first stage of nickel processing.

[31] Laterite deposits involve strip mining which is associated with high environmental costs (large areas of land need to be uncovered, and large range of fauna and flora are affected in often highly biodiverse regions such as tropical forests of Indonesia) while sulphide deposits are typically mined in open pit mines which are smaller and tend to be located in deserted cold areas, e.g. the Arctic, northern Canada, northern Russia).

[32] Note that, while this might still be happening, this practice was officially banned by Indonesia in 2021 (See, for example, Standards and Poors Global (2023[78]).

declined by some 70% in the aftermath.[33] The large changes in historical nickel prices suggests that nickel mining and financing decisions are characterised by particularly high uncertainty. Market participants are more sensitive to price changes and, at the same time, this sensitivity results in a lot of “noise” in nickel prices. In such an environment, even small policy changes may lead to strong market reactions. In combination with high degrees of concentration of production, high price volatility for a mineral translates into higher investment risk and poorer financing options for new mining operations, notably for private investors.[34]

3.1.2. Nickel production, international trade and use of export restrictions

Production

Accounting for some 1.6 mln metric tonnes (or 48%) of global production, Indonesia was the world’s largest primary nickel[35] producer in 2022. In that year, it also produced eight times more primary nickel than a decade earlier (0.2 mln metric tonnes, 8% of global production and 4th largest producer in 2012).

This growth in the volume of production by Indonesia in the 2012-22 period (700%) vastly exceeded overall global production growth (18%) and production growth of other large producers, some of which expanded [New Caledonia (44% increase), China (17%) or Philippines (3%)] and some of which reduced [Russia (-18%), Canada (-39%) and Australia (-47%)] their production levels in the same period. This illustrates the extraordinary nature of Indonesia’s expansion as a primary nickel producer, accompanied by crowding out of some of the other large traditional producers but also diversification towards other efficient primary nickel producers (Figure 3.1, Panel A).[36]

Overall, over the course of the last ten years, Indonesia’s production of primary nickel has expanded considerably both in terms of volumes of production and shares in global production. This has clearly driven the increase in the global concentration of primary nickel production as measured by the HHI (Figure 3.1, Panel B). In 2022, the HHI for global primary nickel production reached 0.25 – a level that suggests a relatively high overall concentration of global production,[37] while global concentration at the beginning of the period was relatively low (0.05). To the extent that Indonesia’s trade policies contributed to the expansion of its production of primary nickel, they appear to have also significantly contributed to global concentration of nickel supply (Figure 3.1, Panel B).

[33] At the beginning of 2022, following Russia’s invasion of Ukraine, the disorderly nickel price movements resulted in temporary suspension of trade of this metal on the London Metal Exchange (LME) and cancelling of trades worth some USD 12 bln which led legal challenges against LME. Several market commentators suggested that these price fluctuations were not necessarily fundamentally due to Russia’s invasion as Russia’s shares in world nickel markets were too small to cause such price fluctuations.

[34] This is especially the case for private investors consider rates of expected return and investment risks, while state investors (with fewer constraints in terms of the cost of capital) may pursue strategic investments and have different attitude towards returns and risks.

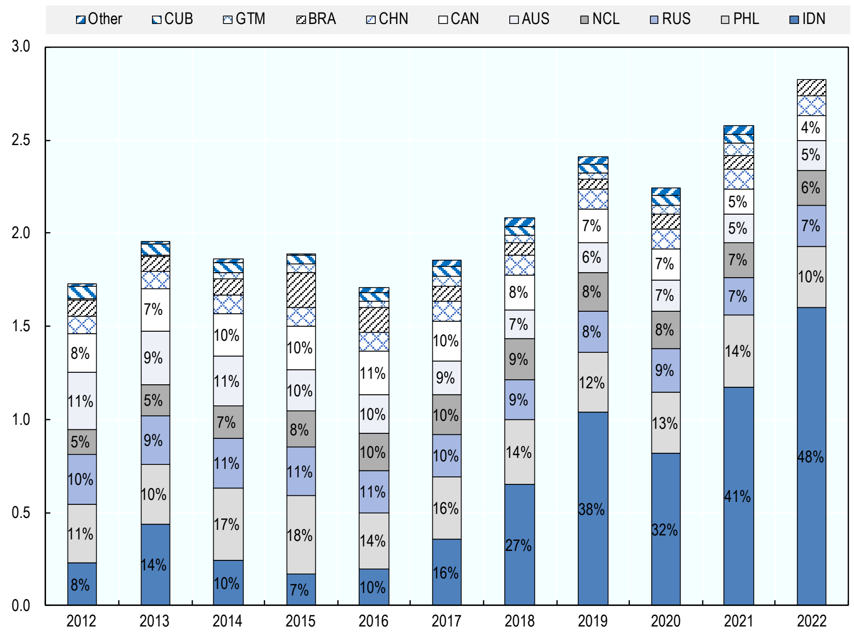
[35] According to the USGS, which is the source of the reserves and production data used in this analysis, primary nickel refers to nickel recovered from ores (both sulfide ores and laterite ores) through mining and refining, as opposed to secondary nickel recovered from scrap.

[36] This comparison does not distinguish between the mining of laterite and sulphide ores – Canada has mainly sulphide resources while Australia is an important holder of both sulphide and lateritic resources. The exposure to competition from Indonesia of Canada and Australia may therefore be different.

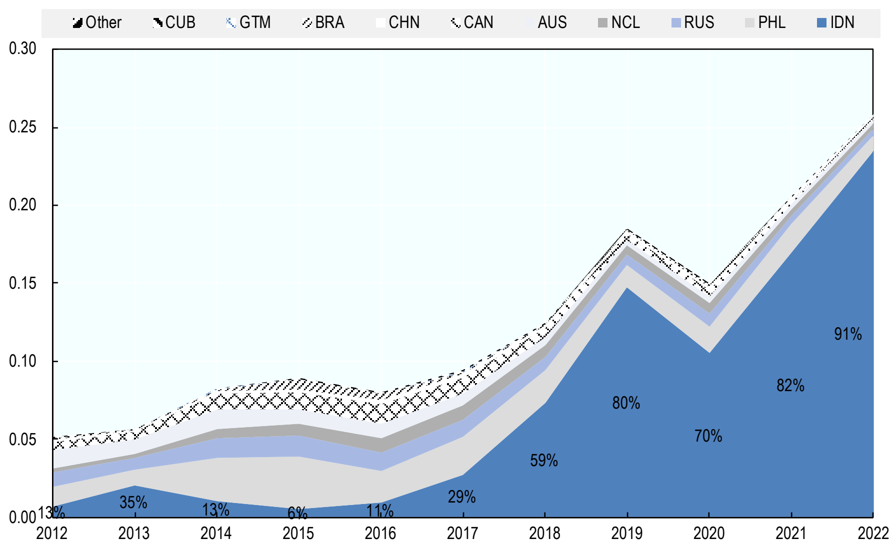
[37] While the value of HHI of 0.25 can be obtained with many different market shares of major producers, a helpful illustrative interpretation of the value of the HHI index of 0.25 is that it is equivalent to the value that would be obtained if there were only four equal-sized players in the market, while the interpretation of the value of the HHI index of 0.50 would be that there were only two equal-sized suppliers.

Figure 3.1. Indonesia dominates production of primary nickel

Panel A. Production of primary nickel by country (mln metric tonnes)



Panel B. The corresponding HHI index of global production concentration (and % contribution to that index of production share of Indonesia)



Source: USGS.

Trade

As for cobalt, nickel products are relatively well tracked in the HS nomenclature, with 6-digit level classifications enabling identification of: ores and concentrates (OM); chemical compounds (CC) (e.g. nickel oxides, hydroxides, chlorides and sulphates); intermediate nickel products referred to in this section as non-ferrous base metals of nickel (NFBM) (e.g. mattes, powders and flakes); as well as waste and scrap (WS) (Table 3.1). An additional consideration, not unique to nickel, is how far down the value chain a mineral should be tracked to be able to trace the effects of export restrictions. In the case of Indonesia, one of the stated objectives of the export restrictions on unprocessed forms of nickel introduced in the mid-1990s was to promote domestic smelting and further processing, initially by the stainless-steel industry (which was at the time a primary user of primary nickel mined in and exported from Indonesia). The analysis of the possible trade effects of export restrictions in Section 4.3. includes also a range of nickel-containing stainless-steel products which can be identified readily in the HS nomenclature. The greater granularity in the HS codes of G7 countries also enables analysis of their participation in trade in nickel products (Box A D.1 in Annex D).

Table 3.1. Nickel products covered in the HS 2007 trade nomenclature

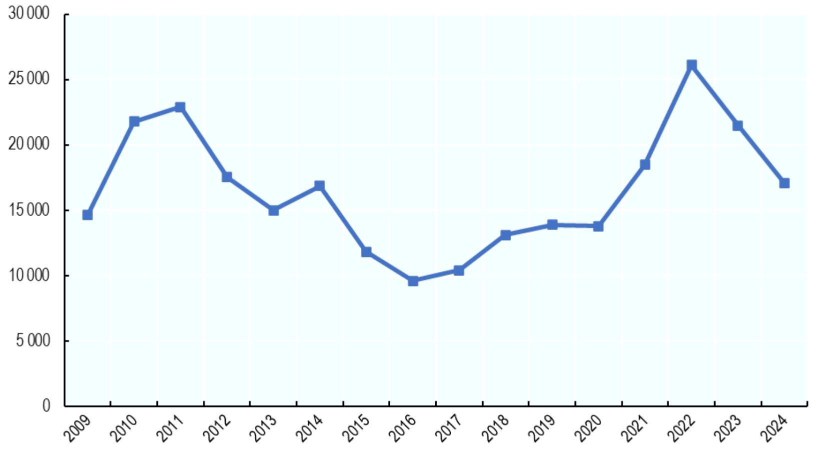
HS2007 code	HS title	Processing stage
260400	Nickel ores & concentrates	OM: ores and concentrates
282540	Nickel oxides & hydroxides	CC: chemical compounds
282735	Nickel chlorides	CC: chemical compounds
283324	Sulphates of nickel	CC: chemical compounds
750110	Nickel mattes	NFMB: metals and semi-manufactured articles
750120	Nickel oxide sinters & other intermediate products of nickel metallurgy	NFMB: metals and semi-manufactured articles
750210	Nickel, not alloyed, unwrought	NFMB: metals and semi-manufactured articles
750220	Nickel alloys, unwrought	NFMB: metals and semi-manufactured articles
750400	Nickel powders & flakes	NFMB: metals and semi-manufactured articles
750511	Bars, rods, profiles & wire, of nickel, not alloyed	NFMB: metals and semi-manufactured articles
750512	Bars, rods, profiles & wire, of nickel alloys	NFMB: metals and semi-manufactured articles
750521	Wire of nickel, not alloyed	NFMB: metals and semi-manufactured articles
750522	Wire of nickel alloys	NFMB: metals and semi-manufactured articles
750610	Plates, sheets, strip & foil, of nickel, not alloyed	NFMB: metals and semi-manufactured articles
750620	Plates, sheets, strip & foil, of nickel alloys	NFMB: metals and semi-manufactured articles
750300	Nickel waste & scrap	WS: waste and scrap

Source: OECD (2025[1]).

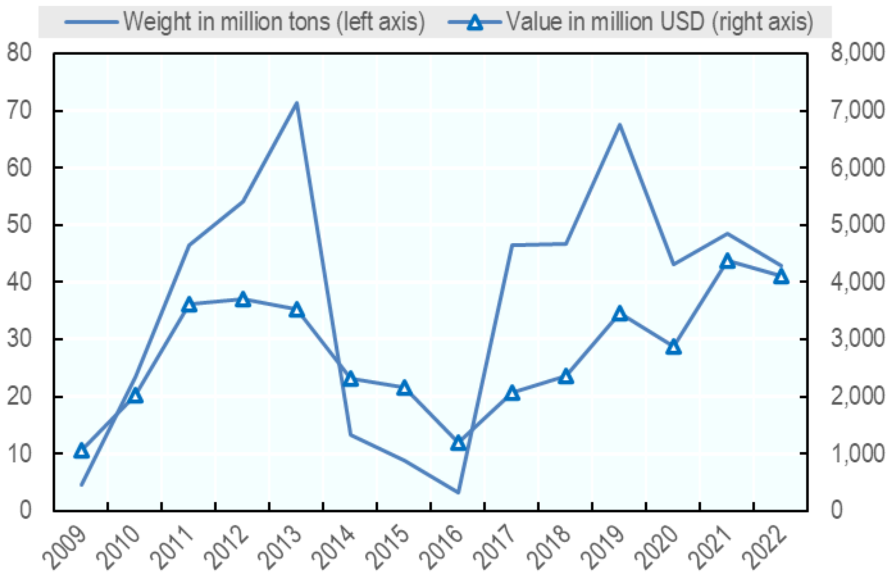
The value of global exports of primary nickel has changed significantly since 2010 (Figure 3.2, Panel B) and these changes appear to reflect not so much the fluctuations of global nickel prices (Figure 3.2, Panel A) but rather the shifts in exports by Indonesia (Figure 3.2, Panel C). Movements in global prices cannot, for example, account for the decline in the global value and volume of exports between 2013 and 2016 and for the slowdown in growth of the value of global exports after 2019. They are, however, closely correlated with significant decreases in the value (and shares) of exports from Indonesia starting first in 2014 and in 2019; that is, in periods when Indonesia introduced significant export restrictions. In 2013, Indonesia accounted for some 35% of world exports of nickel ores and concentrates; in 2015 and 2016, it ceased to export these products almost completely; and in 2019, it again accounted for 32% or world nickel exports. In contrast, in the same period, the values of exports and export market shares of the second and third largest exporters – Philippines and New Caledonia – held up or expanded. These developments show that exports of primary nickel products by Indonesia – which were themselves shaped by the country's use of export restrictions – have shaped the availability of primary nickel in world markets to an important extent.

Figure 3.2. World exports of HS*260400 - Nickel ores & concentrates* have been correlated with nickel prices and Indonesia's policies

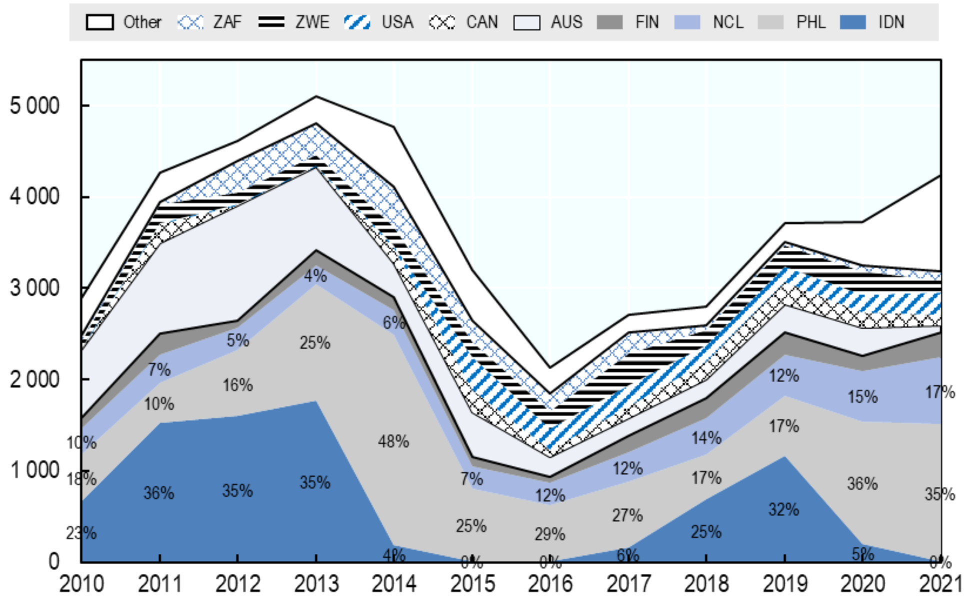
Panel A. Price USD/tonne



Panel B. Volume and value of world exports



Panel C. Value of world exports (USD mln) and selected country shares



Source: OECD calculations based on BACI.

In value terms, primary nickel is not—and has not been in the past—the main type of nickel exported globally (Table 3.2). The share of exports of primary nickel in exports of all nickel products has increased from 5% on average in the period 2002-04 to some 14% in 2020-21, but intermediate nickel products (HS Chapter 75, Table 3.1) accounted for more than 80% of the value of global nickel exports in the most recent period. Yet, Indonesia was not only not a major player in these markets but its share appears to have even fallen (from 6% of the global value of exports of intermediate nickel in 2010 to 4% in 2016 and 5% in 2022) (Figure 3.3).[38] International markets for some of these intermediate products also appear to be relatively more concentrated than those for primary nickel although exports of nickel ores and concentrates have become progressively more concentrated (Figure 3.4). Chemical compounds of nickel accounted for relatively small shares of global nickel exports (Table 3.2) and Indonesia was also a small player in these markets (Figure 3.4, Panel B).

[38] Tritto (2023[37]) suggests that a relatively low level of processing of nickel in Indonesia has been a result of the high environmental costs of doing so from laterite ores, which discourages investors notwithstanding growing demand. Tesla, for example, is reported not have pursued nickel refining investments in Indonesia for environmental reasons. The same author reports, however, that despite the heavy environmental footprint, some of the refining facilities commissioned in Indonesia have become operational in recent years with refined products destined for the Chinese EV industry.

Table 3.2. The value of world exports of nickel products

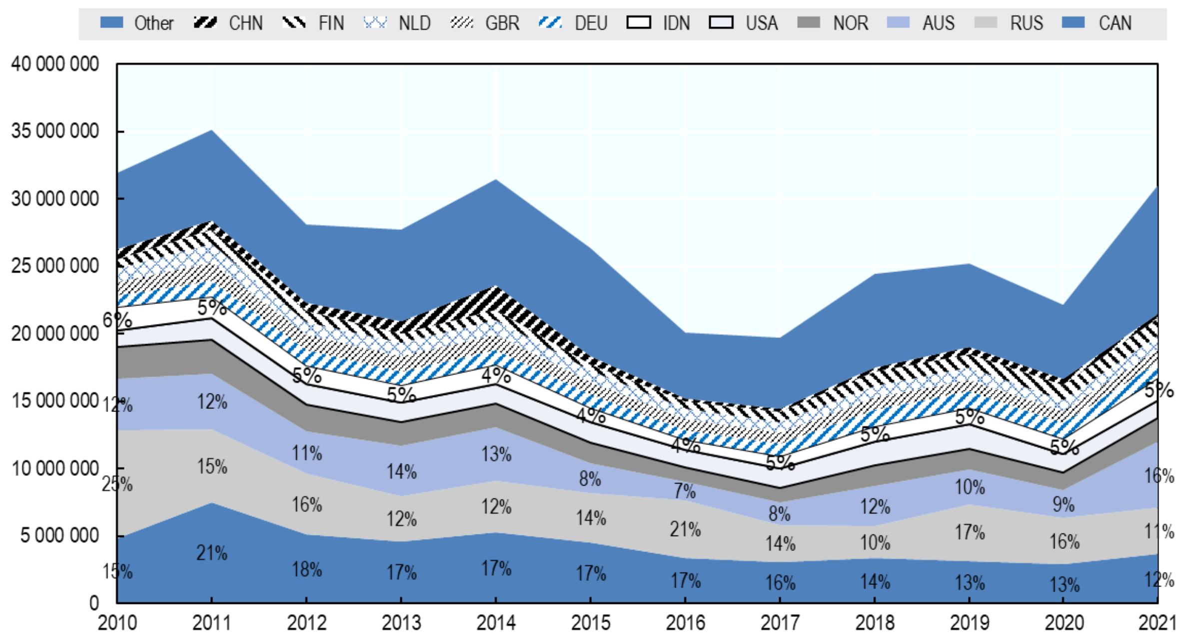
Value (USD mln) and shares

	2002-04		2012-14		2020-21	
750210-Nickel, not alloyed, unwrought	8,132,024	*50%*	16,086,324	*43%*	13,759,124	*40%*
750110-Nickel mattes	1,858,686	*11%*	4,971,208	*13%*	4,725,775	*14%*
260400-Nickel ores & concentrates	850,877	*5%*	4,997,856	*13%*	4,093,543	*12%*
750120-Nickel oxide sinters & other intermediate products of ni	1,743,516	*11%*	2,348,691	*6%*	3,129,615	*9%*
750512-Bars, rods, profiles & wire, of nickel alloys	444,180	*3%*	1,790,982	*5%*	1,605,690	*5%*
750400-Nickel powders & flakes	458,299	*3%*	966,265	*3%*	1,550,213	*5%*
750620-Plates, sheets, strip & foil, of nickel alloys	490,493	*3%*	1,151,263	*3%*	1,178,195	*3%*
750522-Wire of nickel alloys	316,385	*2%*	845,101	*2%*	1,055,073	*3%*
750220-Nickel alloys, unwrought	523,696	*3%*	1,726,457	*5%*	1,013,066	*3%*
750300-Nickel waste & scrap	395,230	*2%*	836,472	*2%*	908,208	*3%*
283324-Sulphates of nickel	168,187	*1%*	347,322	*1%*	827,364	*2%*
750610-Plates, sheets, strip & foil, of nickel, not alloyed	472,027	*3%*	694,191	*2%*	270,310	*1%*
282540-Nickel oxides & hydroxides	180,711	*1%*	287,593	*1%*	159,984	*0%*
750521-Wire of nickel, not alloyed	43,162	*0%*	72,878	*0%*	68,704	*0%*
282735-Nickel chlorides	43,872	*0%*	61,292	*0%*	54,665	*0%*
750511-Bars, rods, profiles & wire, of nickel, not alloyed	285,190	*2%*	611,200	*2%*	44,572	*0%*

Source: OECD calculations based on BACI.

Figure 3.3. Indonesia is not a major exporter of non-fabricated basic metal (NFBM) products of nickel

Value (USD mln) and selected country shares

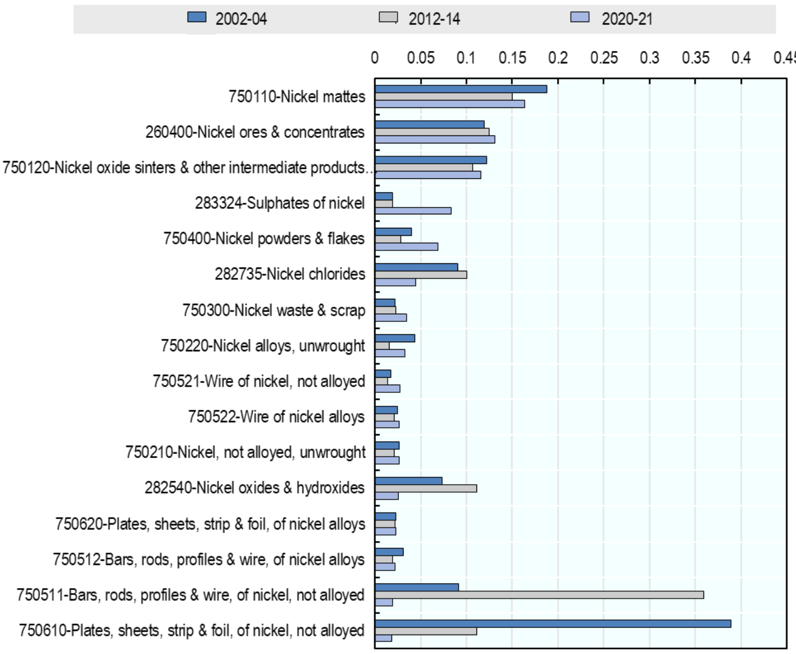


Source: OECD calculations based on BACI.

Figure 3.4. Global export concentrations of nickel products

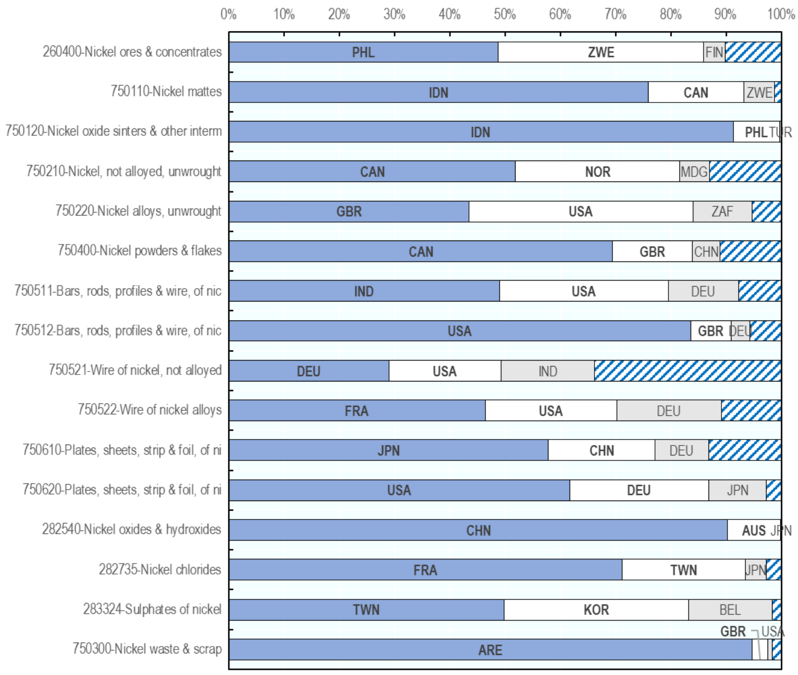
HHI index across exporters

a. Evolution in time



Source: OECD calculations based on BACI.

b. Exporting country shares, 2022



Notes: ARE: United Arab Emirates, AUS: Australia, BEL: Belgium, CAN: Canada, CHN: China, DEU: Germany, FIN: Finland, FRA: France, GBR: United Kingdom, IDN: Indonesia, IND: India, JPN: Japan, KOR: Korea, MDG: Madagascar, NOR: Norway, PHL: Philippines, TUR: Türkiye, TWN: Chinese Taipei, USA: United States, ZAF: South Africa, ZWE: Zimbabwe.
Source: UN Comtrade.

Export restrictions

Use of restrictions on exports of nickel products has grown considerably since the OECD started tracking these measures in its Inventory in 2009 (Figure 3.5). While, compared to cobalt, nickel faces, on average, fewer export restrictions both by number of restrictions and share of trade facing at least one restrictions [see Figures 4.3, 4.4 in Kowalski and Legendre (2023[2]).], for primary forms of these minerals (ores and concentrates - OM), nickel faces more export restrictions than cobalt (*ibid*).[39]

Nickel waste and scrap had the most export restrictions in 2022[40], likely reflecting both the fact that nickel is fully recyclable, and recycling is the most cost-efficient way of obtaining it, as well as environmental considerations regarding international trade of waste and scrap.[41]

Primary nickel products saw a visible growth in, and recorded the second highest count of, export restrictions in 2022 after nickel waste and scrap, followed by intermediate nickel products and chemical compounds (see *HS260400 – Nickel ores and concentrates*, Figure 3.5 Panel A and *Nickel – OM* in Figure 3.5, Panel B).

A marked step increases in the global counts of export restrictions across the different processing stages can be observed particularly around the years 2012-2014 and then around 2020. This may reflect the use of export restrictions by Indonesia as well as other nickel producing countries, possibly in reaction to Indonesia's restrictions, in order to secure more affordably priced nickel supplies for downstream domestic users in the context of falling global supply and rising prices.[42]

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[39] This measure counts export restrictions. Note also that the production and exports of primary cobalt are more concentrated than production and exports of primary nickel.

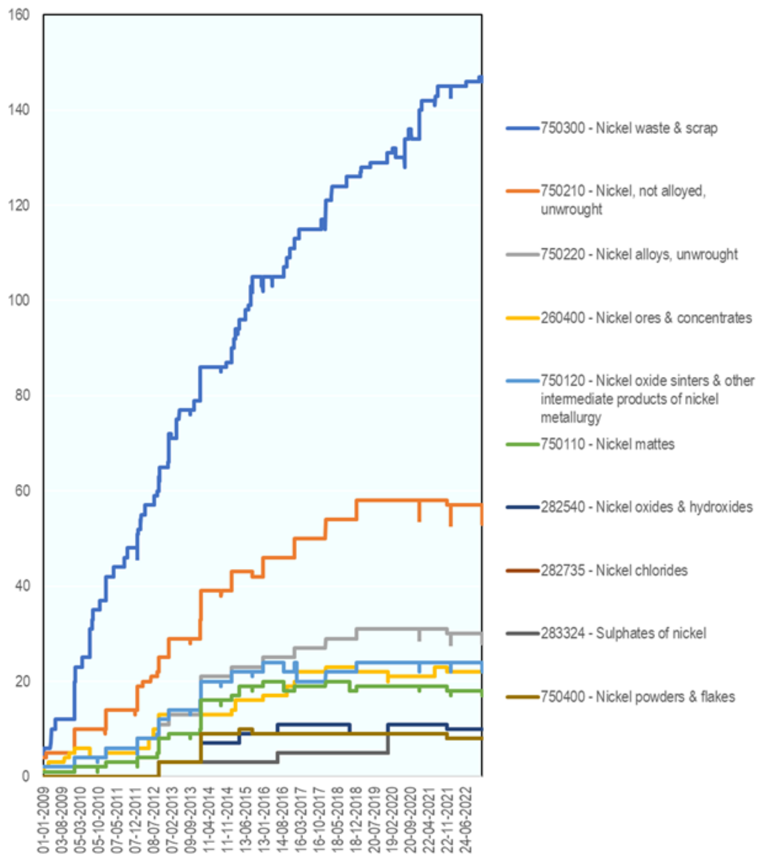
[40] This is the case also when export restrictions on waste and scarp of other products are a point of comparison [see Figure A D.4 in Kowalski and Legendre economy (2023[2]).].

[41] HS lines capturing trade of waste and scrap of various materials generally record high incidence and highest growth rates in export restrictions in place over the last decade economy (Kowalski and Legendre, 2023[2]).

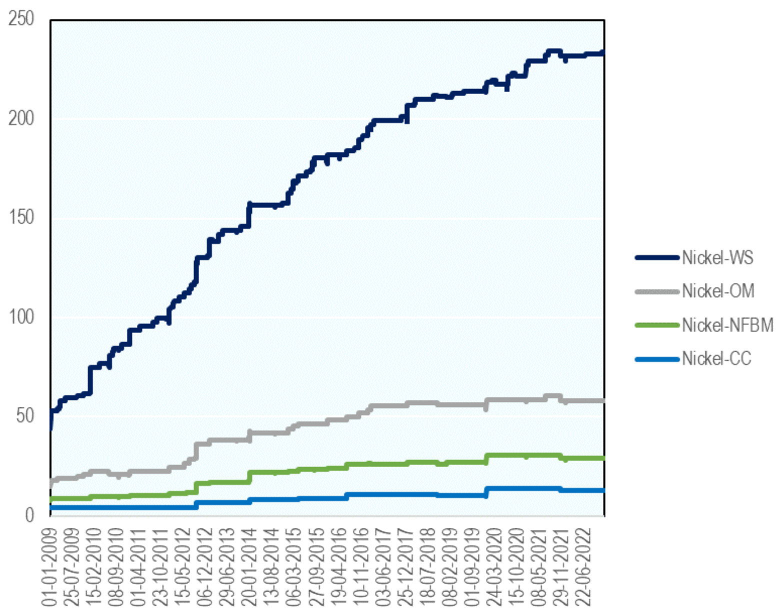
[42] Note the spike in the price of nickel and a decrease in world exports of primary nickel around just after the introduction of export bans by Indonesia in 2014 (Figure 3.4, Panel A) which is also discussed in more detail in Section 3.3.1 below.

Figure 3.5. Global incidence of export restrictions on exports of nickel products

Panel A. Number of measures in force in all countries



Panel B. By stage of processing, per number of HS6 “tariff lines” belonging to each category



Source: OECD (2025[1]).

3.2. The potential impact of export restrictions in Indonesia on trade in nickel products

3.2.1. Towards a better understanding of the potential impact of export restrictions on exports from Indonesia

Indonesia has accounted for significant shares of global export restrictions on nickel throughout the investigated period (Figure 3.6). This has been particularly the case for chemical compounds (Indonesia accounted for around 60% of global export restrictions on this nickel product category at the end of 2022), but also for ores and concentrates and intermediate nickel products (respectively approximately 30 and 20% of global measures, Figure 3.6). Indonesia's shares in global nickel export restrictions have been falling, but this does not mean that Indonesia has been using progressively fewer of these measures (to the contrary), but it rather reflects the growing use of these measures by other countries.

Indonesia's use of export restrictions on nickel has been highly publicised in market and policy analysis and their economic and wider effects remain highly debated (Evenett and Fritz, 2023[40]; IRENA, 2023[41]; Tritto, 2023[37]).

Quantitative restrictions on exports (quotas, export bans) are seen as inconsistent with WTO rules (Pauwelyn, 2020[42]) and some of Indonesia's export restrictions of this kind have been challenged at the WTO.[43] Other export restrictions (e.g. export taxes, licensing requirements) are allowed by the WTO under certain conditions, but they must not cause unjustifiable trade distortions.[44]

According to some business commentators and development policy analysts, Indonesia's use of export restrictions is a successful example of how a resource-rich country with ambitious development goals may use its natural endowments and trade policy instruments to incentivise development of downstream processing industry (Cust, 2023[43]). Downstream processing industry engages in concentrating, smelting or further processing of nickel ores to obtain purer nickel products or produce products which contain significant amounts of nickel (e.g. stainless steel or EV batteries), within the domestic economy. Its development is therefore deemed by some a successful example of maximising gains from natural resource extraction.

Those supportive commentators appear to accept that nickel export restrictions turned out to be a positive policy for Indonesia and instead ask to what extent the country could apply the same model to its other natural resource endowments or to what extent other resource-rich developing countries can emulate this "success" (e.g. Botswana's diamonds has been cited as a potential follower (Cust, 2023[43])). In 2024, Indonesia adopted a new suite of export restrictions on copper, iron, lead, zinc concentrates[45] and anode slime from copper concentrates, which prohibited exports of these products as of the beginning of 2025.[46]

Having said that, while this does appear to be a leading view, some analysts present elements of evidence which cast doubt on whether Indonesia's export restrictions were a success for the country. Evenett and Fritz (2023[40]) argue for example that it is hard to establish whether export restrictions alone would have resulted in the expansion of the downstream industry. The same authors argue also that is unclear whether the tax incentives provided to the downstream industry, in addition to export bans, generated the effects that outweighed the fiscal outlays, especially in the wider context of the generally rapid growth of the Indonesian economy in this period.

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[43] In 2019, the EU launched a WTO case against Indonesia's restrictions on exports of raw materials. The Panel report issued at the end of 2022 found against Indonesia and recommended that it bring its measures into conformity with its obligations under GATT 1994. Indonesia notified an appeal; however, (the WTO Appellate Body is currently non-operational. See DS592: Indonesia — Measures Relating to Raw Materials.

[44] As stipulated by Article VIII of the GATT.

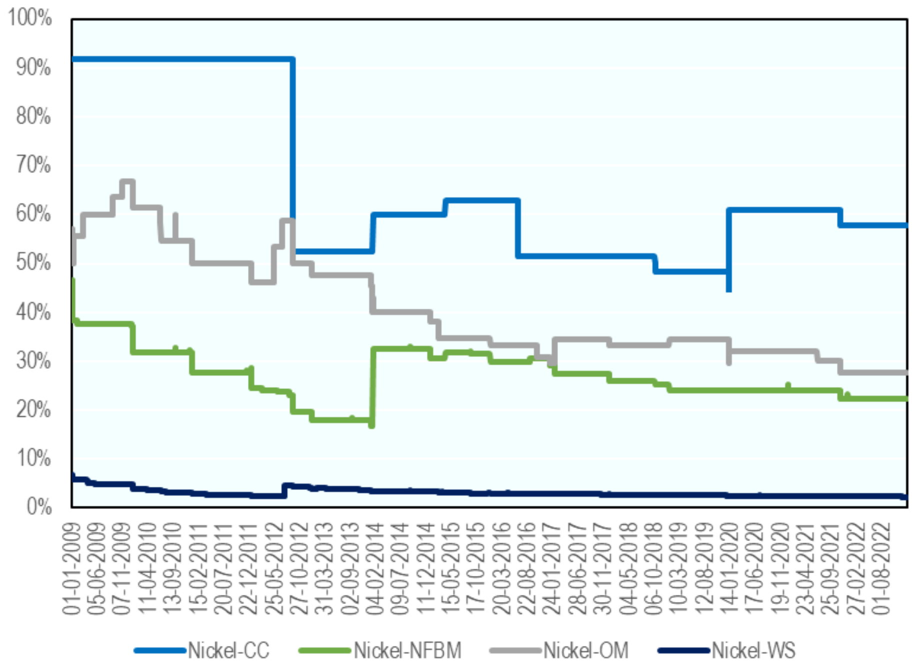
[45] The regulations cited in the footnote below specify specific types of concentrates (according to the content of the associated mineral) which are covered by these measures.

[46] Regulation No. 6/2024 and Regulation No. 10/2024 first extended the permitting period for the export licensing of these products through to the end of 2024, after which exports of these products were fully banned.

In addition, even some of the proponents of the use of export restrictions in this specific case point to a number of other, not necessarily less important, factors that may have enabled a successful development of downstream industry in Indonesia. These include, for example, the exceptionally high share – and thus high degree of influence – of the country in the global market for laterite nickel ore-related products, the relatively low costs of production of processed nickel products domestically (allegedly half of the cost in China), low costs of the much needed (largely in Indonesia coal-based) energy which is needed for processing of nickel (Cust, 2023[43]) as well as the reportedly hefty tax incentives to foreign firms investing in downstream activities handed out at the time of ramping up export restrictions (Cust, 2023[43]; Evenett and Fritz, 2023[40]). Some of the critics also argue that, by using export restrictions, Indonesia was 'pushing water down the hill' and that investments and development of the downstream segments would have happened anyway due to the high attractiveness of Indonesian laterite deposits and competitive costs of processing. In other words, they point to an uncertain balance of costs and benefits of these measures for Indonesia and to largely negative impacts on international markets.

Figure 3.6. Indonesia's contribution to the global number of measures in force

Percentage share of the count of Indonesian measures in force in the overall measures in force in all countries



Source: OECD (2025[1]).

This provides an interesting policy context for an assessment of the effects of Indonesia's nickel export restrictions on the country's economic development and international markets, and such studies have been undertaken recently [e.g. Evenett and Fritz (2023[40])]. Assessing economic impacts of export restrictions is not straightforward, especially in a value chain context which is characterised by vertical links between different but related products, but some broad metrics which allow an appraisal of the overall balance of effects for Indonesia and international markets can be considered.

By restricting exports of primary nickel Indonesia risked harming its mining activities and, by extension, the global supply of primary nickel. The risk would materialise if the boost to the downstream processing industry stemming from lower raw material inputs or other support measures was not large enough to generate the kind of investment in new processing activities that would boost domestic demand for primary nickel by enough to compensate for the foregone exports. However, even if this negative impact on Indonesia's mining of nickel materialised, the balance of effects for Indonesia of upstream and downstream effects in the nickel value chain could have still been positive if the value added generated in the downstream industries would have more than compensated for the value lost in the upstream mining industry. In such a case, depending on whether the new processing capacity would add to global capacity, or whether it would crowd out capacity in trading partners, the global effects could also be either positive or negative.[47]

Descriptive statistical analysis of developments in global and Indonesia's production and trade of primary and processed nickel products is therefore the approach in the remainder of this case study. The analysis first outlines the key developments regarding Indonesia's use of export restrictions on different nickel products and compares them with corresponding developments in Indonesia's and global trade of these products.

Evolution of Indonesian export restrictions on nickel products

The evolution of Indonesian export restrictions on nickel products is presented in Figure 3.6 and Annex Figures A D.1 through A D.5. First, these figures show entries into force and exits of some of the less harmful export restricting measures such as the trade finance requirements captured in "other measures'[48] or "export taxes".[49] Second, they also show that some of the strictest measures on exporting nickel – labelled in Annex Figures A D.1 through A D.5 "export prohibitions" entered into force first in 2014, and again in 2020.

Indonesia's export prohibitions, not only on nickel ores and concentrates but also on nickel oxides and hydroxides, nickel mattes, nickel oxide sinters & other intermediate products of nickel metallurgy, as well as unalloyed and unwrought nickel, nickel alloys, and nickel powders and flakes, were introduced in January 2014, and they were left unchanged until 2020 when they were tightened. In 2020, chemical compounds of nickel (i.e. nickel chlorides and sulphates), were also covered by the export prohibition. Introductions of many of Indonesia's licensing requirements applying to exports of nickel products also broadly followed this timeline.

3.2.2. Impact of Indonesian export restrictions on the country's production and exports

Indonesia's production of primary nickel fell temporarily in 2014 and 2015—just after the country introduced a suite of restrictions on exports of nickel products around 2013-14, before expanding somewhat in 2016 and, more significantly, in 2017 and thereafter. This means that, after the initial decrease, the production of primary nickel took two to three years to recover after the introduction of export restrictions (Figure 3.2, Panel C). This correlates closely with the evolution of the value of Indonesia's exports of ores and concentrates of nickel, which were eliminated almost completely in 2015 and 2016, grew progressively in the period 2017-2019 and were eliminated again in the period 2020-2021 (Figure 3.7, Panel A). Similar, but less marked, u-shaped developments can be observed for intermediate products of nickel.

[47] This could be the case, for example, if the effect of lower domestic prices of primary nickel in Indonesia would generate a large enough boost to the downstream industry, for example through economies of scale or through boosting onshore investments in downstream industries. If the latter effects were large enough, they could also have in principle have positive effects on global supply of processed/final nickel products (e.g. stainless steel, or EV batteries).

[48] Some of the earliest export restrictions on primary nickel, captured under the label "other measures", which were introduced in 2009, were requirements for domestically issued letter of credits for exports. These requirements were eliminated in 2010 (Figure 3.8). See OECD Inventory of Export Restrictions on Industrial Raw Materials.

[49] A 20% export tax, along with new export licensing requirements, was introduced in 2012 and eliminated in January 2014 when new export prohibitions entered into force.

In the same period, Indonesia recorded a significant expansion of exports of stainless-steel products[50] - from almost non-existent in 2010 to USD 8 bln in 2021, destined mainly for China (Figure 3.8). However, summing the values of exports of all types of primary nickel and stainless steel (Figure 3.7, Panel B) shows that the introduction of export restrictions in 2014 was followed by a decrease in the overall value of exports of nickel-related products for about four years: this fell progressively in 2014, 2015 and 2016 and started recovering in 2017, before reaching a new high in 2018 and expanding further in 2019. Coinciding with the introduction of new quantitative export restrictions, 2020 saw another temporary reduction in the overall value of exports (and this was clearly due to the reduction in the value of exports of nickel ores).

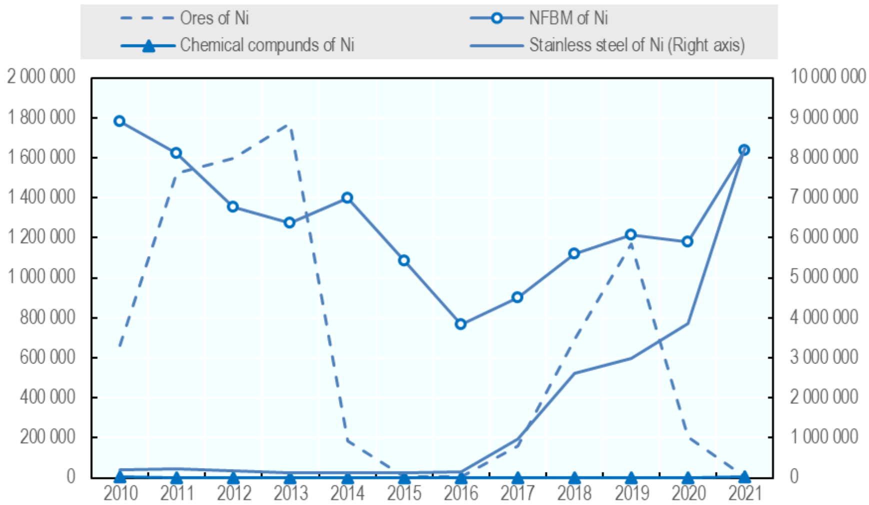
Given the unequal developments in the values of exports of different nickel-related products, which could reflect the time lags necessary for production and exports of downstream products to increase after the introduction of export restrictions on upstream products, it is interesting to ask whether Indonesia may have been overall better or worse off in terms of the overall value of exports of nickel-related products cumulated since the introduction of the restrictions in 2014. Answering this question can be attempted by making assumptions regarding the evolution of exports in a hypothetical scenario in which export restrictions would not have been introduced. For example, adopting a simple "business as usual" assumption of a continued average growth of exports of nickel-related products from 2014 onwards, equivalent to the average performance for the period 2011-13, and comparing it to the actual performance, reveals that from 2014 until 2020, the overall effect of export restrictions on the value of exports was negative and that it became positive only in 2021 (Figure 3.7, Panel C). This suggests that, for several years following the export bans, Indonesia's strategy, while possibly successful in terms of boosting the exports of its stainless-steel industry, may have not been beneficial from a broader, cross-sectoral, perspective.

Moreover, while the positive result in 2021 might suggest that the strategy eventually paid off, this may be misleading, for two reasons. First, it is possible that the exceptional performance of exports of intermediate nickel and stainless-steel products in 2021 was temporary and reflected mainly the large shifts in the product composition and geographical directions of trade observed in 2021 when the global economy and global trade were recovering, albeit unevenly, from the COVID-19 pandemic. Second, and more importantly, it is not clear whether the benefits generated from exports of stainless-steel products actually accrued to Indonesia and its citizens. These considerations are outlined in the following sub-section.

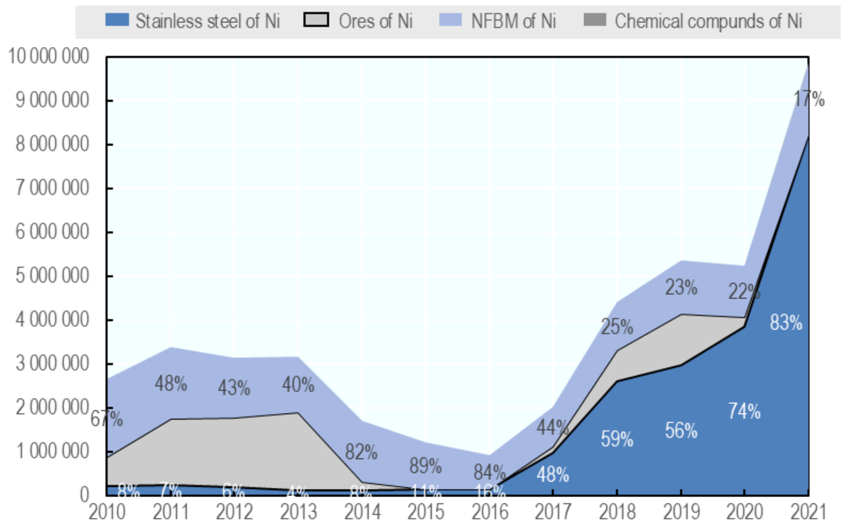
[50] The list of HS4 and HS6 codes included in this analysis of stainless steel is provided in Annex Table A D.1. Note that this list is based on the 2007 version of the HS nomenclature (i.e. HS2007). This nomenclature is used for collecting data included on the OECD Inventory on Export Restrictions. The same nomenclature was used for the analysis of trade developments in this paper. One important limitation of using HS2007 in the context of nickel is that it does not cover trade in ferro-nickel, one of Indonesia's leading nickel-related exports, produced from laterite nickel ores. Ferro-nickel was only added as a distinct product in the 2012 edition of the HS nomenclature (HS 720260). Production capacity has been developed through significant investments and thanks to technological contributions from mainly large metallurgical and mining companies from China—and have been, in part, a response to export policies of nickel-rich countries like Indonesia—ferro-nickel is an alloy of iron and nickel used as a key ingredient in production of stainless steel and it is typically produced by smelting laterite nickel ores in electric arc furnaces.

Figure 3.7. Combined value of Indonesia's exports of nickel ores and key nickel-containing products

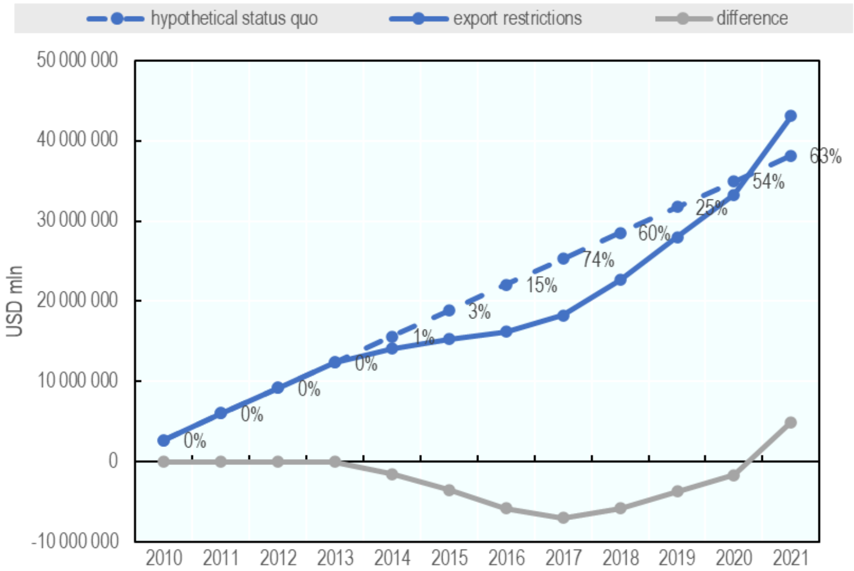
Panel A. Value of exports of individual nickel product types (USD mln)



Panel B. Compound value of exports of individual nickel product types and shares



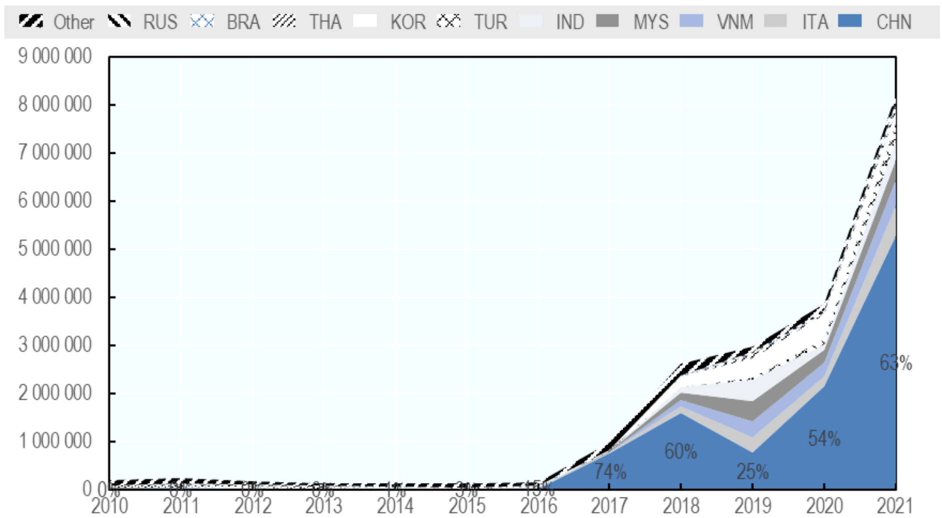
Panel C. Comparison of cumulated values of exports in a hypothetical status-quo and the actual export restrictions scenarios*



Note: *the hypothetical "status-quo" scenario assumes that from 2014 value of exports of Indonesia for each product category would have been equal to the average value of exports in the years 2011, 2012 and 2014. Cumulated values are sums of exports of all categories of products since 2010 until the year in question.
Source: OECD calculations based on BACI.

Figure 3.8 Exports of Indonesian stainless steel by partner

Ten top destinations (USD '000)



Source: OECD calculations based on BACI.

As summarised in Annex Box A D.2 which discusses the ESG effects of Indonesia's export restriction, the expansion of Indonesia's nickel industry, particularly following the 2020 export ban on raw nickel, has been closely linked to Chinese investment, with projects such as the Morowali Industrial Park largely financed and prioritized under China's Belt and Road Initiative. While some economic benefits have accrued to Indonesia, a significant share appears to have gone to Chinese investors and state-linked entities. The rapid growth of Chinese-controlled nickel refining in Indonesia has raised multiple ESG concerns, including deforestation, water contamination, health impacts on local communities, poor labour conditions, and limited local economic integration. Weak enforcement of environmental and social standards, combined with centralised governance and limited oversight, has further exacerbated these challenges, highlighting both the strategic influence of China in the sector and the associated policy and governance risks for Indonesia.

3.3. Nickel: Key findings and tentative conclusions

Perhaps more than any other instance of export restrictions introduced in recent years, Indonesian restrictions on exports of primary nickel have been touted as having positive impacts on expansion of downstream industries and on the country's overall economic development. The descriptive statistical analysis presented in this chapter suggests that the use of export restrictions indeed had a significant impact on the composition and evolution of Indonesian production and exports of nickel-related products but also that their overall impact on the Indonesian economy is still not conclusively positive.

Reductions of exports of primary nickel products have coincided with a significant expansion of exports by Indonesia of stainless-steel products which now dwarf exports of any other nickel-related products. A number of significant downstream projects tied to nickel and started after the export restrictions were implemented. These include plants that process nickel into intermediate products, chemicals, and components for EV batteries. This expansion was, however, not immediate and, for a number of years following the export bans, Indonesia may have been (and may still even be) worse off. In addition, the emergence of the domestic stainless – steel industry – towards which the previously exported nickel ore was re-directed –has been controlled by investors from China to a significant degree. These investors have likely captured portions of value added generated by the growing stainless-steel industry.

Overall, the evidence presented in this chapter suggests that the effects of Indonesian nickel export restrictions are far from straightforward to assess and that their assessment requires a multidisciplinary analysis. A possible positive interpretation is that, while costly for the prior foreign users of Indonesian primary nickel products, the policy was beneficial to Indonesia's economic development. A less positive version is, however, that the massive and to a large extent Chinese state-supported investments in Indonesia's stainless-steel industry, amounted merely to a strategic geopolitical relocation of Chinese-controlled industrial assets, with arguable economic benefits to Indonesia and a host of related ESG concerns.

4. Conclusions and emerging observations

This paper presents the main insights from case studies on selected economic effects of export restrictions on cobalt, lithium and nickel introduced by the Democratic Republic of the Congo, Argentina and Zimbabwe, and Indonesia. While not exhaustive, a descriptive statistical analysis of the evolution of production and international trade of primary and processed forms of these minerals in these countries, combined with insights from the literature on accompanying investments in capacity, ownership and environmental and social aspects of mineral extraction and processing, illustrate some of the key challenges of deepening and diversifying the international markets for CRMs. The exploratory nature of this work and its primary focus on the more direct trade and production effects of export restrictions call for some caution in identifying policy implications, but some broad suggestions can be put forward for consideration by policymakers.

Expanding domestic processing and capturing larger portions of value added in renewable energy and digital technology supply chains were some of the main stated policy objectives of the export bans introduced by the DRC, Zimbabwe and Indonesia. Argentina's tax on exports of lithium was part of a broader tax package intended to raise fiscal revenues from exports across a wider range of products. Bearing in mind the limitations of the simple descriptive statistical analysis employed in this work,[51] in all

[51] The investigation of effects of export restrictions undertaken could be improved, for example, by considering a wider range of performance indicators (e.g. upstream and downstream value added, employment, as well as environmental, social indicators) and employing econometrics to separate out confounding factors and investigate causality.

cases studied, the use of export restrictions appears to have delivered on some of the stated domestic policy objectives but also appears to have had negative effects on international markets through reduced exports.

The bans introduced by the DRC and Indonesia appear to have resulted in a significant re-direction of exports of primary forms of cobalt and nickel from international markets towards domestic processing and were correlated with investments in domestic downstream capacity leading to larger downstream production. However, there were also limitations in terms of the impact of these measures: while export restrictions may have played a role in helping the DRC and Indonesia shift away from exports of primary products, the downstream impacts tended to be concentrated in the immediate next processing stages and involved mainly Chinese investors who already controlled these segments of the value chain. Argentina's export taxes appear to have helped the country raise the tax revenue, but they also seem to have resulted in lower exports.

These findings also provide an illustration of the beggar-thy-neighbour nature of export restrictions: it is important to highlight that the (actual or perceived) positive domestic effects in restriction-using countries are likely to occur at the expense of trading partners. Since no country possesses all the minerals needed for the full range of industrial applications, and all are affected—at least to some extent—by disruptions to the energy and digital transitions, every nation is vulnerable to the negative spillover effects of export restrictions. This suggests the continued relevance of co-operative plurilateral or multilateral solutions to restrain the use of export restrictions. In this context, the relatively limited scope of the current WTO disciplines on export restrictions[52] and the growing use of such measures (OECD, 2025[1]; Kowalski and Legendre, 2023[2]) underscore a need for a better understanding of the motivations of countries using these measures, and of alternative approaches to address their concerns.

While not the central focus of the analysis undertaken for this report, it is striking that, in three cases studied in this report, ownership and control changes associated with the introductions of export restrictions complicate assessment of economic effects in countries imposing those export restrictions. However, they also provide a revealing illustration of some of the mechanisms which have led to extremely high global share of China in CRM extraction and processing. China's shares in investments in downstream processing in both the DRC and Indonesia are also extremely high and, in some cases, may have also benefitted from support by China's strategic state policies. The effects of export bans on primary lithium by Zimbabwe are more recent, but they were also accompanied by announcements of increased Chinese investments in downstream processing.

Prior to the export bans, China already had a strong presence as an owner of mining facilities (DRC) and key importer and processor of raw materials (DRC, Indonesia, and Zimbabwe), and Chinese investors were therefore natural candidates for participation in downstream industry expansions. In addition, amounting to several USD billions in each case, the investments in downstream processing activities were also very large and, due to the high commodity price volatility, also very risky. Views shared by practitioners from the mining sector[53] suggests that typical commercial financial institutions which operate in OECD countries face greater challenges in the financing of such sizable and risky projects than state policy-supported Chinese actors. While evidence gathered at this stage is mostly anecdotal, and requires further investigation, some assessments also suggest that only a part of the value added from the expansion of downstream activities accrued to the host economies. In an already challenging context, some assessments also suggest that some of these projects had mixed ESG results.

The development and strategic dimensions, long time horizons, high risk and extraordinary financing needs, and sustainability dimensions of new CRM projects, appear to imply a strong case for inter-government co-operation, as well as public-private partnerships. Some of these implications are already reflected in various recent policy initiatives aiming to develop diversified and sustainable CRM supply chains. For example, the European Union's raw materials diplomacy partnerships and policy dialogues

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[52] Under WTO rules, quantitative restrictions on exports are generally prohibited while export taxes are not (e.g. Mendez Parra, Schubert and Brutschin (2016[82]) and Pauwelyn (2020[42])), but they must not cause unjustifiable trade distortions.

[53] This issue has been identified as one of the main challenges for Western firms active in the CRM mining and processing industries during recent discussions at the OECD Forum on Critical Supply Chains held in March 2024.

with several resource-rich developing and emerging economies, which accompany the EU's Critical Raw Materials Act and aim to promote promising projects involving foreign partners, explicitly acknowledge both the economic development needs of resource-rich countries and the energy transition aspirations of users as motivations underpinning these strategic co-operations.[54] These partnerships also make a strong reference to investment and high ESG standards including those developed at the OECD.[55] Similarly, the Minerals Security Partnership, a collaboration of 13 OECD Members,[56] India and the European Union, aims to accelerate the development of sustainable CRM supply chains through co-operation between governments and industry, to facilitate targeted financial and diplomatic support for strategic CRM projects.[57] In its Principles for Responsible Critical Mineral Supply Chains, the Partnership also explicitly makes reference to high internationally recognised ESG standards — including in reference to OECD guidelines — and calls for the promotion of local value addition and uplifting of local communities, also recognising that all countries can benefit from the global energy transition.[58] Moreover, in September 2024 members advanced efforts by establishing a Finance Network, aiming to maximize the outcomes of the Partnership's ambitious agenda and accelerate the achievements of the targeted projects.[59] In addition, since April 2024, the MSP contains an additional pillar called MSP Forum which incorporates developing countries and compliments the project work of the MSP with a policy dialogue In September 2024, the membership of the MSP Forum was enlarged by seven new countries, including DRC, thus broadening cooperation and synergies with other major leaders in the CRM sector.[60]

Overall, the growing use of export restrictions on CRMs, their concentration across the different raw materials and countries, and the evidence from the three case studies presented in this report highlight that the effects of some of these measures on international trade and domestic economy may be significant. They also imply that new policy responses may be needed to reduce concentration and promote sustainability of CRM supplies, and that these responses should integrate the interests of resource-rich developing countries.

The case studies presented in this paper also highlight that much remains unknown about export restrictions on CRMs, their success or failure, and that there are promising areas for potential future analytical work on this topic at the OECD. In this context, in addition to the annual update of the OECD Inventory on Export Restrictions on Industrial Raw Materials and its dissemination, the OECD's work continues filling existing evidence gaps by expanding data collection and strengthening analysis of the market impacts of export restrictions by setting them in a wider policy context. Following the approach of this report, future work could continue to focus on selected CRMs that are of particular interest from the perspectives of energy and digital transitions, broader economic security, resilience or geopolitical considerations. These materials could then be studied in greater detail through event studies and more rigorous statistical and econometric analyses. Future work could also include a holistic exploration of the extent to which export restrictions have delivered on stated domestic policy objectives, as opposed to these outcomes being achieved through other means—such as domestic support for mineral processing—by

[54] For the list of partnerships established so far, see Raw materials diplomacy - European Commission (europa.eu).

[55] For example, the strategic Partnerships on Sustainable Raw Materials Value Chains between the European Union and Uzbekistan and Rwanda make a specific reference to OECD Guidelines for Multinational Enterprises.

[56] OECD partners of the Minerals Security Partnership are Australia, Canada, Estonia, Finland, France, Germany, Italy, Japan, Norway, Korea, Sweden, the United Kingdom, and the United States.

[57] See Minerals Security Partnership - United States Department of State.

[58] See MSP-Principles-for-Responsible-Critical-Mineral-Supply-Chains-Accessible.pdf

[59] See Joint Statement on Establishment of the Minerals Security Partnership Finance Network - United States Department of State.

[60] In September 2024, the MSP Forum membership was enlarged to the Democratic Republic of the Congo, the Dominican Republic, Ecuador, the Philippines, Serbia, Türkiye, and Zambia. These countries joined Argentina, Greenland, Kazakhstan, Mexico, Namibia, Peru, Ukraine, and Uzbekistan, who were announced as Forum members at the inaugural high-level MSP Forum event held in July 2024. See EU and US welcome new members to Minerals Security Partnership - European Commission.

disentangling the effects of export restrictions from those of domestic policy support and positive domestic market developments.

First, along the lines of the discussions of the cases of cobalt and lithium in this report, future work could include assessment of the quality of coverage of trade in these minerals in international and national trade statistical nomenclatures and thus the ability to map key features of the related supply chains. Second, as this report demonstrates, export restrictions are often combined with other, technically domestic but practically trade-related, policy measures (production licences, tax incentives) and the effects of the different measures are sometimes confounded. OECD's future work could therefore seek to provide a more systematic, comprehensive and consolidated documentation of different kinds of production, and trade-related policies for the selected CRM value chains, accounting for different stages of processing and different uses of these materials. This work could usefully incorporate information on other relevant trade policy measures which are currently not systematically documented (tariffs and NTMs, government support as well as bilateral, plurilateral and regional partnerships and agreements related to CRMs).

Third, as demonstrated by the approach in this paper, which examines various stages of processing of CRMs, concentration is a feature in the supply of many CRMs, not only for CRMs in their primary forms, but also in more processed forms. Indeed, a number of OECD countries are importers of processed versions of CRMs rather than primary forms, meaning that they are more likely to be affected by indirect or second-round effects of export restrictions taken by source countries, or by measures taken by processing countries. Given the concentration of global processing of CRMs in very few countries, notably China, this warrants further study. Moreover, in a number of cases where source countries have themselves moved into further processing, the more processed CRM are still largely exported to the same economy (China, sometimes reflecting the role of investment by Chinese firms in creating processing capacity). This issue also warrants further study in the context of resilience of global supply of CRMs.

Fourth, beyond the strict domain of trade policy, OECD governments and businesses have been developing a myriad of ESG standards and ways of scrutinising and enforcing them domestically and in foreign activities of multinational enterprises. Some of these take the form of legislation which apply not only to domestic, but also increasingly to foreign, businesses and their supply chain partners. There are also a variety of voluntary standards developed by the industry[61] which are used by mining companies and projects and which that cover a range of ESG issues, including those covered and not covered by existing national/subnational legislation. These standards can go a long way towards ensuring sustainability and positive local development effects of new CRM mining and processing projects. Building on the expertise and experiences of member countries, the OECD, with its multidisciplinary directorate and committee structures, representation the OECD private sector through BIAC, and several existing streams of work on CRMs[62], offers a cross-government perspective, and is well positioned to help OECD Members exchange on, co-ordinate and develop more comprehensive and coherent approaches to engaging resource-rich developing countries and facilitating OECD countries' sustainable access to CRMs in the future.

[61] Many companies are choosing to voluntarily adopt ESG standards for their mining/processing operations because they see their intrinsic value, and more importantly, this is something their customers [e.g. downstream Original Equipment Manufacturers (OEMs)] are looking for. OEMs themselves are responding to more ESG-conscious consumers and investors, which guide their decisions to adopting their own ESG targets and sourcing from suppliers with similar values. As such, compliance to higher ESG standards in mining/processing operations, though it may increase the price of products compared to alternative from countries where these standards are not used, is an important part of their value proposition and offers a competitive advantage to customers specifically seeking products produced in sustainable ways.

[62] OECD's existing workstreams related to CRMs encompass such diverse areas as: facilitation of technological innovation and deployment of existing mining and processing technologies; environmental, social and broader developmental aspects (benefit sharing, indigenous rights) of existing and new technologies; multiple aspects of financing new mining and processing projects; due diligence; and development and sustainability of mining regions.

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Annex A. OECD Inventory of Export Restrictions on Industrial Raw Materials

Initiated by the OECD's Trade Committee in 2010, OECD Inventory of Export Restrictions on Industrial Raw Materials is meant to build a factual inventory of border and domestic measures that restrict the export of industrial raw materials, as part of a larger effort to take stock of such measures in the raw materials sector. From its conception, key aims of the Inventory were to improve the transparency of governments' practises in this area and to serve as a data bank for empirical analysis advancing the understanding of the economic effects of export restrictions (Fliess and Mård, 2012[44]).

Meaningful product and country coverage, and consistency across time, have been principal methodological standards of the Inventory, but the approach to data collection and verification by countries, as well as product and country coverage of the Inventory, has evolved over time in order to improve the quality of the data.

Data collection process

At the beginning of construction of the Inventory, the OECD was first searching online for national official information on the use of export restrictions and subsequently verifying the information through requests for clarifications and further inputs addressed to officials of relevant governments. Quality of information about export restrictions available on government websites varied across countries and, in regard to the verification process, responses to requests for additional information or clarifications were difficult to obtain form some governments. As a result, at the time the quality of data tended to vary across countries and materials covered (Fliess and Mård, 2012[44]).

Since 2018, the database is updated annually or biannually using a structured questionnaire accompanied by detailed guidelines and instructions, for OECD countries, by national contact points identified for this purpose by these countries, and for non-OECD countries, by dedicated external consultants. The information obtained by consultants is later also checked and completed by the OECD.

In the latter case, the information on export restrictions is collected from official websites and documents issued by the governments of the producing countries. These can include Ministries in charge of economy, trade, industry, mining, forestry or foreign affairs as well as customs agencies. Sources of information on policies that restrict exports include legal acts, rules, regulations, public notices, circulars and notifications by ministries published on their websites. Secondary sources, such as news articles, were used to identify export measures applied by a country or changes made to measures during the survey year. However, only measures that can be substantiated from official sources are entered into the database.

Product coverage

From the beginning of the data collection efforts, the exercise covered most commodities in their unprocessed as well as in their semi-processed form, focusing on industrial raw materials that were considered strategic or critical, as in the EU Raw Materials Initiative or listed in the US Strategic and Critical Materials 2013 report, and including also waste and scrap of metals. Included are products in their raw and semi-processed forms from the following HS chapters (2-digit of HS 2007 classification) and their sub-chapters (4-digit HS):

- HS25 (Salt; sulphur; earths and stone; plastering materials, lime and cement) – all HS 6-digit codes
- HS26 (Ores, slag and ash) – all HS 6-digit codes
- HS27 (Mineral fuels, mineral oils and products of their distillation; bituminous substances; mineral waxes): only HS270112 (Bituminous coal, whether/not pulverised but not agglomerated) and HS270400 (Coke & semi-coke of coal/lignite/peat, whether/not agglomerated; retort carbon)
- HS28 (Inorganic chemicals; organic or inorganic compounds of precious metals, of rare-earth metals, of radioactive elements or of isotopes) – all HS 6-digit codes

- HS31 (Fertilisers): only HS310420 (Potassium chloride), HS310430 (Potassium sulphate) and HS310490 (Mineral/chemical fertilisers, potassic (excl. of 3104.20 & 3104.30)
- HS44 (Wood and articles of wood; wood charcoal): all HS6codes in the following sub-chapters: HS4403 (Wood in the rough, whether or not stripped of bark or sapwood, or roughly squared); HS4407 (Wood sawn or chipped lengthwise, sliced or peeled, whether or not planed, sanded or end-jointed, of a thickness exceeding 6mm) and HS4412 (Plywood, veneered panels and similar laminated wood)
- HS71 (Natural or cultured pearls, precious or semi-precious stones, precious metals, metals clad with precious metal, and articles thereof; imitation jewellery; coin) – all HS 6-digit codes
- HS72 (Iron and steel) – all HS 6-digit codes
- HS73 (Articles of iron or steel) – all HS 6-digit codes
- HS74 (Copper and articles thereof) – all HS 6-digit codes
- HS75 (Nickel and articles thereof) – all HS 6-digit codes
- HS76 (Aluminium and articles thereof) – all HS 6-digit codes
- HS78 (Lead and articles thereof) – all HS 6-digit codes
- HS79 (Zinc and articles thereof) – all HS 6-digit codes
- HS80 (Tin and articles thereof) – all HS 6-digit codes
- HS81 (Other base metals; cermets; articles thereof) – all HS 6-digit codes

In 2015, the scope of the Inventory was revised to ensure a systematic overview of export-restricting measures by metals and minerals exporting countries. In 2020, the scope of products included for each country in the Inventory was updated to ensure the product scope reflected current patterns of production and exports. In 2021, the country coverage of the inventory was extended to cover one additional product, bismuth (and also seven new countries).

Currently, the list of covered materials encompasses 65 strategic, industrial commodities, including 58 mineral and metals, six wood products, and metallic waste and scrap for all minerals and metals covered in the Inventory.

Minerals and Metals (57)

Aluminium	Antimony	Arsenic	Barytes	Bentonite
Beryllium	Bismuth	Borates	Cadmium	Chromium
Cobalt	Coke	Coking coal	Copper	Diamonds
Diatomite	Feldspar	Fluorspar	Gallium	Garnet
Germanium	Gold	Natural graphite	Gypsum	Indium
Iron and steel	Kaolin	Lead	Limestone	Lithium
Magnesite	Magnesium	Manganese	Mercury	Molybdenum
Nickel	Niobium	Perlite	Phosphates	Pig iron
Platinum group metals (PGMs)[1]	Potash	Rare earths (REE)	Rhenium	Selenium
Silica	Silicon	Silver	Strontium	Talc
Tantalum	Tellurium	Tin	Titanium	Tungsten
Vanadium	Zinc	Zirconium		

1. Platinum group metals (PGM) that include platinum, palladium, and all other PMG metals (rhodium, iridium, osmium and ruthenium).

Wood (6)

Industrial roundwood coniferous	Sawnwood coniferous
Industrial roundwood non-coniferous non-tropical	Sawnwood non-coniferous non-tropical
Industrial roundwood non-coniferous tropical	Sawnwood non-coniferous tropical

Other

Metal waste and scrap for all metals and minerals included in the Inventory

Export restrictions are recorded in the database at the 6-digit level using the Harmonised System (HS) 2007 nomenclature.[63] If a measure is applied at the HS8 or HS10 digit level and the information is available in the data source, this information is recorded in the respective field in the Inventory.

See Annex B for the list of HS 2007 codes included in the Inventory along with the associated product name.

Country coverage

In the first versions of the database, countries were included in the Inventory if they either produced at least 3% of 2012 global production (in volume terms) or were among the top five producers of any of the raw materials covered in the Inventory. The country scope was reviewed in 2021 to include seven additional countries which fit the 3% global production criterion in 2018. This determined the scope of countries surveyed. For each of the countries in the scope of the Inventory, other commodities on the raw materials list where these countries produced at least 1% of global production were surveyed for the existence of export-restricting measures.

The information for countries and products not covered in the earlier versions of the Inventory has been supplemented accordingly for the earlier periods so that the product, country and time coverage in the current database is consistent.

In this way the country coverage, while not full, allows monitoring of export restrictions of all significant raw materials producers around the world. Eighty OECD and non-OECD countries producing industrial raw material commodities are currently being surveyed. These countries accounted for 97% of the world production of minerals and metals and 82% of world production of wood in 2020.

Types of measures covered

The Inventory records measures known or suspected to restrain export activity. These measures typically increase the relative price of exported products, decrease the quantity supplied or change the terms of competition among suppliers. The list of surveyed measures is relatively exhaustive, including export taxes, prohibitions and non-automatic licensing requirements, and any other export restricting measure. Definitions of the measures covered are provided below in Table A A.1.

Export controls that are in place to comply with international conventions and agreements limiting trade in certain goods such as the Kimberley Process for diamonds, the Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES) for some wood products, the Rotterdam Convention as regards some chemical products and the Basel Convention regarding metallic waste and scrap are not included in the Inventory of export restrictions on industrial raw materials. Note that the export controls in place in the context of these agreements and conventions are generally expressed at a more detailed level than HS 6-digit products, and most national regulations indicate that such restrictions are in place in order to comply with these international agreements.

[63] The Inventory uses the HS 2007 nomenclature. If a country has moved to using the HS 2012 nomenclature, the 6-digit level code is converted back to the HS2007 nomenclature. Among the products in the scope of the Inventory, only two were revised between the two HS classification versions: borates and mercury.

Dual-use items are goods, software and technology that can be used for both civilian and military applications. Many countries control exports of dual-use items which are included in international treaties, conventions and resolutions such as UN Security Council Resolution 1540, the Nuclear Non-Proliferation Treaty, the Chemical Weapons Convention, the Wassenaar Arrangement, the Nuclear Suppliers Group and the Missile Technology Control Regime. Export licenses and bans on goods that can be identified within the scope of these regimes are not included in the OECD Inventory of export restrictions. Similarly, the Minamata Convention regulates trade in mercury so those export controls should not be recorded.

Export controls that are implemented in the context of bilateral sanctions, in particular in the context of United Nations monitored trade with select countries, are not included in the Inventory of export restrictions on industrial raw materials.

Table A A.1. Measures restricting exports and their definitions

Export restriction	Definition
Export tax	A tax collected on goods or commodities at the time they leave a customs territory. This tax can be set either on a *per unit* basis or an *ad valorem* (percentage of value) basis. Other terms equivalent to export tax: *export tariff, export duty, export levy, export charge*.
Fiscal tax on exports	A tax not paid at the border, but which applies only or discriminates against goods or commodities intended for export. An example is when the *sales tax* which a government charges is higher for goods or commodities intended for export than when these goods or commodities are offered for sale in the domestic market. Another term equivalent to fiscal tax on exports: *export royalty*.
Export surtax	A tax collected on goods or commodities at the time they leave a customs territory, and which is applied in addition to the normal export tax rate. They can be part of a progressive tax system or can be adapted to price trends and thus be of a temporary nature. Example: a USD 10 surcharge is applied on each tonne of a commodity exported when the world price of this commodity exceeds USD 1 800 a ton. Other terms equivalent to export surtax: *export surcharge.*
Export quota	A prescribed maximum volume of permitted exports.
Export prohibition	No exports are permitted. Exceptions may be granted through export licences. Other terms equivalent to export prohibition: *export ban, export embargo*.
Non-automatic export licensing requirement (M6)	Exporters must obtain prior approval, in form of a license, to export a good or commodity. This practice requires submission of an application or other documentation as a condition for being authorised to export. Although export licensing regimes may vary in their impact on exports, with some regimes that have a fairly small economic impact, at the very least, such regimes increase the amount of time needed to engage in trade. Licensing schemes can operate on the basis of product lists of various types, usually lists of restricted products that require licences be applied to restrict exports by destination (e.g. specific countries) or have other conditions attached, such as a requirement that exportation may only be for a specified purpose. Other terms equivalent to non-automatic licensing: *export permit*.
Minimum export price / price reference for exports	A minimum allowable price for a good being exported. This practice is often used in conjunction with export taxes in order to prevent under-invoicing and can be used as a base to calculate export taxes. In some cases, minimum export prices are not binding but are used as reference prices. Other terms equivalent to minimum export price: *administered pricing*.
VAT tax rebate reduction / withdrawal	Most countries with a VAT system will rebate the VAT on exports. By denying VAT reimbursement in whole or part, it is relatively less advantageous to export a product than to sell it domestically. This measure is usually used to encourage downstream production of products produced locally that use the raw material input. A variant is the removal or reduction of rebate from *other sales taxes* on exports of a product.
Restriction on customs clearance point for exports	The government specifies ports or customs offices through which export of a good or commodity is to be channeled.
Qualified exporters list	The right to export a certain commodity are allocated to specific companies by the government, through a process of application and registration.
Domestic market obligation	The requirement for producers to allocate a proportion of their annual production output for sale to the domestic market. Domestic market obligations are sometimes part of production sharing contracts or contracts allowing extraction by foreign firms.
Captive mining	When a processing company is required to own the mine, which produces its inputs or has been awarded captive mining rights with the intent that the company will mine the commodity for use in its own domestic processes and not trade it. Captive mining is a form of government support for firms with access to captive supplies, as well as a means to control the price and availability of a commodity. When captive mining concessions increase (as a share of production), exports are likely to fall.
Other export measures	Measures not elsewhere specified, but which influence *de jure* or *de facto* the level or direction of exports of industrial raw materials.

Source: Adapted from OECD, Analysis of non-tariff measures: the case of export restrictions (TAD/TC/WP(2003)7/FINAL, 4 April 2003, p.8; Joanna Bonarriva et al, Export controls: An overview of their use, economic effects, and treatment in the global trading system, *Office of Industries Working Paper* No. ID-23, US International Trade Commission, August 2009, p. 2; Jeonghoi Kim, Recent trends in export restrictions on raw materials, *OECD Trade Policy Working Papers*, No. 101, OECD Publishing, 2010, p 6 and 12.

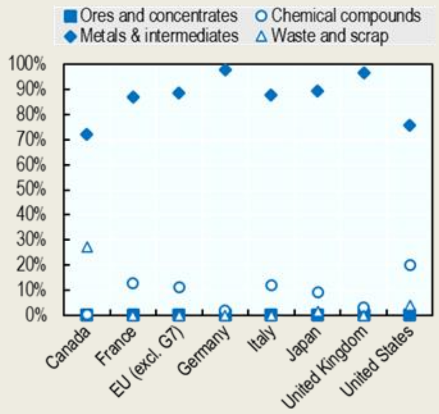
Annex B. Cobalt: Supporting material

Box A B.1. Participation of G7 countries in cobalt value chains

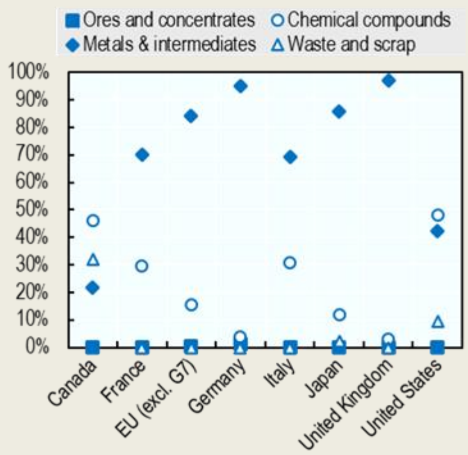
The granular classification of cobalt presented in Table 1.1 in the main text allows to more accurately identify how G7 countries participate in international trade of cobalt, and hence how they may be exposed to export restrictions. G7 Members are, in large part, importers of intermediate and metal forms of cobalt (e.g. unwrought cobalt, powders, mattes), which comprised between 70% and 98% of their total value of imports of cobalt, with some variation across G7 countries, in 2022 (Box figure below). This is followed by imports of chemical compounds of cobalt – the visibility of which varies across countries (Table 1.1) – and cobalt waste and scrap, which comprised, for instance, 27% of the value of imported cobalt goods in Canada and 4% in the United States. Imports of cobalt ores and concentrates (i.e. the primary forms of cobalt) only comprised about 0.04% of the value and 0.2% of the volume of imports of cobalt in G7 countries on average.

Figure A B.1. G7 countries participate in cobalt trade mainly as importers of intermediate and metal items and chemical compounds

a. Share of different forms of cobalt in the total value of imported cobalt goods (in USD), 2022



b. Share of different forms of cobalt in the total volume of imported cobalt goods (in kg), 2022

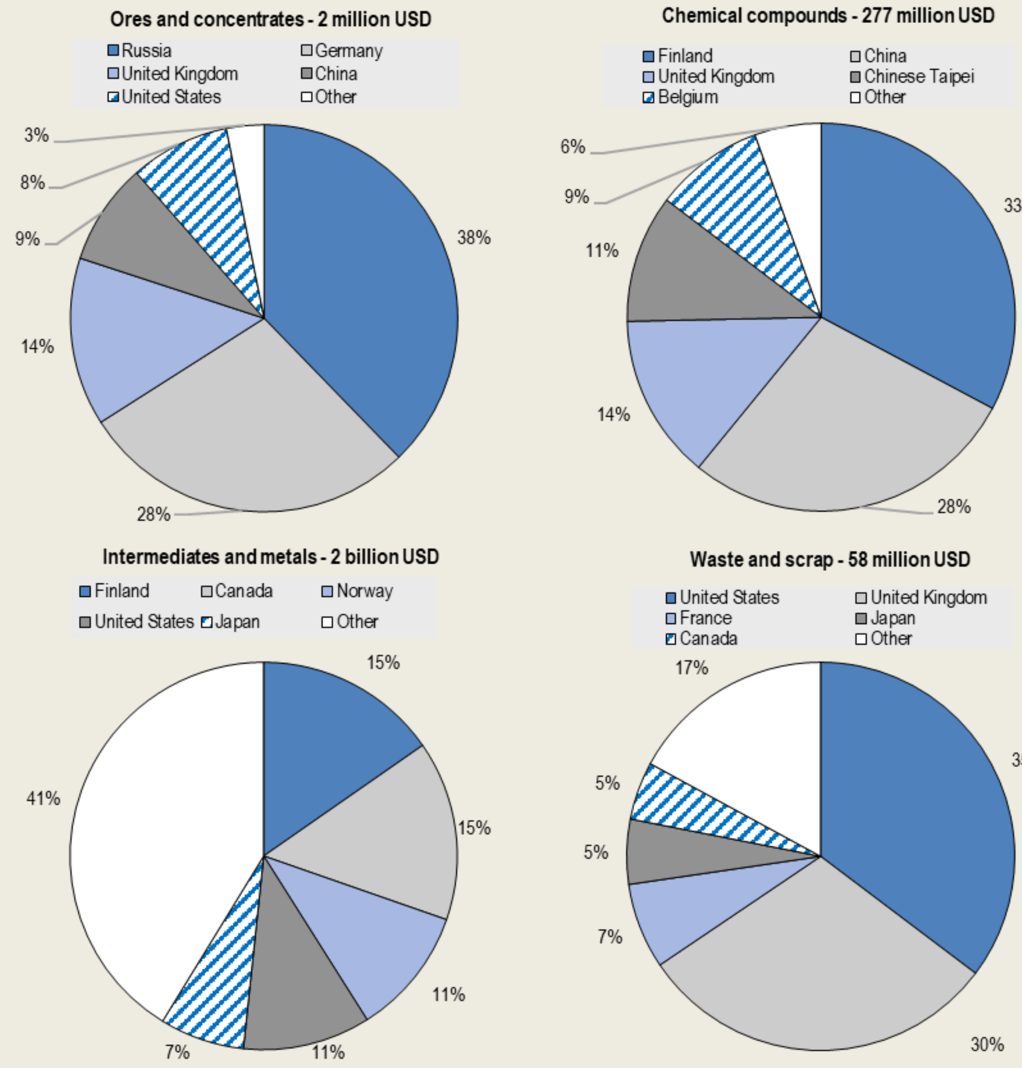


Note: in this Figure cobalt products are classified according to the national or EU customs nomenclatures as summarised in Table 1.1.
Source: Statistics Canada, Statistics of Japan, EUROSTAT COMEXT, UK Office for National Statistics, USITC. Import shares sum to 100%. Trade values transformed in USD using IMF representative exchange rates. The precise product coverage of imports depends on national customs classifications (Table 1).

The geographical origin of these imports also sheds light on the cobalt value chain. While important producers of cobalt ore, such as Russia are main exporters of cobalt in primary forms to G7 economies, the majority of cobalt imports comes from countries that are not major producers of the mineral themselves (e.g. Finland, Canada, Norway and China) – Box figures below. This stands in stark contrast with the geography of extraction and production (Section 1.1 in the main text) – highlighting that G7 countries are much more likely to suffer the impact of exports restrictions through second-round and indirect effects or by measures taken in processing countries, rather than directly through measures affecting bilateral trade with cobalt-producing economies.

Figure A B.2. G7 direct imports of cobalt goods concentrate in countries that are not major producers of cobalt ores

Imports of cobalt products. Chart titles indicate the processing stage of imported cobalt and its total value in imports of G7 countries in 2022



Note: Figures exclude intra-EU trade and include imports of non-G7 EU Members.
Source: Statistics Canada, Statistics of Japan, EUROSTAT COMEXT, UK Office for National Statistics, USITC. Trade values transformed in USD using IMF representative exchange rates.

Box A B.2. The export prohibition regime for cobalt ores and concentrates in the DRC

The DRC export prohibition regime for cobalt ores and concentrates is complex. Gulley (2022[9]) traces the adoption of the first export prohibition on cobalt ores and concentrates to 2006, while the same measure also appears to have been included in an Inter-ministerial decree of 2013.[1] However, it is difficult to understand how widely these prohibitions have actually been implemented.

One reported reason for non-application of the measure is the problem of electricity shortages, which create significant barriers to the energy-intensive cobalt refining process, and have provided grounds for the extension of waivers to mining companies.[2] This explains why, the latest version of the prohibition captured in the OECD Inventory – i.e. Article 7 of the Arrêté interministériel n°0129/CAB.MIN/MINES/01/2017 – includes a moratorium on the application of the measure for energy-related reasons.[3] Insufficient smelting capacity also provided grounds for the provision of waivers to the ban for mining companies.[4]

The period 2017-2020 is used in this report – in line with the OECD Inventory – as a period likely to have witnessed further tightening of ore and concentrate exports, due respectively to the adoption of the latest ministerial decree in 2017 and the extension of another moratorium, in 2020, on some mining operations for reasons including the energy deficit.[5] However, it is important to keep in mind that there are challenges to assigning a specific date of adoption and enforcement for this measure, including because of the reportedly important scope that waivers and moratoria have had for its application.

Notes
1. Arrêté interministériel No.122/CAB.MIN/MINES/01/2013 et No.782/CAB.MIN/FINANCES/2013
2. See "Congo delays ban on copper and cobalt exports due to power shortages", by Mining Technology.
3. Article 7 - *Les exportations des concentrés de cuivre et de cobalt sont interdites. Toutefois, un moratoire allant jusqu'à la résolution définitive du déficit énergétique, est accordé à tous les opérateurs miniers qui produisent des concentrés de cuivre et de cobalt. Les teneurs de ces concentrés doivent être conformes à celles reprises dans le tableau en annexe.*
4. See "Congo grants new export ban waivers for copper, cobalt, tin", by Reuters.
5. See Moratoire sur exportation des concentrés. Industry sources (Mining.com) indicate that concentrates faced tightened conditions for exports again in 2021.

Table A B.1. Changes to the regulation of cobalt export restrictions in the DRC over 2009-2022

Export restrictions by cobalt form

Stage of processing	HS6	Type	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022
Ores and concentrates	260500	Export prohibition	.	.	.	.	.	.	.	.	1	1	1	1	.	.
Ores and concentrates	260500	Export surtax	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Ores and concentrates	260500	Export tax	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Ores and concentrates	260500	Fiscal tax on exports	1	1	1	1	1	1	1	1	1	1	c	1	1	1
Ores and concentrates	260500	Licensing requirement	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Ores and concentrates	260500	Other export measures	1	1	1	1	1	1	1	1	1	c	1	1	1	1
Mattes, unwrought cobalt, powders	810520	Export surtax	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Mattes, unwrought cobalt, powders	810520	Export tax	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Mattes, unwrought cobalt, powders	810520	Fiscal tax on exports	1	1	1	1	1	1	1	1	1	1	c	1	1	1
Mattes, unwrought cobalt, powders	810520	Licensing requirement	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Mattes, unwrought cobalt, powders	810520	Other export measures	1	1	1	1	1	1	1	1	1	c	1	1	1	1
Other intermediate cobalt goods	810590	Export surtax	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Other intermediate cobalt goods	810590	Export tax	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Other intermediate cobalt goods	810590	Fiscal tax on exports	1	1	1	1	1	1	1	1	1	1	c	1	1	1
Other intermediate cobalt goods	810590	Licensing requirement	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Other intermediate cobalt goods	810590	Other export measures	1	1	1	1	1	1	1	1	1	c	1	1	1	1
Chemical compounds	282200	Export surtax	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Chemical compounds	282200	Export tax	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Chemical compounds	282200	Fiscal tax on exports	1	1	1	1	1	1	1	1	1	1	c	1	1	1
Chemical compounds	282200	Licensing requirement	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Chemical compounds	282200	Other export measures	1	1	1	1	1	1	1	1	1	c	1	1	1	1
Waste and scrap	810530	Export surtax	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Waste and scrap	810530	Export tax	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Waste and scrap	810530	Fiscal tax on exports	1	1	1	1	1	1	1	1	1	1	c	1	1	1
Waste and scrap	810530	Licensing requirement	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Waste and scrap	810530	Other export measures	1	1	1	1	1	1	1	1	1	c	1	1	1	1

Note: c indicates changes in the scope of the regulation (e.g. increases in export earnings to be repatriated in the DRC from 40% to 60% in 2018, registered under "other export measures").
Source: OECD (2025[1]).

Box A B.3. ESG impacts of the cobalt industry in the DRC

The DRC's "cobalt boom" of the past decade, and the growing geopolitical contest over CRM supplies, has brought with it a unique set of challenges that range from corruption and governance to environmental impacts, the exploitation of labour, and the funding of conflict.

Institutional fragility and corruption are often referred to as a barrier to the attraction of foreign investment in the DRC (Bridle et al., 2021[45]; Church and Crawford, 2018[46]). From a Responsible Business Conduct (RBC) perspective, significant gaps and challenges have also been recognised in due diligence and risk mitigation of adverse impacts by companies sourcing cobalt (as well as copper and other minerals) from the DRC (OECD, 2019[47]). This is due to factors related to both companies and the weak governance environment, such as the lack of legal recognition for artisanal mining, weaknesses in due diligence related to child and forced labour, corruption, tax evasion and a lack of environmental safeguards.

Armed conflict and conflict financing have also become a major concern for the DRC's extractive industry. Since 2022, renewed violence with non-state armed groups, in eastern DRC has heightened scrutiny on whether minerals, including cobalt, are being used to directly fund and perpetuate conflict (Le Monde, 2024[48]). The DRC's recognition as a Conflict Affected High-Risk Area (CAHRA) also bears implications for trade in critical minerals, as rigorous due diligence procedures, such as those defined in the OECD Due Diligence Guidance for Responsible Supply Chains of Minerals from Conflict-Affected and High-Risk Areas (OECD, 2016[7]), are required to minimize the risk of including "conflict minerals" in supply chains. Even with these safeguards, the race for minerals poses a threat to peace and stability in the DRC (Rupert, 2024[49]). For companies, failure to adhere to due diligence standards risks infringing on international law as well as reputational damage. A recent case highlights this point, where in 2024 the DRC Government publicly accused Apple of using "illegally exploited" minerals from conflict zones (Le Monde, 2024[48]). Lastly, it is important to emphasize that requirements for adherence to OECD Due Diligence Guidance for Responsible Business Conduct should not discourage engagement with countries like the DRC, but rather improve conditions and foster responsible development.

In terms of the environmental cost, while the DRC has sought to create extra value addition domestically through a push to curtail the export of unprocessed cobalt ores, and a shift in its exports to intermediate products, the growth of a domestic refining industry has also added to the already significant impact which primary extraction poses for local communities and ecologies such as deforestation and water contamination.

The development of further processing and refining capacity underlines the need for capacity building and assistance to promote environmental regulation and more effective governance of companies involved in processing activities to enable strategies to mitigate pollution from tailings and run-off. The processes involved in refining cobalt-bearing ores into even crude cobalt hydroxide, the DRC's largest exported form of cobalt, can require the input of multiple highly toxic and environmentally harmful chemicals. Processes such as froth flotation, depending on the method employed, and later the acid-leaching of cobalt ores can involve such chemicals as xanthates, and sulphuric acid (Shengo et al., 2019[50]) which have been shown to exacerbate Acid Mine Drainage (AMD) (Yuan et al., 2024[51]). Water contamination and acidification from AMD and other steps in the cobalt extraction process, have already had impacts on health, crop fertility and biodiversity and pose long-term risks to local communities (Balasha and Peša, 2023[52]; RAID UK, 2024[53]).

A key method of addressing these risks posed by the mining industry in the DRC, and more broadly, has been the development of mining governance initiatives tasked with holding firms to standards of accountability and transparency. Chinese companies have emerged as the dominant force in the DRC's cobalt industry, and additional scrutiny has been applied to China's approach to supply chain due diligence and transparency. Disclosure and transparency policies remain voluntary and unstandardised with regard to ESG impacts, making it challenging to identify ethical breaches and to establish a chain of accountability. A state-driven response to the lack of a unified and legally binding standard has been announced with the Chinese Government with the aim of establishing an ESG reporting procedure for firms by 2030 (Xue, 2024[54]).

On an individual firm level, some Chinese organisations have taken steps to establish norms of transparency and accountability, an example being the Responsible Critical Minerals Initiative (RMI), which was established in consultation with OECD RBC Guidelines (Park, 2023[55]). Many Chinese firms also individually participate in governance regimes. This is part of a broader push on the part of China and Chinese firms, to both integrate into governance initiatives developed by OECD countries, and to take an active role in governance development. Park (2023) maps China's growing engagement with transnational extractive governance but points out the preference for "thin" transparency initiatives, as compared to OECD counterparts.[1]

Note

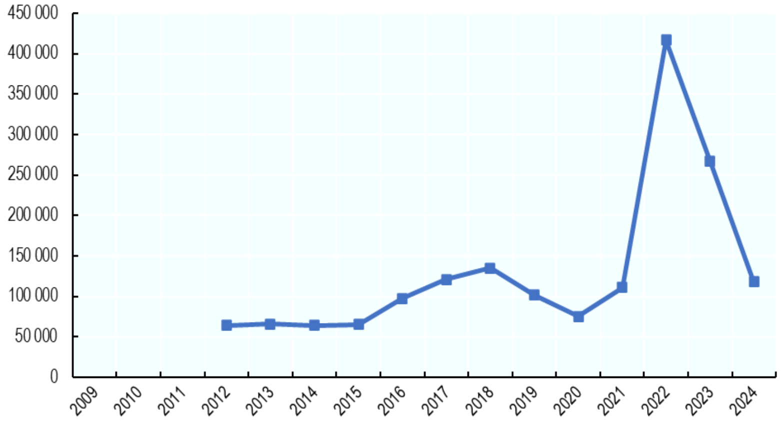
1. Thin transparency initiatives refer to those generally lacking systems of verification and with limited public access, or "market-focused accountability over democratic accountability in global civil society" (Park, 2023[56]).

Annex C. Lithium: Supporting material

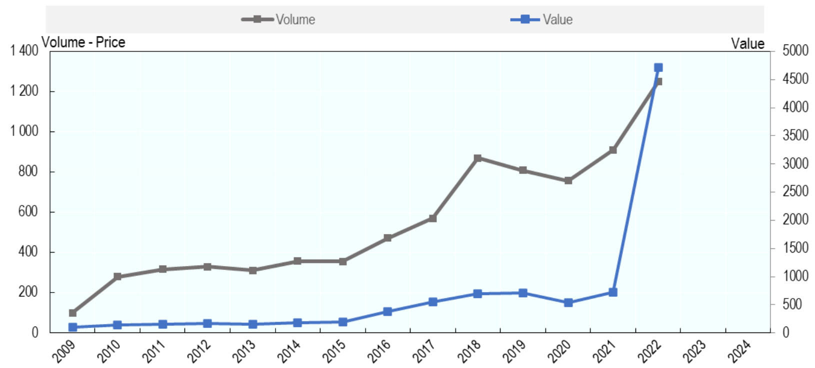
While global trade of lithium ores is difficult to track due the fact that they are not separately identifiable in the HS classification (see Section 3.2 of the main paper), the evolution of international lithium prices quoted on the London Metal Exchange shows some correspondence to the evolution of global trade in lithium chemical compounds. Driven by the growing demand in the renewable energy storage sector and the rapid expansion of EV market, boosted by the substantial investments in lithium mining and production capacity in China (estimated to account for 70% of global lithium-ion batteries production (Sanin, 2022[57])), the price of lithium and quantity of internationally-traded lithium compounds have been increasing since 2015, and both reached their first peak in 2018 before declining in 2019 and 2020. The price decline in 2019, which was mirrored by a drop in volumes of exported lithium compounds, was reportedly due to oversupply and a market correction. The continuation of this downward trend in 2020 was further attributed to an economic slowdown associated with the outbreak of the COVID-19 pandemic (Carbon Credits, 2024[58]).

Figure A C.1. The increase in lithium prices in 2021-22 followed a tight supply of the product due to restrictive export measures

a. Historical prices - battery grade lithium prices (USD/tonne)



b. Index of global trade value and volume (2012=100)



Note: The combined figure presents the price of lithium metal (Li) - 99% pure, battery grade, along with the volume and the value of global exports of lithium chemical compounds. Due to limitation trade data ends in 2022. Index for the year 2012 equals 100.
Source: London Metal Exchange and UN COMTRADE.

In contrast, 2021, and particularly 2022, were marked by a dramatic increase in lithium prices which were much larger than the contemporaneous increases in traded volumes. This price spike has been attributed to supply shortages due to lithium supply agreements, high concentrations of key sections of the supply chain in Chile, Australia, Argentina, China, and policy interventions adopted by governments (Yao, 2022[59]). In the first half of 2023, lithium prices, then saw a considerable decline due to slowing growth of EV sales in China (Dempsey and Fildes, 2024[60]). The falls continued through 2024 and lithium prices stabilised at multi-year lows in the first half of 2025. Considering the small share of global lithium trade covered by export restrictions in place, the price fluctuations observed in the period 2009-22 appear unrelated to the use of these measures. However, they do show sensitivity to demand and supply changes and suggest potentially significant impacts should more lithium-producing countries resort to trade restrictive measures.

Box A C.1. G7 participation in lithium value chains

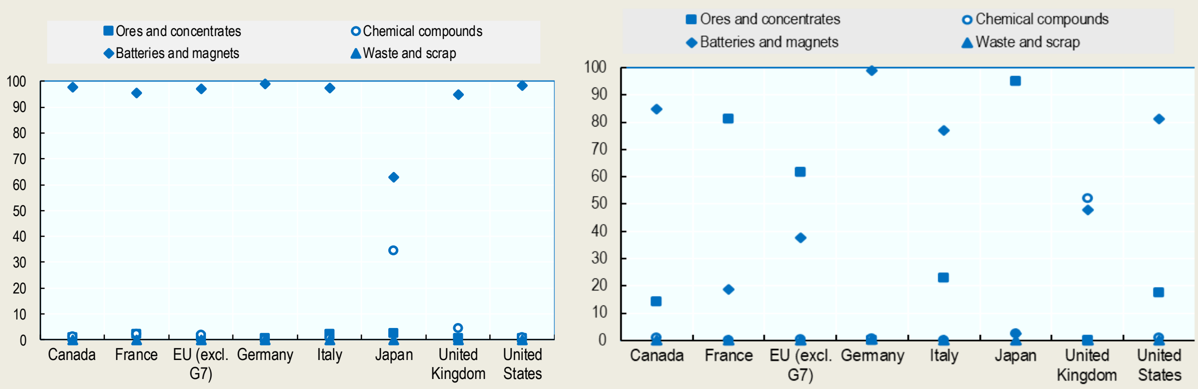
The listed classification of lithium products provided in Table 2.1 enables an approximate breakdown of how G7 countries engaged in international trade of lithium and the extent to which they may have been affected by export restrictions imposed by other nations. At a glance, it is evident that G7 members mainly imported more processed forms of lithium products, particularly batteries and magnets (Box Figure, Panel A). These imports represented between 60% and 99% of the total value of the relevant products. However, variations existed across countries with Japan ranking lowest in terms of imports of these processed products, with a share of almost 63%, and Germany leading with nearly 99% of imports by value.

That said, G7 countries also imported unprocessed mineral products, including lithium – indicated as ores and concentrates – with these imports often ranking a close second to more processed forms. (Box Figure, Panel B). Lastly, the United Kingdom's imports of chemical compounds accounted for 34.5% of the total value of all lithium-related imports, being the largest importer of this type of lithium product, while imports of waste and scrap recorded very small import values and quantities.

Figure A C.2. G7 countries participate in lithium trade mainly as importers of ores and concentrates and manufactured items

a. Share of different forms of lithium in the total value of imported lithium goods (in USD), 2022

b. Share of different forms of lithium in the total volume of imported lithium goods (in kg), 2022

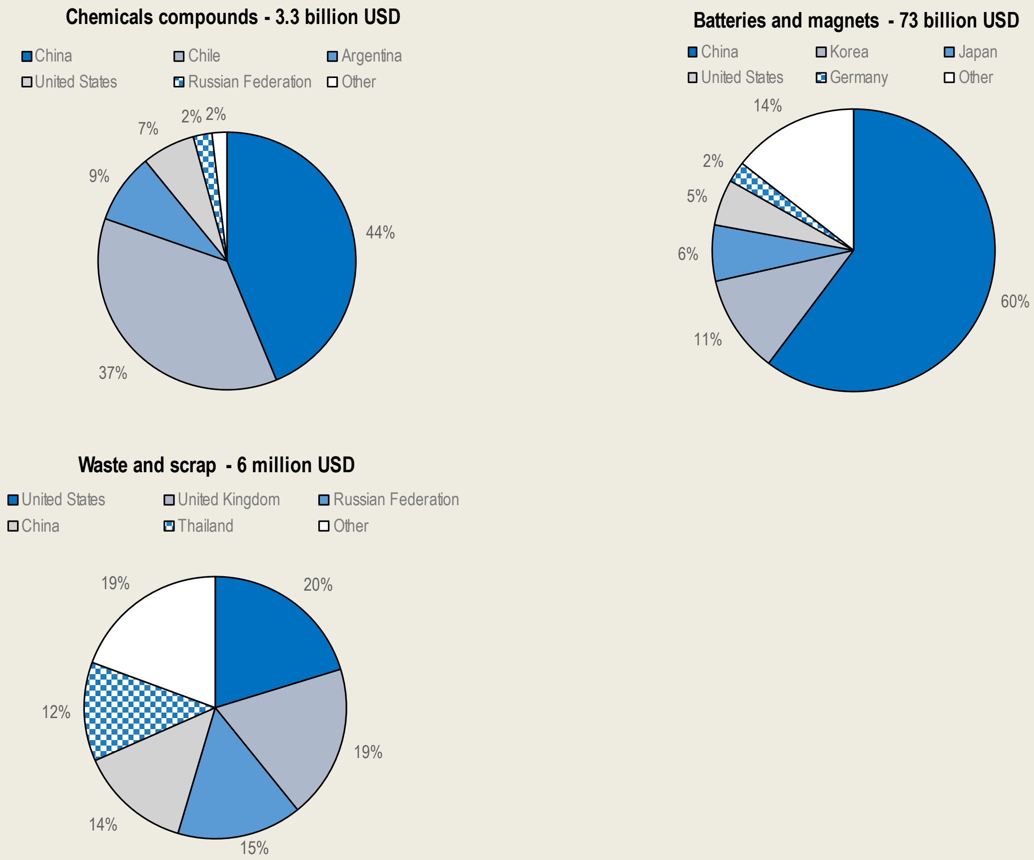


Note: HS-6-digit codes for ores and concentrates (OM) are: 253090; chemical compounds (CC): 283691, 282520, 290433; batteries and magnets (BM), 850650, 850760; and waste and scrap (WS): 854913, 854914.
Source: UN Comtrade.

The geographical distribution of the source of these imports highlights key trends in the global lithium value chain (Box figures below). China emerges as the dominant source of both raw and more processed lithium products, such as chemical compounds, batteries and magnets, with a significant share of G7 imports at 44% and 60%, respectively. That said, an important part of the imports of lithium waste and scrap was sourced from other G7 countries. Despite their significant lithium reserves, the share of lithium chemical compounds imported from Chile and Argentina was lower than that of China, underscoring the country's dominant position in refining capacity and manufacturing and highlighting the potentially negative impact of any export restrictions adopted by China on G7 economies. At the battery and magnets manufacturing stage, Korea has experienced fast growth and emerged as the second-largest supplier of batteries and magnets to G7 countries. In 2022, G7 imports of waste and scrap of batteries were still very small and sourced from a range of countries, none of which were big producers or miners of lithium products.

Figure A C.3. G7 direct imports of lithium goods are concentrated in countries that are not producers of the lithium mineral

Imports of lithium products. Chart titles indicate the processing stage of imported lithium and its total value in imports of G7 countries in 2022.



Note: Figures exclude intra-EU trade and include imports of non-G7 EU Members.
Source: UN Comtrade.

Box A C.2. On potential ESG effects of Argentina's export taxes

As with the export bans on cobalt in the Democratic Republic of the Congo (Chapter 1), lithium in Zimbabwe (discussed later in Chapter 2), and nickel in Indonesia (Chapter 3), Argentina's export taxes can also, in principle, indirectly influence environmental and social ESG outcomes—such as water and ecosystem degradation, impacts on local communities, and governance.

Similar to the other cases of export restrictions examined in this report, increased export taxes on primary lithium products may also in principle create incentives for investment in downstream processing facilities, which could also generate additional ESG impacts. Conversely, low tax rates can encourage the rapid expansion of lithium extraction, as companies face fewer financial constraints—potentially leading to increased water consumption in fragile high-altitude salt flats. Moreover, limited tax revenues may reduce the financial capacity of provincial and federal authorities to enforce environmental regulations, invest in sustainable water management, or compensate affected local communities. As a result, weak fiscal leverage through export taxes may contribute to a governance environment where social and ecological costs are insufficiently mitigated.

However, in contrast to the other export restriction cases discussed in this report, Argentina's export taxes remained at a moderate 5% for most of the period under review, although they fluctuated significantly—dropping to 0% for three years and rising to 12% and 8% in two others. As documented earlier in this chapter, these tax fluctuations appear to have had both positive and negative effects on the value and volume of lithium exports, as well as on fiscal revenues. These developments suggest that the relationship between Argentina's export taxes and ESG outcomes during the period investigated is, at best, weak and largely indirect. Furthermore, Argentina is a relatively smaller player in global lithium markets, with more limited potential to attract significant downstream investment through export restrictions, although some such investments have reportedly occurred in recent years.[1]

These factors may help explain why, to the best of our knowledge, existing research does not establish a direct link between Argentina's export taxes and ESG performance. Instead, most analyses tend to examine broader policy regimes that intersect with ESG challenges in Argentina's lithium sector, typically emphasising low taxation and weak regulatory enforcement. For example, while export taxes on lithium may have contributed to increased government revenues, several recent studies and reports highlight that weak environmental regulations and limited royalty frameworks often result in minimal benefits for local communities—who may nonetheless bear significant social and environmental costs [e.g. Johnson et al. (2024[61]) and Pelletieri (2024[62])]. This is not to suggest that Argentina's export tax policies are irrelevant to the country's ESG outcomes, but rather that the relationship between the specific measures examined in this chapter and ESG impacts appears to be less direct and less robust than in the other cases investigated.

Note:

[1] For instance, Chinese companies CATL and BYD were reported to have secured agreements to purchase 85 000 tonnes of lithium hydroxide from Argentine producers in 2025, accompanied by USD 740 million in investments for local conversion facilities. See, for example, Argentina official says lithium, copper to drive metal exports to $10 billion by 2027 | Reuters.

Box A C.3. ESG impacts of Zimbabwe’s lithium mining industry

As in other developing countries and regions which experienced mining booms, there are concerns that the pursuit of profits associated with mining and processing of lithium will come at a cost of environment, social and governance standards. (Chitagu, 2024[63]) reports for example that the on-going expansion of the sector in Zimbabwe faces insufficient access to basic resources, such as electricity, and is associated with human rights abuse and neglect the working and living conditions of the population. In Bikita District in Masvingo Province for instance, the Chinese company “Sinomine Resource Group” was instructed to halt operations until it complies with legal requirements, including those related to labor, environmental and immigration laws (Mkodzongi, 2024[64]). Thousands of artisanal miners are reported to operate under hazardous conditions, with allegations of use of child labour and fatalities due to mine collapses (Chitagu, 2024[63]). There are also reports that populations have been forcibly displaced from their own land and minerals deposits there have been confiscated (Matanzima, 2024[65]).

The introduction of Zimbabwe’s ban on lithium exports in 2022 is too recent to allow an assessment of its impact the country’s environmental, social and governance performance, although the commentary that has been emerging following the policy change signals concerns about impacts on local communities – in particular artisanal miners – environment and corruption.

For example, there are concerns that local artisanal miners, which extract lithium and – not having the capacity to process it – usually sell it directly for exports, may be excluded when exports are banned and domestic production is channeled towards local processing (Global Witness, 2023[66]). However, it must be also noted that the output of artisanal mining often feeds into illicit exports (mainly to China (Matiashe, 2024[67])). In addition, concerns about the future of artisanal mining in Zimbabwe have to be put into the larger context of developmental objectives of the country which can arguably be supported by an expansion of a competitive and formal mining and processing industry capable of making the necessary large capital and infrastructural investments (Sellwood, Hodgkins and Hirschel-Burns, 2023[68]).

Environment-related challenges remain a source of conflict and of insecurity for local communities in the areas surrounding mines. The pollution of the water and the loss of biodiversity are the most important among others. One example is Bikita mine-- Zimbabwe's largest lithium mining operation and one of the world's top producers of lithium-bearing minerals--which was accused of extracting vast amounts of water from a dam that was mainly reserved for the local people and where hundreds of people were reported to have faced water contamination with toxic substances and the sustainability of the ecosystem was undermined (Chitagu, 2024[69]).

There have also been allegations of corruption concerning the mines processing lithium. Questions have also raised about the transparency of the licensing procedures laws (Mkodzongi, 2024[64]). In early 2023, for example, companies, including the Sandawana mine, were revealed to have close ties to Zimbabwe’s ruling party, the military, or other entities sanctioned by the United States or the European Union (Global Witness, 2023[70]). According to some commentators, the lack of a robust governance rules, appears to favor large Chinese corporations and their affiliates, who, leveraging their political connections, continue to export lithium and transport large amounts of ores out of the country even after the imposition of the export ban (Global Witness, 2023[71]).

Annex D. Nickel: Supporting material

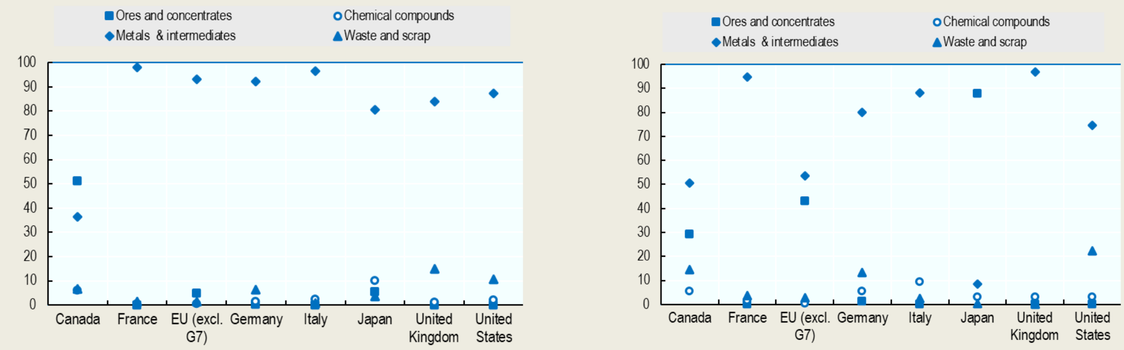
Box A D.1. G7 Participation in nickel value chains

The breakdown of nickel classifications provided in Table 3.1 helps to more accurately assess G7 countries' participation in international nickel trade, and further, how they might be affected by export restrictions. Metals and intermediates, the products classified under NFMB in Table 4.1 in the main text, represent the largest share of G7 Members' nickel imports reaching between 80% and 98% of total value of nickel imports, with the exception of Canada for which the figure is 36% as shown in the figure below. Canada is marked by a high share of ore and concentrate imports at 51% of the total value of nickel imports, however, for the remaining G7 countries the value of this import is marginal, at between 0% and 6%. Waste and scrap as well as chemical compounds comprise the remainder of nickel products and represent between 1% and 16% and between 0.4% and 10% respectively.

Figure A D.1. G7 countries participate in nickel trade mainly as importers of metals and intermediates

a. Share of value of imported nickel goods (in USD), 2022 of different forms of nickel in the total value of imported nickel goods (in USD), 2022

b. Share of different forms of nickel in the total volume of imported nickel goods (in kg), 2022

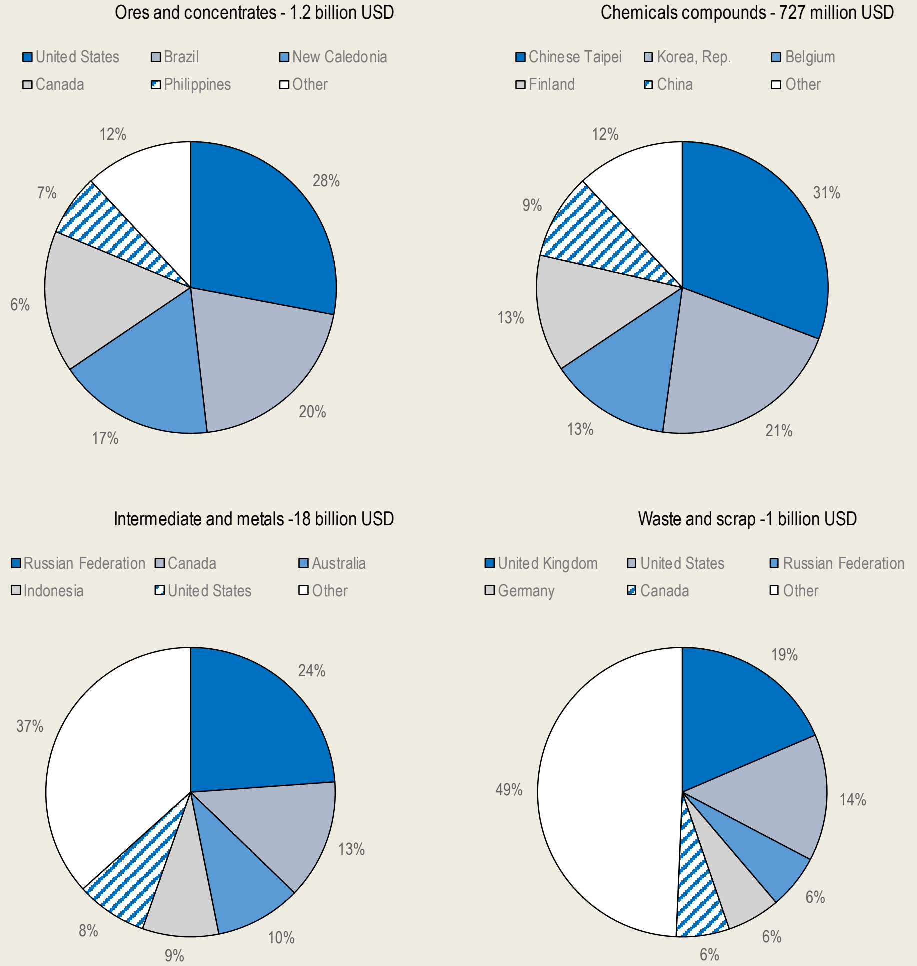


Note: HS-6-digit codes for products: ores and concentrates (OM), 260400; chemical compounds (CC) 282540, 282735, 283324; Metals and intermediates (NFMB), 750110, 750120, 750210, 750220, 750400, 750511, 750512, 750521, 750522, 750610, 750620; Waste and scrap (WS) 750300.
Source: UN COMTRADE.

The geographic origins of G7 nickel products imports, while overall more diverse than for lithium and cobalt, still indicates a degree of concentration. Imports of intermediates and metals were the largest nickel imports, in both volume and value terms for G7 countries in 2022. In 2022, just under 25% of these imports were sourced from Russia in trade worth roughly USD 4.5 billion as shown in the figure below. The other three categories display comparable levels of concentration; however, their combined value is only 16% of the value of intermediates and metal imports. Overall, there is significant inter-G7 trade for nickel, with the United States, Canada, France (New Caledonia), the United Kingdom, and EU countries all being major points of origin. While not as highly concentrated as other critical minerals discussed in this report, there is still exposure to supply risk, such as with the Russian position in intermediates and metals. That said, these figures predate sanctions on Russian nickel imposed by G7 countries in 2023, which will likely have reshaped G7 imports since 2022. In addition, as foreshadowed in the discussion of the recent nickel price fluctuations, it is not clear if Russia's shares in the world nickel markets observed, even prior to sanctions, were significant enough for Russia's supply fluctuations to cause fluctuations in international nickel prices.

Figure A D.2. The origins of G7 imports of nickel products are more diverse than for lithium and cobalt

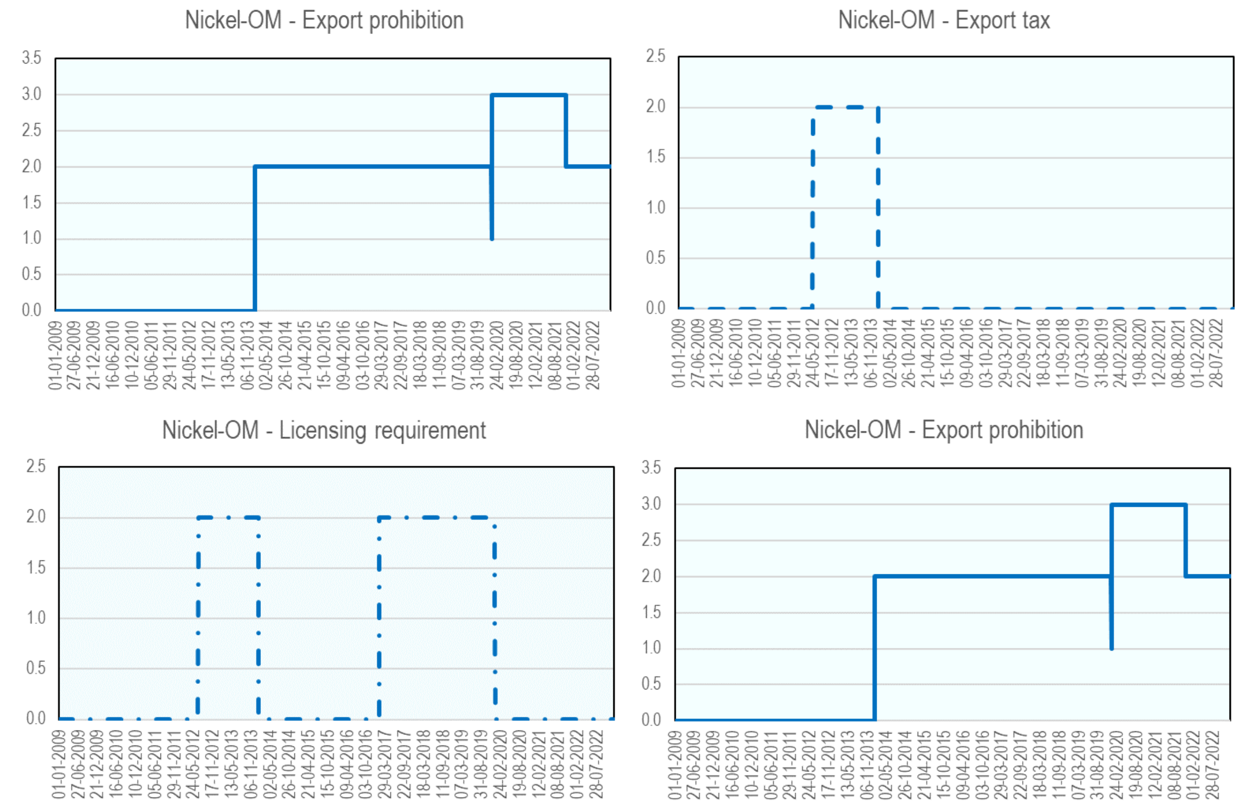
Imports of nickel products by origin. Chart titles indicate the processing stage of imported nickel and its total value in imports of G7 countries in 2022



Note: Figures exclude intra-EU trade and include imports of non-G7 EU Members.
Source: UN COMTRADE.

Figure A D.3. Indonesia's export restrictions on nickel ores and concentrates (OM), by type of export restriction

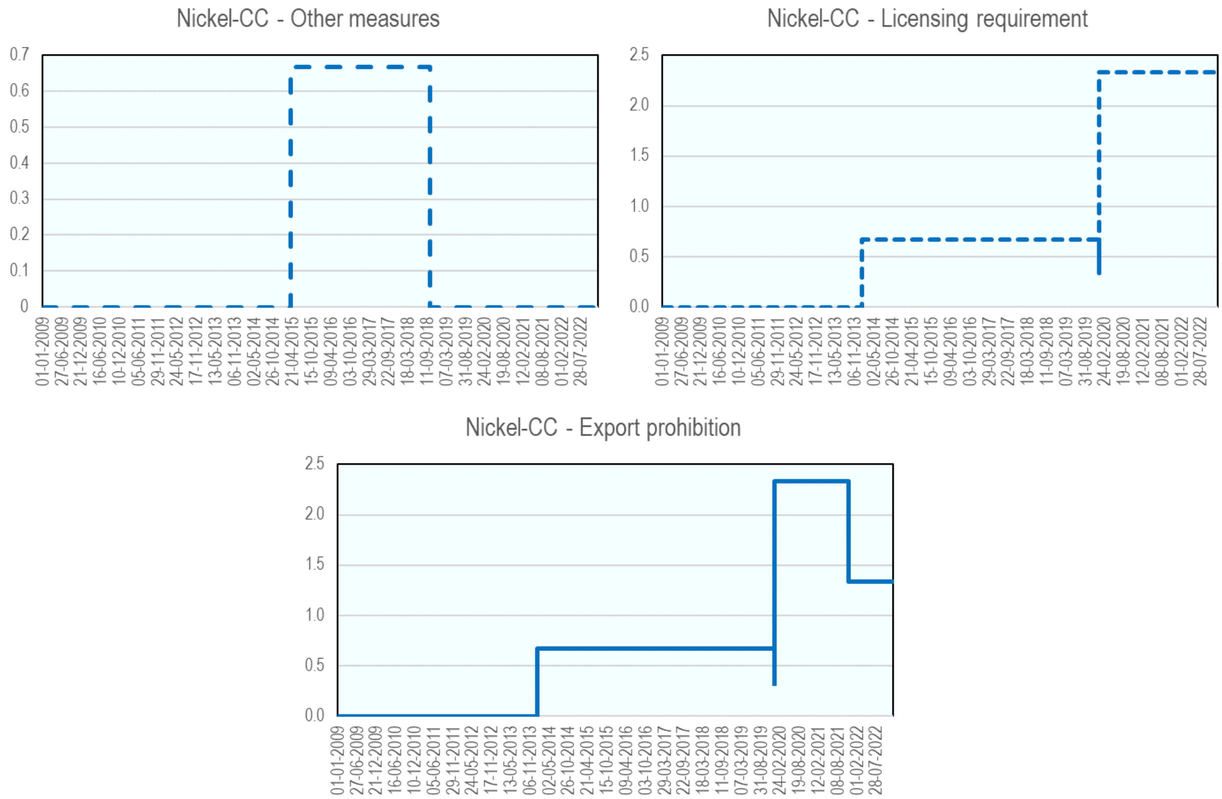
Average number of measures in force, per number HS6 "tariff lines" belonging to each category



Source: OECD (2025[1]).

Figure A D.4. Indonesia’s export restrictions on nickel chemical compounds (CC), by type of export restriction

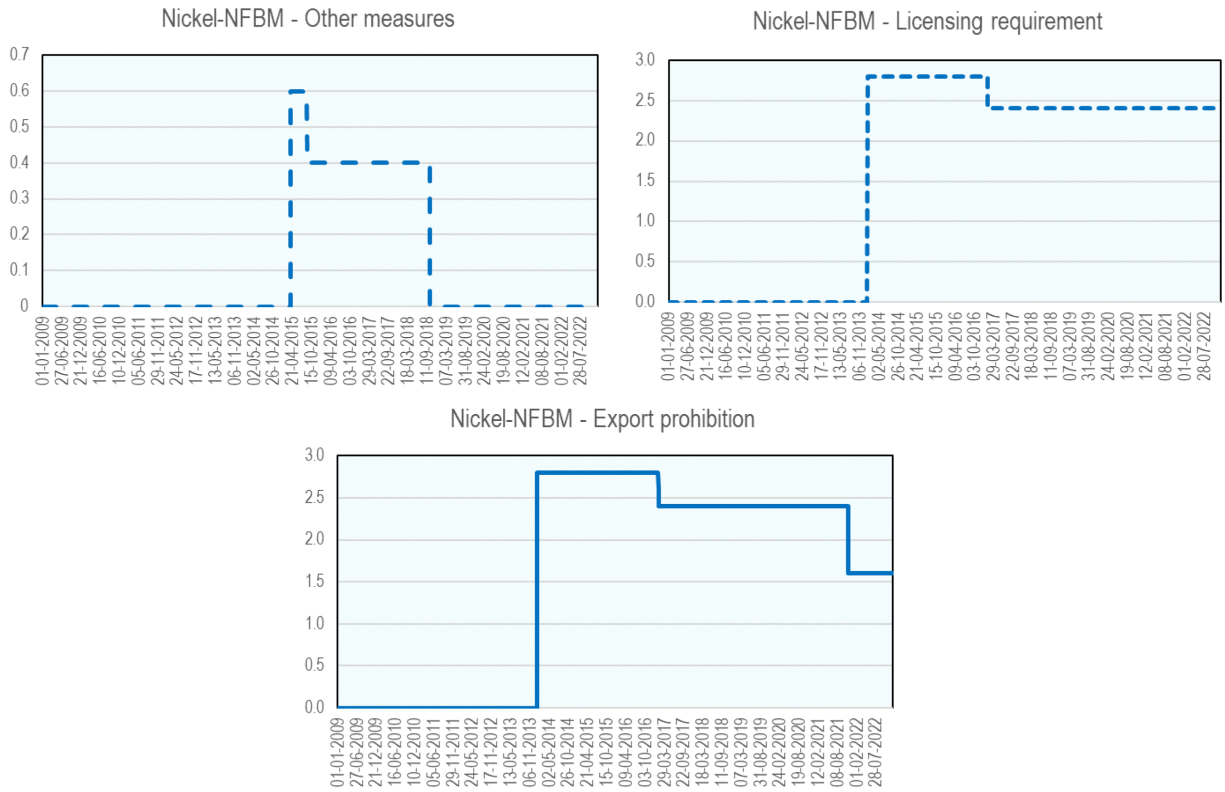
Average number of measures in force, per number HS6 “tariff lines” belonging to each category



Source: OECD (2025[1]).

Figure A D.5. Indonesia's export restrictions on non-fabricated basic metal (NFBM) products of nickel, by type of export restriction

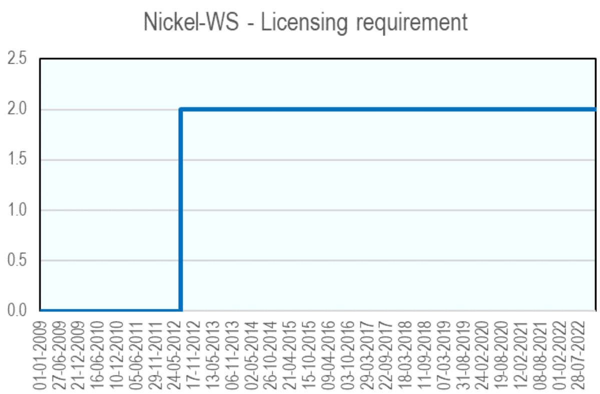
Average number of measures in force, per number HS6 "tariff lines" belonging to each category



Source: OECD (2025[1]).

Figure A D.6. Indonesia's export restrictions on waste and scrap of nickel (WS), by type of export restriction

Average number of measures in force, per number HS6 "tariff lines" belonging to each category



Source: OECD Inventory on Export Restrictions of Industrial Raw Materials.

Box A D.2. Broader economic and ESG impacts of the export restrictions in Indonesia

It has been widely reported that the Indonesian ban on exports of nickel ores was a part of long-lasting efforts of Indonesian authorities to attract Chinese investment [e.g. Tritto (2023[37])]. Immediately prior to the ban, China was the market for 82% of Indonesian nickel ore exports which were destined for use in China's stainless steel and other industries. At that time, nickel mining in Indonesia was not dominated by Chinese investors (Leruth et al., 2022[72]). The Indonesian stainless steel industry boom on the other hand, which materialised soon after the introduction of the export ban on unprocessed nickel[1] but also in presence of tax incentives and permitting simplifications, was driven almost entirely by Chinese investments (Tritto, 2023[37]). An assessment of the exceptional performance of exports of stainless steel from Indonesia, which were incidentally also directed mainly to China (Figure 3.8), therefore needs to consider the extent to which the benefits from this activity may have remained in Indonesia.

While a thorough cost-benefit analysis of this kind is beyond the scope of this report, existing literature suggests that while some benefits from such investments accrued to Indonesia and its population, an important share may have gone to Chinese stakeholders, both private investors and the Chinese state.

The example of Indonesia Morowali Industrial Park (IMIP) in Central Sulawesi, one of the largest Chinese nickel processing-related investments in Indonesia since the introduction of the export ban, as documented in detail by (Tritto, 2023[37]), is one such argument in this respect. While apparently propelled ultimately by Indonesia's ban on exports of raw materials, the USD 8 bln project would likely not have materialised without its elevation to a strategic China's national priority within its Belt and Road Initiative and support through very large Chinese state-linked funding from the China-ASEAN Investment Corporation Fund, a quasi-sovereign wealth fund linked to the Export-Import Bank of China, as well as loans from China's largest state policy bank China Development Bank (Tritto, 2023[37]).

According to Tritto (2023[37]), IMIP appears also to have had several traits of a typically export growth-oriented Special Economic Zone from China which are often disconnected from the local economy, as suggested by the reported presence in the park of illegal Chinese labour, discrimination in pay and treatment between Chinese and local workers, and higher incidence of occupational accidents and environmental damages.

Other positions in recent press and literature on the ESG impacts of Indonesia's nickel industry also highlight several critical concerns and underscore the need for comprehensive reforms.

For example, Jong (2025[73]), citing a recent report by C4ADS (2025[74]) which documents Indonesia's nickel refining capacity being controlled by Chinese companies, many of which are reported to have ties to the Chinese Government, argues that nickel mining operations have led to significant deforestation, particularly on the islands of Sulawesi and Halmahera and that this deforestation threatens biodiversity and disrupts local ecosystems. Wahil and Milko (2024[75]) report that, in regions like Kabaena and Raja Ampat, mining activities have resulted in water contamination with hazardous substances such as nickel, lead, and cadmium. This pollution has reportedly adversely affected local fisheries and agriculture, leading to economic losses and health issues among residents.

Jong (2025[73]) argues also that the rapid expansion of the nickel industry has been accompanied by reports of poor labour conditions ,with workers often endure long hours with inadequate safety measures, leading to numerous fatalities and injuries, with incidents often going unreported. Wahil and Milko (2024[75]) raise concerns about displacement and health problems of indigenous communities, such as the Bajau in Kabaena, due to mining activities where contaminated water sources have reportedly led to dermatological and respiratory issues among residents, and disruption of traditional livelihoods like fishing.

Other recent publications emphasised governance and policy challenges related to the rapid expansion of Indonesia's nickel sector. For example, PRAKARSA (2024[76])—an Indonesian institution for research and policy advocacy—pointed out that recent policy shifts, such as the 2020 Minerba Law and the Omnibus Law, have centralised authority over mining operations, reducing the role of local governments and civil society in decision-making processes. This centralisation has reportedly hindered effective oversight and accountability, leading to governance challenges in the nickel sector. The National Resource Governance Institute (2025[77]) argues also that, despite the establishment of initiatives like the Indonesia Sustainable Finance Taxonomy and the SIMBARA system for commodity traceability, enforcement of ESG standards remains weak and that lack of stringent regulations and oversight has allowed companies to operate with minimal accountability, exacerbating environmental and social issues.

1. Tritto (2023[37]) discusses some earlier attempts by Indonesian authorities to attract Chinese investments in the stainless-steel capacity in Indonesia which nevertheless did not come to fruition.

Table A D.1. List of stainless-steel products included in the analysis of nickel export restrictions by Indonesia

HS4 code	HS6 product code	HS6 product code description
7218	721810	Steel, stainless: ingots and other primary forms
7218	721890	Steel, stainless: semi-finished products
7219	721911	Steel, stainless: flat-rolled, width 600mm or more, hot-rolled, in coils, of a thickness exceeding 10mm
7219	721912	Steel, stainless: flat-rolled, width 600mm or more, hot-rolled, in coils, of a thickness of 4.75mm or more but not exceeding 10mm
7219	721913	Steel, stainless: flat-rolled, width 600mm or more, hot-rolled, in coils, of a thickness of 3mm or more but less than 4.75mm
7219	721914	Steel, stainless: flat-rolled, width 600mm or more, hot-rolled, in coils, of a thickness of less than 3mm
7219	721921	Steel, stainless: flat-rolled, width 600mm or more, hot-rolled, (not in coils), of a thickness exceeding 10mm
7219	721922	Steel, stainless: flat-rolled, width 600mm or more, hot-rolled, (not in coils), of a thickness of 4.75mm or more but not exceeding 10mm
7219	721923	Steel, stainless: flat-rolled, width 600mm or more, hot-rolled, (not in coils), of a thickness of 3mm or more but less than 4.75mm
7219	721924	Steel, stainless: flat-rolled, width 600mm or more, hot-rolled, (not in coils), of a thickness of less than 3mm
7219	721931	Steel, stainless: flat-rolled, width 600mm or more, cold-rolled or cold-reduced, of a thickness of 4.75mm or more
7219	721932	Steel, stainless: flat-rolled, width 600mm or more, cold-rolled, of a thickness of 3mm or more but less than 4.75mm
7219	721933	Steel, stainless: flat-rolled, width 600mm or more, cold-rolled, of a thickness exceeding 1mm but less than 3mm
7219	721934	Steel, stainless: flat-rolled, width 600mm or more, cold-rolled, of a thickness of 0.5mm or more but not exceeding 1mm
7219	721935	Steel, stainless: flat-rolled, width 600mm or more, cold-rolled, of a thickness of less than 0.5mm
7219	721990	Steel, stainless: flat-rolled, width 600mm or more, n.e.s. in heading no. 7219
7220	722011	Steel, stainless: flat-rolled, width less than 600mm, hot-rolled, of a thickness of 4.75mm or more
7220	722012	Steel, stainless: flat-rolled, width less than 600mm, hot-rolled, of a thickness of less than 4.75mm
7220	722020	Steel, stainless: flat-rolled, width less than 600mm, cold-rolled
7220	722090	Steel, stainless: flat-rolled, width less than 600mm, nes in heading no 7220
7221	722100	Steel, stainless: bars and rods, hot-rolled, in irregularly wound coils
7222	722210	Steel, stainless: bars and rods, hot-rolled, hot-drawn or extruded
7222	722220	Steel, stainless: bars and rods, cold-formed or cold-finished
7222	722230	Steel, stainless: bars and rods, n.e.s. in heading no. 7222
7222	722240	Steel, stainless: angles, shapes and sections
7223	722300	Steel, stainless: wire

Source: OECD calculations based on BACI and HS2007.